

OPTIMAL LOCAL LAWS AND CLT FOR THE CIRCULAR RIESZ GAS

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ABSTRACT. We study the long-range one-dimensional Riesz gas on the circle, a continuous system of particles interacting through a Riesz kernel. We establish near-optimal rigidity estimates on gaps valid at any scale. Leveraging these local laws together with Stein's method, we prove a quantitative Central Limit Theorem for linear statistics. The proof is based on a mean-field transport and a fine analysis of the fluctuations of local error terms using various convexity and monotonicity arguments. Using a comparison principle for the Helffer-Sjöstrand equation, the method can handle very singular test functions, including characteristic functions of intervals.

CONTENTS

1. Introduction	1
2. Preliminaries	12
3. The Helffer-Sjöstrand equation	18
4. Near-optimal rigidity	26
5. Optimal rigidity for singular linear statistics	41
6. Central Limit Theorem	57
Appendix A. Well-posedness of the H.-S. equation	61
Appendix B. Auxiliary estimates	64
References	74

1. INTRODUCTION

1.1. **Setting of the problem.** In this paper, we study the one-dimensional Riesz gas on the circle. We denote $\mathbb{T} := \mathbb{R}/\mathbb{Z}$. For a parameter $s \in (0, 1)$, let us consider the Riesz s -kernel on $\mathbb{T}$, defined by

$$g(x) = \lim_{n \rightarrow \infty} \left(\sum_{k=-n}^n \frac{1}{|k+x|^s} - \frac{2}{1-s} n^{1-s} \right) = \zeta(s, x) + \zeta(s, 1-x), \quad (1.1)$$

where $\zeta(s, x)$ stands for the Hurwitz zeta function [Ber72]. Note that g is the fundamental solution of the fractional Laplace equation on the circle

$$\begin{cases} (-\Delta)^{\frac{1-s}{2}} g = c_s(\delta_0 - 1) \\ \int g = 0, \end{cases} \quad (1.2)$$

with c_s given by

$$c_s = \frac{\Gamma(\frac{1-s}{2})}{\Gamma(\frac{s}{2})} \frac{\sqrt{\pi}}{2^{1-s}}. \quad (1.3)$$

We endow $\mathbb{T}$ with the natural order $x < y$ if $x = x' + k$, $y = y' + k'$ with $k, k' \in \mathbb{Z}$, $x', y' \in [0, 1)$ and $x' < y'$ and work on the set of ordered configurations

$$D_N = \{X_N = (x_1, \dots, x_N) \in \mathbb{T}^N : x_2 - x_1 < \dots < x_N - x_1\}.$$

On D_N let us consider the energy

$$\mathcal{H}_N : X_N \in D_N \mapsto N^{-s} \sum_{i \neq j} g(x_i - x_j), \quad (1.4)$$

where g is given by (1.1). The circular Riesz gas, at the inverse temperature $\beta > 0$, is defined by the Gibbs measure

$$d\mathbb{P}_{N,\beta} = \frac{1}{Z_{N,\beta}} \exp(-\beta \mathcal{H}_N(X_N)) \mathbb{1}_{D_N}(X_N) dX_N,$$

where $Z_{N,\beta}$ is the normalization constant, called *partition function*, given by

$$Z_{N,\beta} = \int_{D_N} e^{-\beta \mathcal{H}_N(X_N)} dX_N.$$

Throughout the paper, s is a fixed parameter in $(0, 1)$.

The choice of normalization in the definition of the energy (1.4) is a natural choice because it makes β the effective inverse temperature governing the microscopic scale behavior.

The model described above belongs to a family of interacting particle systems called *Riesz gases*. On $\mathbb{R}^d$, these are associated to a kernel of the form $|x|^{-s}$ with $s > 0$. The Riesz family also contains in dimensions 1 and 2 the so-called *log-gases* with kernel $-\log|x|$. For $d \geq 3$ and $s = d - 2$, $|x|^{-s}$ is the fundamental solution of the Laplace equation on $\mathbb{R}^d$ and therefore corresponds to the Coulomb interaction. The parameter s determines the singularity as well as the *range* of the interaction. When $s > d$, the interaction is short-range and the system, referred to as *hypersingular Riesz gas*, resembles a nearest-neighbor model. For $s \in (0, d)$ or $s = 0$ and $d = 1, 2$, Riesz gases are long-range particle systems, which, as such, have attracted much attention both in mathematical and physical contexts.

The 1D log-gas, also called the β -ensemble, has been extensively studied in the last few decades, partly because of its connection to random matrix theory. Indeed, it corresponds, in the cases $\beta \in \{1, 2, 4\}$, to the distribution of the eigenvalues of $N \times N$ symmetric/Hermitian/symplectic random matrices with independent Gaussian entries (see the original paper of Dyson [Dys62]). The 1D log-gas also appears in many other contexts, such as zeros of random polynomials, zeros of the Riemann zeta function, and is conjectured to be related to the eigenvalues of random Schrödinger operators [Lew22]. The 2D Coulomb gas is another fundamental model. Among many other examples, it is connected to non-Hermitian random matrices, Ginzburg-Landau vortices, Fekete points, complex geometry, the XY model, and the KT transition [Ser18]. For other values of s , let us mention that the case $s = 2$ in dimension 1 is an integrable system, called *classical Calogero-Sutherland model*. The study of minimizers of Riesz interactions is also a dynamic topic [HS05, CSW21] and is the subject of long-standing conjectures related to sphere packing problems [BL15]. From a statistical physics perspective, even in dimension 1, the Riesz gas is not fully elucidated, since the classical theory of the 60-70s [Rue74] cannot be applied due to the long-range nature of the interaction. The reader may refer to the nice review [Lew22], where an account of the literature and many open problems on Riesz gases are given.

As the number of particles N tends to infinity, the empirical measure

$$\mu_N := \frac{1}{N} \sum_{i=1}^N \delta_{x_i}$$

converges almost surely under $\mathbb{P}_{N,\beta}$ (in a suitable topology) to the uniform measure on the circle. This result can be obtained through standard large deviations techniques (see, for instance, [Ser14, Ch. 2] for the case of Riesz gases on the real line, which adapts readily to the periodic setting). In the large N limit, particles tend to spread uniformly in the circle, which suggests that particles spacing (or gaps) $N(x_{i+k} - x_i)$ concentrate around the value k . The first goal of this paper is to quantify the fluctuations of the gaps around their mean. We establish the optimal size of the fluctuations of $N(x_{i+k} - x_i)$, which turns out to be in $O(\beta^{-\frac{1}{2}} k^{\frac{s}{2}})$, as conjectured in the recent physics paper [SKA+21]. This type of result is referred to in the literature as a rigidity estimate. It was intensively investigated for β -ensembles (see, for instance, [BEY12, BEY+14b, BEY14a, BMP22]), but the correct observable in that case is $x_i - \gamma_i$, where x_i is the i -th particle and γ_i the classical location of the i -th particle, that is, the corresponding quantile of the equilibrium measure arising in the mean-field limit.

A complementary way to study the rigidity of the system is to investigate the fluctuations of linear statistics of the form

$$\text{Fluct}_N[\xi(\ell_N^{-1} \cdot)] := \sum_{i=1}^N \xi(\ell_N^{-1} x_i) - N \ell_N \int_{\mathbb{T}} \xi, \quad (1.5)$$

where $\xi : \mathbb{T} \rightarrow \mathbb{R}$ is a given measurable test function and (ℓ_N) a sequence of numbers in $(0, 1]$. For smooth test functions, many central limit theorem (CLT) results are available in the literature on 1D-log gases, including [Joh98, Shc13, BG13, BEY12, BEY⁺14b, BLS⁺18, BMP22, HL21]. For 1D Riesz gases with $s \in (0, 1)$, to our knowledge, no prior results on CLT for linear statistics are known. In this paper, we obtain a *quantitative* CLT for (1.5), which is valid at all scales (ℓ_N) down to microscopic scales $\ell_N \gg \frac{1}{N}$. A major direction in random matrix theory is to establish CLTs for (1.5) allowing for test functions which are *as singular as possible*. Indeed, it is a natural question to capture the fluctuations of the number of points and of the logarithmic potential, which are key observables for the log-gas. The question of optimal regularity of the test function has also drawn some interest because it encapsulates non-universality phenomena in the context of Wigner matrices. One of the goals of this paper is to provide a robust method that allows singular test functions to be treated in a systematic way. The main question we investigate therefore is *regularity issue*. Using new concentration inequalities, we can treat singular test functions, including characteristic functions of intervals and inverse power functions up to the critical power $\frac{s}{2}$. In particular, we obtain a CLT for the number of points, which extends to a Riesz setting some of the recent results of [BMP22].

Let us now introduce the main tools and objects used in the proofs. For any reasonable Gibbs measure on D_N (or $\mathbb{R}^N$), the fluctuations of any (smooth) statistic $F : D_N \rightarrow \mathbb{R}$ are related to the properties of a partial differential equation called the Helffer-Sjöstrand (H.-S.) equation, which is sometimes referred to as a Witten Laplacian (on 1-forms). This equation appears in [Sjo93a, Sjo93b, HS94]. It is more extensively studied in [Hel98b, Hel98a, NS97], where it is used to establish correlation decay, uniqueness of the limiting measure, and log-Sobolev inequalities for models with convex interactions. The purpose of the present paper is to show how the analysis of Helffer-Sjöstrand equations provides powerful tools to study the fluctuations of linear statistics with singular test functions.

The proof of the near-optimal rigidity is essentially similar to [BEY⁺14b]. It exploits the convexity of the interaction and is thus very specific to 1D systems. The method is mainly based on a concentration inequality for divergence-free functions and on a key convexity result due to Brascamp and Lieb [BL02].

The method of proof of the CLT for linear statistics starts with performing the mean-field transport argument usually attributed to Johansson [Joh98]. When studying the Laplace transform of linear statistics $\text{Fluct}_N[\xi]$, this consists in applying a well-chosen change of variables on each point, depending only on its position, to transport the uniform measure on the circle to the perturbed equilibrium measure (perturbed by the effect of adding $t\text{Fluct}_N[\xi]$ to the energy). In this paper, variances are computed instead of Laplace transforms and the implementation of the transport of [Joh98] takes the form of an integration by parts. This argument is a variation of the so-called loop equations (see [BEY⁺14b] for various comments on this topic). It is the starting point of many CLTs in Coulomb gases and β -ensembles, but received a more systematic analysis in the series of works [LS18, BLS⁺18, Leb18, LZ20, Ser20]. One can also interpret this transport as a mean-field approximation of the H.-S. equation associated with (the gradient of) linear statistics.

Since the transport is an approximate solution (a mean-field approximation) of the H.-S. equation, it creates an error term, sometimes called a loop equation term, which is essentially a local weighted energy and the heart of the problem is to estimate its fluctuations. In contrast, in [LS18, BLS⁺18, Leb18, Ser20] the typical size of this error term is evaluated in the sense of large deviation, rather than the size of its fluctuations. However, we point out that the latter seems intractable in dimension $d \geq 2$ because of the lack of convexity. The core of this paper is about the control of the fluctuations of this loop equation term through the analysis of the related H.-S. equation. Our proof is based on the Brascamp-Lieb inequality, the near-optimal rigidity estimates on gaps and nearest-neighbor distances, and a comparison principle for the Helffer-Sjöstrand equation.

The use of the comparison principle mentioned above (also known in [Car74, HS94]) is one of the main novelties of the article. This is the key technical tool for treating linear statistics with singular test functions. Indeed, after performing loop equation techniques, we will study singular local quantities for which standard concentration inequalities, such as the Brascamp-Lieb or log-Sobolev inequalities, do not give the right order of fluctuations.

The central limit theorem is then obtained from a rather straightforward application of Stein's method. We show how the mean-field transport naturally leads to an approximate Gaussian integration by parts formula. As a result, quantifying normality reduces to controlling the variance of the loop equation term.

1.2. Main results. Throughout, we fix a long-range parameter $s \in (0, 1)$.

1.2.1. Rigidity of gaps and local particle numbers. Our first theorem extends the rigidity estimate of [BEY+14b, Thm. 3.1] from the log gas to the one-dimensional Riesz gas.

Theorem 1 (Near-optimal rigidity). *For $\varepsilon > 0$ set $\delta(\varepsilon) := \frac{\varepsilon}{4(s+2)}$. Then, there exists $\varepsilon_0 > 0$ such that for every $\varepsilon \in (0, \varepsilon_0)$ the following holds. There exist constants $c > 0$ and $C > 0$ depending on β , s and ε such that:*

(1) (Gap fluctuations) For every $i \in \{1, \dots, N\}$ and $k \in \{1, \dots, \lfloor \frac{N}{2} \rfloor\}$,

$$\mathbb{P}_{N,\beta}(|N(x_{i+k} - x_i) - k| \geq k^{\frac{s}{2}+\varepsilon}) \leq Ce^{-ck^{\delta(\varepsilon)}}. \quad (1.6)$$

(2) (Discrepancy in an interval) For every $a \in \mathbb{T}$ and $\ell_N \in (0, \frac{1}{4}]$,

$$\mathbb{P}_{N,\beta}\left(\left|\sum_{i=1}^N \mathbb{1}_{(a-\ell_N, a+\ell_N)}(x_i) - 2N\ell_N\right| \geq (N\ell_N)^{\frac{s}{2}+\varepsilon}\right) \leq Ce^{-c(N\ell_N)^{\delta(\varepsilon)}}. \quad (1.7)$$

1.2.2. Singular linear statistics: variance expansion. Next, we prove sharp variance estimates, valid at any scale, for possibly singular test functions satisfying the following assumption:

Assumption 1.1 (Admissible test function). *Let $\ell_N \in (0, 1]$ and let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$. Let ψ be given by*

$$\psi' = (-\Delta)^{\frac{1-s}{2}} \xi \quad \text{and} \quad \int \psi = 0.$$

(A1) **Global Hölder regularity.** *There exists $\varepsilon > 0$ such that $\xi' = \chi$ for some $\chi \in \mathcal{C}^{1-s+\varepsilon}(\ell_N^{-1}\mathbb{T})$.*

(A2) **Piecewise $\mathcal{C}^{2-s}$ regularity.** *For some integer $p \geq 0$ and points $a_1 < \dots < a_p < 1$, letting $a_{p+1} := a_1$, we have that ψ is $\mathcal{C}^2$ on every open arc (a_j, a_{j+1}) .*

(A3) **Controlled algebraic singularities.** *There exists a constant $C > 0$ and an exponent $\alpha \in [s-1, \frac{s}{2})$ such that*

$$|\psi''(x)| \leq C \left(\sum_{j=1}^p \frac{1}{|x - a_j|^{2-s+\alpha}} \mathbb{1}_{|x| \leq 1} + \frac{1}{|x|^{3-s}} \mathbb{1}_{|x| \geq 1} \right), \quad x \in \ell_N^{-1}\mathbb{T} \setminus \{a_1, \dots, a_p\}. \quad (1.8)$$

(A4) **Support.** *Either $\ell_N \equiv 1$, or $\ell_N \rightarrow 0$ and, in this case, ξ is supported in $(-\frac{1}{2}, \frac{1}{2})$.*

Remark 1.2. *We can observe that, given $\alpha > 0$, the function $x \in \mathbb{T} \setminus \{0\} \mapsto |x|^{-\alpha}$ satisfies Assumption 1.1 if and only if $\alpha < \frac{s}{2}$. Moreover, the function $\mathbb{1}_{(a,b)}$ satisfies Assumption 1.1 (as opposed to the case where $s = 0$).*

Let us recall the definition of the fractional Sobolev seminorm on the circle $\|\cdot\|_{H^\alpha(\mathbb{T})}$, for $\alpha > 0$. Let $h \in L^2(\mathbb{T})$ with Fourier coefficients $\hat{h}(k)$, $k \in \mathbb{Z}$. Whenever it is finite, we call $\|h\|_{H^\alpha(\mathbb{T})}^2$ the quantity

$$\|h\|_{H^\alpha(\mathbb{T})}^2 := \sum_{k \in \mathbb{Z}} |k|^{2\alpha} |\hat{h}(k)|^2.$$

Similarly, the fractional Sobolev seminorm of a function $h \in L^2(\mathbb{R})$, denoted $\|h\|_{H^\alpha(\mathbb{R})}$, is defined by

$$\|h\|_{H^\alpha(\mathbb{R})}^2 := \int_{\mathbb{R}} |\xi|^{2\alpha} |\hat{h}(\xi)|^2 d\xi, \quad (1.9)$$

where $\hat{h}$ stands for the Fourier transform of h .

Let (ℓ_N) be a sequence taking values in $(0, 1]$. For any function $\xi \in H^{\frac{1-s}{2}}(\ell_N^{-1}\mathbb{T})$, define

$$\sigma_{\ell_N}^2(\xi) := \frac{(2\pi)^{1-s}}{2\beta c_s} \ell_N^{-s} \|\xi(\ell_N^{-1}\cdot)\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2. \quad (1.10)$$

If $\xi \in H^{\frac{1-s}{2}}(\mathbb{T})$ is supported on $(-\frac{1}{2}, \frac{1}{2})$, let $\xi_0 : \mathbb{R} \rightarrow \mathbb{R}$ be given for all $x \in \mathbb{R}$ by

$$\xi_0(x) = \begin{cases} \xi(x) & \text{if } |x| \leq \frac{1}{2} \\ 0 & \text{if } |x| > \frac{1}{2} \end{cases} \quad (1.11)$$

and

$$\sigma_0^2(\xi) := \frac{(2\pi)^{1-s}}{2\beta c_s} \|\xi_0\|_{H^{\frac{1-s}{2}}(\mathbb{R})}^2. \quad (1.12)$$

Theorem 2 (Variance of singular linear statistics). *Let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$ and (ℓ_N) satisfy Assumption 1.1. Let $\alpha \in [s-1, \frac{s}{2})$ be such that (1.8) holds.*

For all $\varepsilon > 0$, there exists a constant $C > 0$ depending on β, s, ε and ξ such that

$$\left| \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[\xi(\ell_N^{-1}\cdot)]] - (N\ell_N)^s \sigma_{\ell_N}^2(\xi) \right| \leq C(N\ell_N)^{\max(2s-1, 2\alpha)+\varepsilon}. \quad (1.13)$$

Moreover, there exists a constant $C > 0$ depending on s and ξ such that

$$|\sigma_{\ell_N}^2(\xi) - \sigma_0^2(\xi)| \leq C\ell_N^{2-s}.$$

Since $\alpha < \frac{s}{2}$, note that $\xi \in H^{\frac{1-s}{2}}(\ell_N^{-1}\mathbb{T})$ and that the error term in (1.13) is $o((N\ell_N)^s)$.

Theorem 2 implies that the number variance of the 1-D Riesz gas grows like N^s , the same order as for any smooth linear statistic. This growth rate interpolates smoothly between:

- the log-gas case $s = 0$, where smooth statistics have variance of order 1 but where interval counts fluctuate like $\log N$ because $\mathbb{1}_{(-a,a)} \notin H^{1/2}(\mathbb{T})$ [BLS⁺18], and
- the Poisson-like case $s = 1$, where the variance is linear in N .

Thus for every $s \in (0, 1)$ the Riesz gas is *hyperuniform* in the sense of [Tor16]: the number-variance is much smaller than for i.i.d variables.

1.2.3. Central limit theorem. Let d_{BL} be the bounded-Lipschitz distance between probability measures on $\mathbb{R}$, defined for all μ and ν by

$$d_{\text{BL}}(\mu, \nu) := \sup_{f \in \mathcal{C}^1(\mathbb{R})} \left\{ \int f d(\mu - \nu) : \|f\|_{L^\infty(\mathbb{R})} \leq 1, \|f'\|_{L^\infty(\mathbb{R})} \leq 1 \right\}.$$

We establish the following result:

Theorem 3 (CLT for singular linear statistics). *Let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$ and (ℓ_N) satisfy Assumption 1.1. Let $\alpha \in [s-1, \frac{s}{2})$ be such that (1.8) holds. Assume in addition that $\ell_N \gg N^{-1}$. Let $Z_N \sim \mathcal{N}(0, \sigma_{\ell_N}^2(\xi))$ and for $\ell \in \{0, 1\}$, let $Z_\ell \sim \mathcal{N}(0, \sigma_\ell^2(\xi))$.*

- (1) *The random variables $(N\ell_N)^{-s/2} \text{Fluct}_N[\xi(\ell_N^{-1}\cdot)]$ converge in distribution to Z_ℓ , where $\ell := \lim_{N \rightarrow \infty} \ell_N \in \{0, 1\}$.*
- (2) *For every $\varepsilon > 0$, there exists a constant $C > 0$ depending on β, s, ε and ξ such that*

$$d_{\text{BL}}(\text{Law}((N\ell_N)^{-s/2} \text{Fluct}_N), \text{Law}(Z_N)) \leq C(N\ell_N)^{\varepsilon + \max(-\frac{s}{2}, \frac{s-1}{2}, \alpha - \frac{s}{2})}.$$

Theorem 3 can be interpreted as the convergence of the field

$$\sum_{i=1}^N g(x_i - \cdot) - N \int g(x - \cdot) dx$$

to a fractional Gaussian field. Observe that if ξ and χ are smooth test functions with disjoint support, then the asymptotic covariance is, in general, not equal to 0, meaning that the corresponding fractional field does not exhibit spatial independence. This reflects the non-local nature of the fractional Laplacian $(-\Delta)^{\frac{1-s}{2}}$ for $s \in (0, 1)$.

1.2.4. Examples. By Remark 1.2, Theorem 3 gives a CLT for the number of points in an interval and for the observable gaps.

Corollary 1 (CLT for the number of points in an interval). *Let $a \in (0, \frac{1}{2})$ and (ℓ_N) be a sequence taking values on $(0, 1]$ such that $\ell_N \gg \frac{1}{N}$. We have*

$$\frac{1}{\sqrt{N^s \zeta(-s, 2a\ell_N)}} \left(\sum_{i=1}^N \mathbb{1}_{(-a\ell_N, a\ell_N)}(x_i) - 2aN\ell_N \right) \xrightarrow[\text{Law}]{} \mathcal{N}(0, \sigma^2),$$

where

$$\sigma^2 := \frac{2\cotan(\frac{\pi}{2}s)}{\beta 4^s \pi s}. \quad (1.14)$$

Let (k_N) be a sequence taking values in $\left\{1, \dots, \lfloor \frac{N}{2} \rfloor\right\}$ such that $k_N \rightarrow \infty$ as $N \rightarrow \infty$. We have

$$\frac{1}{\sqrt{N^s \zeta(-s, \frac{k_N}{N})}} (N(x_{1+k_N} - x_1) - k_N) \xRightarrow[\text{Law}]{} \mathcal{N}(0, \sigma^2),$$

where σ^2 is as in (1.14).

Corollary 1 is an extension of the results on the fluctuations of single particles in the bulk for β -ensembles, see [Gus05, CFLW21] for the GUE case and [BMP22].

Corollary 2 (CLT for power-law observables). *Let $\alpha \in (0, \frac{s}{2})$. Then,*

$$N^{-\frac{s}{2}} \left(\sum_{i=1}^N \zeta(-\alpha, x_i) - N \int \zeta(-\alpha, x) dx \right) \xRightarrow[\text{Law}]{} \mathcal{N}(0, \sigma^2),$$

where

$$\sigma^2 := (2\pi)^{-1+2\alpha-s} \frac{c_\alpha^2}{\beta c_s} \zeta(1+s-2\alpha),$$

with c_α and c_s are as in (1.3).

Remark 1.3. *The borderline observable $x \mapsto |x|^{-s/2}$ lies exactly at the edge of $H^{\frac{1-s}{2}}(\mathbb{T})$; for $s = 0$ it reduces to the classical cases $\mathbb{1}_{(-a,a)}$ and $-\log|x|$ whose log-correlated CLT is proved in [BMP22].*

1.3. Context, related results, open questions.

1.3.1. Rigidity of β -ensembles. As mentioned in the Introduction, Theorems 1 and 3 are natural extensions to the circular Riesz gas of some known results on the fluctuations of β -ensembles. We refer again to [BEY12, BEY⁺14b, BEY14a, BMP22] for rigidity estimates, to [Joh98, Shc13, BG13, BLS⁺18, LLW⁺19, HL20, Pei22] for CLTs for linear statistics with smooth test functions, and to [HL20, Lam21] for the case of the circular β -ensemble. In the case of the GUE, that is for $\beta = 2$ with a quadratic potential, a CLT for test functions in $H^{\frac{1}{2}}$ is obtained in [SW13] using a Littlewood-Paley-type decomposition argument. However, as observed in [Lam21, Rem. 1.3], the minimal regularity of the test function depends on β . Indeed, for $\beta = 4$, leveraging variance expansions of [JM15], [Lam21] exhibits a test function in $H^{\frac{1}{2}}$ such that the associated linear statistic does not have a finite limit. Since the characteristic function of a given interval is not in $H^{\frac{1}{2}}$, the asymptotic scaling of discrepancies in intervals is not of order 1. It is proved in [Gus05] that for the GUE, eigenvalues x_i in the bulk fluctuate in $O(\sqrt{\log i})$ and that discrepancies are of order $\sqrt{\log N}$. A general CLT for the characteristic functions of intervals and for the logarithm function is given in the recent paper [BMP22]. Concerning the method of proof, let us point out a very similar variation on Stein's method developed in [LLW⁺19], see also [HL20] for a high-temperature regime.

1.3.2. Local laws and fluctuations for the Langevin dynamics. A related and much-studied question concerns the rigidity of the Dyson Brownian motion, an evolving gas of particles whose invariant distribution is given by β -ensemble. The time to equilibrium at the microscopic scale of Dyson Brownian motion was studied in many papers including [ESY11, EY12], see also [Bou21] for optimal relaxation estimates. A central limit theorem at mesoscopic scale for linear statistics of the Dyson Brownian motion is established in [HL16], thus exhibiting a time-dependent covariance structure.

1.3.3. Decay of the correlations and Helffer-Sjöstrand representation. The decay of gap correlations of β -ensembles has been extensively studied in [EY15], where a power-law decay in the inverse squared distance is established. The starting point of [EY15] is based on a representation of the correlation function by a random walk in a dynamic random environment or in other words on a dynamic interpretation of the Helffer-Sjöstrand operator. The paper [EY15] then develops a sophisticated homogenization theory for a system of discrete parabolic equations. In a different context, a more direct analysis of the Helffer-Sjöstrand operator has been developed in the ground work [NS97] to characterize the scaling limit of the gradient interface model in an arbitrary dimension $d \geq 1$. Combining ideas from [NS97] and from quantitative stochastic homogenization, the paper [AW19] then shows that the free energy associated with this model is at least $C^{2,\alpha}$ for some $\alpha > 0$. We also refer to the recent paper [Tho21] which studies in a similar framework the scaling limit of the non-Gaussian membrane model. In non-convex settings,

far fewer results are available in the literature. One can mention the work [DW20] which establishes the optimal decay for the two-point correlation function of the Villain rotator model in $\mathbb{Z}^d$, for $d \geq 3$ at low temperature. It could be interesting to develop a direct method to analyze the large-scale decay of the Helffer-Sjöstrand equation in the context of one-dimensional Riesz gases. We plan to address this question in future work.

1.3.4. Uniqueness of the limiting point process. The question of the decay of the correlations mentioned above is related to a property of uniqueness of the limiting measure. One expects that after rescaling, chosen so that the typical distance between consecutive points is of order 1, the point process converges, in a suitable topology, to a certain point process Riesz_β . For $s = 0$, the limiting point process called Sine_β is unique and universal as proved in [BEY12, BEY⁺14b]. The existence of a limit was first established in [VV09] for β -ensembles with quadratic exterior potential, together with a sophisticated description and in [KS09] for the circular β -ensemble. The Sine_β process has also been characterized as the unique minimizer of the free energy functional governing the microscopic behavior in [EHL18] using a displacement convexity argument. In [Bou22], we prove the existence of a limiting point process Riesz_β for the circular Riesz ensemble.

1.3.5. 1D hypersingular Riesz gases. Although the 1D hypersingular Riesz gas (i.e. $s > 1$) is not hyperuniform, its fluctuations are also of interest. In such a system, the macroscopic and microscopic behaviors are *coupled*, a fact which translates into the linear response associated with linear statistics (in contrast with long-range particle systems, the linear response is a combination of a mean-field change of variables—moving each point according only to its position—and local perturbations). Simple heuristic computations show that the limiting variance is then proportional to an L^2 norm (after subtraction of the mean) with a factor depending on the second-order derivative of the free energy.

1.3.6. Fluctuations of Riesz gases in higher dimension. For $d \geq 1$ and s smaller than d , the existence of a thermodynamic limit for the Riesz gas (after extraction of a subsequence) is delicate as the energy is long-range. It was obtained in [LS17] for $s \in (\min(d-2, 0), d)$, leveraging an electric formulation of the Riesz energy, see [PS17], and a screening procedure introduced in [SS12] and then improved in [RS13, PS17]. The first task to study the fluctuations of higher dimensional long-range Riesz gases is to establish local laws, that is, to control the number of points and the energy in cubes of small scales. This was done for the Coulomb gas in arbitrary dimensions down to the microscopic scale in the paper [AS19] using sub-additive and super-additive approximating energies. Due to the lack of convexity, establishing a CLT or even a Poissonian rigidity estimate for linear statistics of Riesz gases in arbitrary dimensions is a delicate task. In dimension 2, since long-range interactions are overwhelmingly dominant, a CLT for linear statistics with smooth test functions can be proved, see [LS18, BBNY16, Leb17], without proving any “probabilistic cancellation” on local quantities, but only a “quenched cancellation” on some angle term. Let us finally mention the work [Leb21] where the 2D Coulomb gas is shown to be hyperuniform, meaning that the variance of the number of points in a ball scales much smaller than the volume. The paper [Leb21] establishes an important quantitative translation invariance property based on refinements of Mermin-Wagner type arguments, see also [Tho22]. In higher dimensions, far fewer results are available. One can mention the result of [Ser20] which treats the 3D Coulomb gas at high temperature “under a no phase transition assumption”. A simpler variation of the 3D Coulomb gas, named *hierarchical* Coulomb gas, has also been investigated in the work [Cha19], followed by [GS20].

1.4. Outline of the main proofs.

Rigidity. The proof of Theorem 1 is similar to the proof of [BEY⁺14b, Th. 3.1]. Let us explain the main steps.

First, we establish a local law on gaps: for all $\delta > 0$, there exist $\kappa > 0$, $c > 0$, $C > 0$ such that for every $k = 1, \dots, \lfloor N/2 \rfloor$ and every $i = 1, \dots, N$,

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq C e^{-ck^\kappa}. \quad (1.15)$$

This estimate is derived via a bootstrap procedure combined with certain concentration inequalities, whose details we omit here.

Let us explain how to prove Theorem 1 from the estimate (1.15). Let d be the distance on $\{1, \dots, N\}$ given for all $i, j \in \{1, \dots, N\}$ by $d(i, j) = \min(|j - i|, N - |j - i|)$. For every $k = 1, \dots, \lfloor N/2 \rfloor$ and

$i = 1, \dots, N$, define the block of size k centered around i ,

$$I_k(i) := \{j \in \{1, \dots, N\} : d(i, j) \leq k\}$$

and the average of $(x_j)_j$ over the block $I_k(i)$,

$$x_i^{[k]} := \frac{1}{2k+1} \sum_{j \in I_k(i)} x_j.$$

Fix k and i , and let $p \geq 1$ be a large number. Set $\alpha := \frac{1}{p}$. The idea in [BEY⁺14b] is to decompose x_i as follows:

$$x_i = x_i^{[k]} + \sum_{m=0}^{p-1} \left(x_i^{[\lfloor k^{m\alpha} \rfloor]} - x_i^{[\lfloor k^{(m+1)\alpha} \rfloor]} \right).$$

For each $k = 1, \dots, p-1$, denote

$$G_m := N \left(x_i^{[\lfloor k^{m\alpha} \rfloor]} - x_i^{[\lfloor k^{(m+1)\alpha} \rfloor]} \right).$$

One may note that as m decreases, $|\nabla G_m|^2$ becomes larger, while G_m becomes more localized, since it depends only on the variables in $J_{m+1} := I_{[\lfloor k^{(m+1)\alpha} \rfloor]}(i)$. We therefore need to exploit the convexity of the interactions within J_{m+1} . To gain uniform convexity, we consider a smooth, non-negative, convex function $\theta : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $\theta(x) = 0$ for $x \in [0, 1]$ and $\theta(x) = x^2$ for $x > 2$. For some fixed $\varepsilon > 0$, we define the forcing term

$$F_m := 2 \sum_{i, j \in J_{m+1} : i < j} \theta \left(\frac{N(x_j - x_i)}{k^{1+\varepsilon}} \right)$$

and introduce the locally constrained measure

$$d\mathbb{Q}_{N,\beta}(X_N) \propto e^{-\beta(\mathcal{H}_N + F_m)(X_N)} \mathbb{1}_{D_N}(X_N) dX_N.$$

Let μ be the law of $(x_i)_{i \in J_{m+1}}$ when X_N is distributed according to $\mathbb{Q}_{N,\beta}$. Using the Brascamp-Lieb inequality, one can show that the density of μ can be written as

$$d\mu(x) = e^{-\beta(H + F_m(x) + \tilde{H}(x))} \mathbb{1}_{D_{|J_{m+1}|}}(x) dx,$$

with $\nabla^2 \tilde{H} \geq 0$ and H defined by

$$H : x = (x_i)_{i \in J_{m+1}} \in D_{|J_{m+1}|} \mapsto N^{-s} \sum_{i, j \in J_{m+1} : i \neq j} g(x_j - x_i).$$

By construction, for all $U \in \mathbb{R}^{|J_{m+1}|}$, we have

$$\begin{aligned} U \cdot \nabla^2(H + F_m)U &= 2 \sum_{i < j \in J_{m+1}} \left(N^{-s} g''(x_j - x_i) + \theta'' \left(\frac{N}{K}(x_j - x_i) \right) \left(\frac{N}{K} \right)^2 \right) (u_j - u_i)^2 \\ &\geq N^{-s} \left(\frac{k^{(m+1)\alpha(1+\varepsilon)}}{N} \right)^{-(s+2)} \sum_{i, j \in J_{m+1}} (u_j - u_i)^2. \end{aligned} \quad (1.16)$$

The point is that when $\sum_{i \in J_{m+1}} u_i = 0$,

$$\sum_{i, j \in J_{m+1}} (u_j - u_i)^2 = 2|J_{m+1}| \sum_{i \in J_{m+1}} u_i^2.$$

Inserting this into (1.16) yields

$$U \cdot \nabla^2(H + F_m)U \geq c_1 \sum_{i \in J_{m+1}} u_i^2 \quad \text{where } c_1 := 2|J_{m+1}| N^{-s} \left(\frac{k^{(m+1)\alpha}}{N} \right)^{-(s+2)(1+\varepsilon)}.$$

Besides, one can observe that

$$|\nabla G_m|^2 = O \left(\frac{N^2}{k^{m\alpha}} \right).$$

Since H , F_m , $\tilde{H}$ and G_m are independent of $\sum_{i \in J_{m+1}} x_i$, one can argue with the Bakry-Emery criterion that for all $t \in \mathbb{R}$,

$$\log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{tG_m}] = t\mathbb{E}_{\mathbb{Q}_{N,\beta}}[G_m] + O\left(t^2 \frac{N^2}{k^{m\alpha}} c_1^{-1}\right) = t\mathbb{E}_{\mathbb{Q}_{N,\beta}}[G_m] + O(t^2 k^{s+\kappa\varepsilon+2\alpha}).$$

Using the log-Sobolev inequality and the local law (1.15), one can easily show by taking α small enough, that for all $\varepsilon > 0$, $G_m = O(k^{\frac{s}{2}+\varepsilon})$ with high probability (under $\mathbb{P}_{N,\beta}$), which shows that

$$N(x_{i+k} - x_i) = N(x_{i+k}^{[k]} - x_i^{[k]}) + O(k^{\frac{s}{2}+\varepsilon}),$$

with high probability. Using similar arguments, it is easy to prove $N(x_{i+k}^{[k]} - x_i^{[k]}) = k + O(k^{\frac{s}{2}+\varepsilon})$ with high probability, which will conclude the proof of Theorem 1.

We set $\ell_N = 1$ for simplicity and focus on the fluctuations of the linear statistic $\text{Fluct}_N[\xi]$, where $\xi \in L^2(\mathbb{T})$ satisfies Assumption 1.1. Let $\delta > 0$ be such that $\xi \in H^{\frac{1-s}{2}+\delta}(\mathbb{T})$ and let $\alpha \in [s-1, \frac{s}{2})$ be such that the singularity condition (1.8) holds.

With these hypotheses in place, we now describe how to derive both the variance expansion formula (Theorem 2) and the central limit theorem (Theorem 3).

Mollification. Let K be a smooth kernel supported on $(-\frac{1}{2}, \frac{1}{2})$ and let $K_\ell := \ell^{-1}K(\ell^{-1}\cdot)$. We consider the smooth test function $\xi_{\text{reg}} := \xi * K_{1/N}$. We first show that for all $\varepsilon > 0$, there exists $C > 0$ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[\xi - \xi_{\text{reg}}]] \leq CN^{\varepsilon+\max(2\alpha,0)}. \quad (1.17)$$

The Helffer-Sjöstrand equation. Let $F : D_N \rightarrow \mathbb{R}$ be sufficiently smooth. The fluctuations of F are related to a partial differential equation via the representation

$$\text{Var}_{\mathbb{P}_{N,\beta}}[F] = \mathbb{E}_{\mathbb{P}_{N,\beta}}[\nabla F \cdot \nabla \phi], \quad (1.18)$$

where $\phi \in H^1(\mathbb{P}_{N,\beta})$ solves the Poisson equation

$$\mathcal{L}\phi = F - \mathbb{E}_{\mathbb{P}_{N,\beta}}[F], \quad (1.19)$$

where $\mathcal{L}$ denotes the generator

$$\mathcal{L} := \beta \nabla \mathcal{H}_N \cdot \nabla - \Delta.$$

Note that (1.18) follows directly by integration by parts, once it is known that (1.19) admits a solution. Differentiating (1.19) yields the so-called Helffer-Sjöstrand equation:

$$\begin{cases} A_1 \nabla \phi = \nabla F & \text{in } D_N, \\ \nabla \phi \cdot \vec{n} = 0 & \text{on } \partial D_N, \end{cases} \quad (1.20)$$

where A_1 is formally given by

$$A_1 := \beta \nabla^2 \mathcal{H}_N + \mathcal{L} \otimes I_N.$$

When F is a function of the gaps, or when g is replaced by a bounded kernel, the existence and uniqueness of a solution to (1.19) roughly follow from the Lax-Milgram lemma. Indeed, under such an assumption, the Poincaré inequality holds, which ensures the coercivity of the associated bilinear form. To analyze the solution of (1.20), we will employ various tools based on mean-field approximations, convexity, and monotonicity.

Since $\partial_{ij} \mathcal{H}_N \leq 0$ for every $i \neq j$, it is standard that $\mathbb{P}_{N,\beta}$ satisfies the FKG inequality, meaning that the covariance between any two increasing functions is non-negative. This can be formulated by stating that $\mathcal{L}^{-1}$ preserves the cone of increasing functions, or equivalently that A_1 preserves the cone of gradients of increasing functions: if $\nabla F \geq 0$ (coordinate-wise), then $\nabla \phi := A_1^{-1} \nabla F \geq 0$. A useful consequence is the following: if $F, G : D_N \rightarrow \mathbb{R}$ are such that $|\nabla F| \leq \nabla G$, then

$$\text{Var}_{\mathbb{P}_{N,\beta}}[F] \leq \text{Var}_{\mathbb{P}_{N,\beta}}[G]. \quad (1.21)$$

This comparison principle can be extended to non-gradient vector fields, at the price of working with the variables $(x_i - x_1)_{2 \leq i \leq N}$. This upgraded comparison principle will serve as a key tool for analyzing the fluctuations of certain complicated singular functions.

Mean-field transport. It turns out that when F is a linear statistic, then the solution $\nabla\phi$ of (1.20) can be approximated by a transport Ψ_N of the form

$$\Psi_N : X_N \in D_N \mapsto \frac{1}{N^{1-s}}(\psi(x_1), \dots, \psi(x_N)),$$

for some well-chosen map $\psi : \mathbb{T} \rightarrow \mathbb{R}$.

Let $F := \text{Fluct}_N[\xi_{\text{reg}}]$, where we recall that $\xi_{\text{reg}} = \xi * K_{1/N}$. Letting

$$\text{Diag}_{\mathbb{T}^2} := \{(x, y) \in \mathbb{T}^2 : x = y\},$$

one may write

$$\nabla \mathcal{H}_N \cdot \Psi_N = N \iint_{\Delta^c} g'(x-y)(\psi(x) - \psi(y)) d\mu_N(x) d\mu_N(y),$$

where $\mu_N := \frac{1}{N} \sum_{i=1}^N \delta_{x_i}$. Let us expand μ_N around the Lebesgue measure dx on $\mathbb{T}$ and denote $\text{fluct}_N := N(\mu_N - dx)$. Noting that the constant term vanishes, one can check that

$$\nabla \mathcal{H}_N \cdot \Psi_N = 2 \int (-g' * \psi) \text{fluct}_N + \frac{1}{N^{1-s}} \text{Loop}[\psi] \quad (1.22)$$

where

$$\text{Loop}[\psi] := \iint_{\text{Diag}_{\mathbb{T}^2}^c} (\psi(x) - \psi(y)) N^{-s} g'(x-y) d\text{fluct}_N(x) d\text{fluct}_N(y). \quad (1.23)$$

If we neglect the sub-leading terms $\text{div} \Psi_N$ and $\text{Loop}[\psi]$, we therefore obtain

$$\nabla \mathcal{H}_N \cdot \Psi_N - \text{div} \Psi_N \simeq F,$$

by choosing ψ given by

$$\psi' = -\frac{1}{2\beta c_s} (-\Delta)^{\frac{1-s}{2}} \xi_{\text{reg}} \quad \text{and} \quad \int \psi = 0.$$

The central task of the paper is to show that for all $\varepsilon > 0$, there exists a constant $C > 0$ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Loop}[\psi]] \leq C N^{\varepsilon + \max(1, 2(\alpha+1-s))}. \quad (1.24)$$

Splitting the variance of the next-order term. Denote

$$\zeta_0 : (x, y) \in \mathbb{T}^2 \mapsto \frac{\psi(x) - \psi(y)}{x - y},$$

so that

$$\text{Loop}[\psi] = \iint_{\text{Diag}_{\mathbb{T}^2}^c} \zeta_0(x, y) N^{-s} \tilde{g}(x-y) d\text{fluct}_N(x) d\text{fluct}_N(y), \quad (1.25)$$

where $\tilde{g} : x \in \mathbb{T} \setminus \{0\} \mapsto x g'(x)$. Note that for every $i \in \{1, \dots, N\}$,

$$\begin{aligned} \partial_i \text{Loop}[\psi] &= \underbrace{2 \int_{y \neq x_i} \partial_1 \zeta_0(x_i, y) N^{-s} \tilde{g}(x_i - y) d\text{fluct}_N(y)}_{\simeq V_i} + \underbrace{2 \int_{y \neq x_i} \zeta_0(x_i, y) N^{-s} \tilde{g}'(x_i - y) d\text{fluct}_N(y)}_{\simeq W_i}. \end{aligned}$$

By sub-additivity,

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Loop}[\psi]] \leq \underbrace{2 \mathbb{E}_{\mathbb{P}_{N,\beta}}[V \cdot A_1^{-1} V]}_{(I)} + \underbrace{2 \mathbb{E}_{\mathbb{P}_{N,\beta}}[W \cdot A_1^{-1} W]}_{(II)}.$$

Control on (II) with Poincaré inequality in gap coordinates. In gap coordinates, the microscopic force $\mathbf{W}$ behaves well: there exists a vector field $\tilde{\mathbf{W}}$ such that for all $U_N \in \mathbb{R}^N$,

$$\mathbf{W} \cdot U_N = - \sum_{i=1}^N \tilde{\mathbf{W}}_i N(u_{i+1} - u_i),$$

and which typically satisfies (i.e., with overwhelming probability), for all $\varepsilon > 0$, the estimate

$$|\tilde{\mathbf{W}}|^2 \leq N^{\varepsilon + \max(1, 2(\alpha+1-s))}. \quad (1.26)$$

By introducing a penalty for configurations with large nearest-neighbor distances, one can modify the Gibbs measure into a new one that is uniformly log-concave with respect to the variables $N(x_{i+1} - x_i)$, for $i = 1, \dots, N$. Applying the Brascamp-Lieb inequality and using (1.26), we obtain that for all $\varepsilon > 0$, there exists $C > 0$ such that

$$(II) \leq N^{\varepsilon + \max(1, 2(\alpha+1-s))}. \quad (1.27)$$

Control on (I) with the comparison principle. In substance, one should think of V as satisfying for every $i = 1, \dots, N$,

$$|V_i| \leq C|\psi''(x_i)| + \text{“Lower order terms”}. \quad (1.28)$$

Note that for instance if $\xi = \mathbb{1}_{(a,b)}$, ψ'' blows like $\max(|x|, \frac{1}{N})^{-(2-s)}$. Therefore for such singular ψ , the Poincaré inequality does not provide satisfactory estimates for (I). If V_i was exactly given by $\psi''(x_i)$ for every $i = 1, \dots, N$, one could bound $\mathbb{E}_{\mathbb{P}_{N,\beta}}[V \cdot A_1^{-1}V]$ by the variance of $\text{Fluct}_N[\psi']$.

The idea is to use the comparison principle (1.21) to compare $\mathbb{E}_{\mathbb{P}_{N,\beta}}[V \cdot A_1^{-1}V]$ to the variance of a linear statistic, which is easier to handle using the rigidity estimates of Theorem 1. Let $\chi : \mathbb{T} \rightarrow \mathbb{R}$ be such that $\chi = C|\psi''|$. Equation (1.28) can be rewritten in the form

$$|V| \leq \nabla \text{Fluct}_N[\chi].$$

Applying the extension of (1.21) to general vector fields, we get that, roughly,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[V \cdot A_1^{-1}V] \leq \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[\chi]] + \text{“Lower order terms”}.$$

By the comparison principle again and the rigidity estimates of Theorem 1,

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[\chi]] \leq N^{\varepsilon + \max(1, 2(\alpha+1-s))}.$$

which yields

$$(I) \leq N^{\varepsilon + \max(1, 2(\alpha+1-s))}. \quad (1.29)$$

Let us emphasize that the bound on V is in fact slightly different and requires working with the variables $(x_i - x_1)_{2 \leq i \leq N}$. Combining (1.27) and (1.29) yields (1.24), thereby completing the proof of Theorem 2.

Central limit theorem. The starting point of the proof of the CLT of Theorem 3 is very similar to [LLW⁺19] and proceeds by Stein’s method. Let $G_N := N^{-\frac{s}{2}} \text{Fluct}_N[\xi_{\text{reg}}]$. We shall prove that for all $\eta \in \mathcal{C}^1(\mathbb{R})$ such that $\|\eta'\|_{L^\infty(\mathbb{R})} \leq 1$ and $\|\eta\|_{L^\infty(\mathbb{R})} \leq 1$,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)G_N] = \sigma_1^2(\xi_{\text{reg}})\mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta'(G_N)] + \text{Error}_N, \quad (1.30)$$

with $\sigma_1^2(\xi_{\text{reg}})$ as in (1.10) and Error_N a small error term. The fundamental observation of Stein is that this approximate integration by parts formula quantifies a distance to normality. Indeed, letting Z be a centered random variable with variance $\sigma_1^2(\xi_{\text{reg}})$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ smooth, one can solve the ODE

$$\sigma_1^2(\xi_{\text{reg}})x\eta(x) - \eta'(x) = h(x) - \mathbb{E}[h(Z)] \quad (1.31)$$

and (1.30) can be written as

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[h(G_N)] - \mathbb{E}[h(Z)] = \text{Error}_N,$$

which shows that G_N is approximately Gaussian. We now explain how to obtain (1.30). Let $\nabla\phi = A_1^{-1}\nabla G_N$. By integration by parts,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)G_N] = \mathbb{E}_{\mathbb{P}_{N,\beta}}[\sigma_1^2(\xi_{\text{reg}})\eta'(G_N)\nabla G_N \cdot \nabla\phi]. \quad (1.32)$$

The goal is then to prove that $\nabla G_N \cdot \nabla \phi$ concentrates around $\sigma_1^2(\xi_{\text{reg}})$. As explained above, $\nabla \phi$ may be approximated by the transport $N^{-1+\frac{s}{2}}\Psi$ where $\Psi : X_N \in D_N \mapsto (\psi(x_1), \dots, \psi(x_N))$. Performing this approximate transport allows one to replace (1.32) by

$$\begin{aligned} & \mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)G_N] - \sigma_1^2(\xi_{\text{reg}})\mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta'(G_N)] \\ &= \underbrace{\mathbb{E}_{\mathbb{P}_{N,\beta}}\left[\eta'(G_N)\left(\frac{1}{N}\sum_{i=1}^N \xi'_{\text{reg}}(x_i)\psi(x_i) - \sigma_1^2(\xi_{\text{reg}})\right)\right]}_{(I)} + \underbrace{\frac{1}{N^{1-\frac{s}{2}}}\text{Cov}_{\mathbb{P}_{N,\beta}}[\eta(G_N), \beta\text{Loop}[\psi] - \text{Fluct}_N[\psi']]}_{(II)}. \end{aligned} \quad (1.33)$$

The error term (I) is controlled using the local laws, while the error term (II) is controlled using (1.24). Together with (1.17), this concludes the proof of the central limit theorem of Theorem 3.

1.5. Structure of the paper.

- Section 2 presents useful preliminaries on the fractional Laplacian on the circle.
- Section 3 establishes the well-posedness of the Helffer–Sjöstrand equation and derives consequences of convexity and monotonicity.
- Section 4 completes the proof of the near-optimal rigidity result (Theorem 1).
- Section 5 proves the variance expansion (Theorem 2).
- Section 6 contains the proof of the central limit theorem (Theorem 3) and its corollaries.
- Appendix A proves the existence and uniqueness results stated in Section 3.
- Appendix B contains proofs of various discrete convolution and energy estimates.

1.6. Notation.

- We denote $d : \{1, \dots, N\}^2 \rightarrow \mathbb{N}$ the symmetric distance defined for each $1 \leq i, j \leq N$ by

$$d(i, j) = \min(|j - i|, N - |j - i|).$$

- We let

$$\text{Diag}_{\mathbb{T}^2} := \{(x, y) \in \mathbb{T}^2 : x = y\}.$$

- For any vector field $\psi : \Omega \rightarrow \mathbb{R}^m$ where Ω is an open set of $\mathbb{R}^n$, we let $D\psi$ be the matrix of partial derivatives of ψ . We also write $\nabla^2 f$ for the Hessian of a real-valued function f .
- For a matrix $M \in \mathcal{M}_N(\mathbb{R})$, we denote by $\|M\|_F$ the Frobenius norm of M

$$\|M\|_F := (\text{Tr}(MM^T))^{\frac{1}{2}}.$$

- For all $\alpha \in (0, 1)$, we let $\mathcal{C}^\alpha(\mathbb{T})$ be the space of α -Hölder continuous functions from $\mathbb{T}$ to $\mathbb{R}$.

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2. PRELIMINARIES

2.1. The fundamental solution of the fractional Laplacian on the circle. We begin by justifying that the fundamental solution of the fractional Laplace equation on the circle is given by (1.1). Roughly, (1.1) corresponds to the periodic summation of $x \mapsto |x|^{-s}$, which is the fundamental solution of $(-\Delta)^{\frac{1-s}{2}}$ on the real line.

For all complex variables s and a such that $\text{Re}(s) > 1$ and $a \neq 0, -1, -2, \dots$, let

$$\zeta(s, a) := \sum_{n=0}^{\infty} \frac{1}{(n+a)^s}. \quad (2.1)$$

Given $a \neq 0, -1, -2, \dots$, one can uniquely extend $\zeta(\cdot, a)$ into a meromorphic function on the full complex plane with a unique pole at $s = 1$, which is simple with a residue equal to 1. This function is called the Hurwitz zeta function [Ber72].

For any integrable function $f \in L^1(\mathbb{T})$, let

$$\hat{f}(k) = \int_{\mathbb{T}} f(y) e^{-2i\pi ky} dy, \quad \text{for all } k \in \mathbb{Z}.$$

Lemma 2.1.

(1) Let g be the solution of (1.2). Let $\zeta(s, x)$ be the Hurwitz zeta function (2.1). Then, for all $x \in \mathbb{T} \setminus \{0\}$,

$$g(x) = \zeta(s, x) + \zeta(s, 1 - x) = \lim_{n \rightarrow \infty} \left(\sum_{k=-n}^n \frac{1}{|k+x|^s} - \frac{2}{1-s} n^{1-s} \right). \quad (2.2)$$

(2) Let $\alpha \in (0, \frac{1}{2})$. For every $f \in \mathcal{C}^\infty(\mathbb{T})$, we have

$$(-\Delta)^\alpha f(x) = C_\alpha \text{ p.v. } \sum_{y \in (0,1)} \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} \frac{f(x) - f(y)}{|x - y - n|^{1+2\alpha}} dy, \quad \text{where } C_\alpha := \frac{2^{2\alpha} \Gamma(\frac{1+2\alpha}{2})}{\sqrt{\pi} |\Gamma(-\alpha)|}. \quad (2.3)$$

The above lemma is standard, but for completeness, we include a proof, following roughly [RS16].

Proof.

Step 1: semi-group representation. Let g be the unique solution of (1.2). We first derive the semi-group representation for $(-\Delta)^{-\frac{1-s}{2}}$. Let $\alpha \in (0, 1)$. Recall

$$\lambda^{-\alpha} = \frac{1}{\Gamma(\alpha)} \int_0^\infty e^{-\lambda t} \frac{dt}{t^{1-\alpha}}, \quad \text{for all } \lambda > 0. \quad (2.4)$$

The fractional Laplacian $(-\Delta)^{-\alpha}$ is defined by the following Fourier multiplier: for all $f \in \mathcal{C}^\infty(\mathbb{T})$ such that $\int f = 0$, letting $h = (-\Delta)^{-\alpha} f$, we have

$$\hat{h}(k) = (2\pi)^{-2\alpha} |k|^{-2\alpha} \hat{f}(k), \quad \text{for all } k \in \mathbb{Z} \setminus \{0\} \quad \text{and} \quad \hat{h}(0) = 0.$$

Let $f \in \mathcal{C}^\infty(\mathbb{T})$ be such that $\int f = 0$ and let $h = (-\Delta)^{-\alpha} f$. Applying (2.4) to $\lambda = |k|^2$ for $k \neq 0$ gives

$$\hat{h}(k) = \frac{1}{\Gamma(\alpha)} \int_0^\infty e^{-|k|^2 t} \hat{f}(k) \frac{dt}{t^{1-\alpha}}, \quad (2.5)$$

which also holds for $k = 0$ since $\int f = \hat{f}(0) = 0$. Let $(W_t)_{t \geq 0}$ be the heat kernel of the Laplacian on $\mathbb{T}$. Recall that the Fourier coefficients of W_t are given by

$$\widehat{W}_t(k) = e^{-4t\pi^2 |k|^2}, \quad \text{for all } k \in \mathbb{Z}, t \geq 0.$$

One may therefore rewrite (2.5) as

$$\hat{h}(k) = \frac{1}{\Gamma(\alpha)} \int_0^\infty \widehat{f * W_t}(k) \frac{dt}{t^{1-\alpha}}.$$

It follows that

$$h(x) = (-\Delta)^{-\alpha} f(x) = \frac{1}{\Gamma(\alpha)} \int_0^\infty f * W_t(x) \frac{dt}{t^{1-\alpha}} \quad \text{for all } x \in \mathbb{T}. \quad (2.6)$$

We now derive an explicit expression for g starting from (2.6). Recall

$$W_t(x) = \sum_{k \in \mathbb{Z}} e^{-4t\pi^2 |k|^2} e^{2i\pi kx}.$$

Recall the Poisson-summation identity: for all Schwartz-class function $\phi : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$\sum_{k \in \mathbb{Z}} \phi(x + k) = \sum_{n \in \mathbb{Z}} \hat{\phi}(n) e^{2i\pi nx} \quad \text{for all } x \in \mathbb{R}.$$

Applying this to $\phi(y) := e^{-4t\pi^2 y^2}$ gives that for all $x \in \mathbb{T}$ and $t > 0$,

$$W_t(x) = \frac{1}{\sqrt{4\pi t}} \sum_{n \in \mathbb{Z}} e^{-\frac{1}{4t} (x-n)^2}. \quad (2.7)$$

Taking $\alpha = \frac{1-s}{2} \in (0, 1)$ in (2.6), we deduce from a regularization argument that

$$g(x) := c_s(-\Delta)^{-\frac{1-s}{2}}(\delta_0 - 1)\Big|_x = \frac{c_s}{\Gamma(\frac{1-s}{2})} \int_0^\infty (W_t(x) - 1) \frac{dt}{t^{\frac{1+s}{2}}}.$$

Then, inserting (2.7), we get

$$g(x) = \frac{c_s}{\Gamma(\frac{1-s}{2})\sqrt{4\pi}} \int_0^\infty \sum_{k \in \mathbb{Z}} \left(e^{-\frac{1}{4t}(x-k)^2} - \int_{\mathbb{T}} e^{-\frac{1}{4t}(y-k)^2} dy \right) \frac{dt}{t^{1+\frac{s}{2}}}. \quad (2.8)$$

Step 2: absolute integrability of $\sum_k u_k^x$. Define the sequence of functions

$$u_k^x : t \in (0, \infty) \mapsto \frac{1}{t^{1+\frac{s}{2}}} \left(e^{-\frac{1}{4t}(x-k)^2} - \int_{\mathbb{T}} e^{-\frac{1}{4t}(y-k)^2} dy \right), \quad k \in \mathbb{Z}^*.$$

Let us prove that $\sum_k \int_0^\infty |u_k^x|(t) dt < \infty$. We have

$$\sum_{k \in \mathbb{Z} \setminus \{0\}} e^{-\frac{1}{4t}(k-x)^2} = 2 \sum_{k=1}^\infty e^{-\frac{1}{4t}(k-x)^2}.$$

Since $k \in \mathbb{N} \mapsto e^{-\frac{1}{4t}(k-x)^2}$ is decreasing on $[1, \infty]$, we get

$$\sum_{k \in \mathbb{N} \setminus \{0\}} e^{-\frac{1}{4t}(k-x)^2} \leq 2 \int_{1-x}^\infty e^{-\frac{1}{4t}y^2} dy \leq \int_{\mathbb{R}} e^{-\frac{1}{4t}y^2} dy = \sqrt{4\pi t}.$$

Similarly,

$$\sum_{k \in \mathbb{N} \setminus \{0\}} e^{-\frac{1}{4t}(-k-x)^2} \leq \sqrt{4\pi t}.$$

It follows that

$$\sum_{k \in \mathbb{Z} \setminus \{0\}} e^{-\frac{1}{4t}(-k-x)^2} \leq 2\sqrt{4\pi t}$$

and in particular,

$$\int_{\mathbb{T}} \sum_{k \in \mathbb{Z} \setminus \{0\}} e^{-\frac{1}{4t}(k-y)^2} dy \leq \sqrt{4\pi t}.$$

Therefore, since $s \in (0, 1)$, we obtain

$$\sum_{k \in \mathbb{Z}} \int_0^1 |u_k^x|(t) dt \leq C \int_0^1 \frac{1}{t^{\frac{1+s}{2}}} dt < \infty. \quad (2.9)$$

We now consider the region $t \geq 1$. By Taylor expansion, there exist constants $C > 0$ and $C' > 0$ such that for all $t > 0$,

$$|u_k^x|(t) \leq \frac{C}{t^{2+\frac{s}{2}}} |(k-x)^2 - k^2| \leq C' \frac{|k| + 1}{t^{2+\frac{s}{2}}}.$$

Thus, there exists a constant $C > 0$ such that for all $t > 0$,

$$\sup_{t \in (0, 1)} \sum_{k: |k| \leq (\sqrt{t})^{1+\frac{s}{4}}} |u_k^x|(t) \leq \frac{C}{t^{1+\frac{s}{2}-\frac{s}{4}}}.$$

Hence, integrating the above between 1 and $+\infty$, we find that

$$\sum_{k: |k| \leq (\sqrt{t})^{1+\frac{s}{4}}} \int_1^\infty |u_k^x|(t) dt < \infty.$$

On the other hand, by comparison with an integral, one can check that there exist constants $C > 0$ and $c > 0$ such that for all $t \in (0, 1)$,

$$\sum_{k: |k| > (\sqrt{t})^{1+\frac{s}{4}}} e^{-\frac{1}{4t}(k-x)^2} \leq C\sqrt{t}e^{-ct^{\frac{s}{4}}}.$$

Thus,

$$\int_1^\infty \left(\sum_{k: |k| > (\sqrt{t})^{1+\frac{s}{4}}} e^{-\frac{1}{4t}(k-x)^2} \right) dt < \infty.$$

Similarly,

$$\int_1^\infty \left(\sum_{k: |k| > (\sqrt{t})^{1+\frac{s}{4}}} \int_{\mathbb{T}} e^{-\frac{1}{4t}(k-y)^2} dy \right) dt < \infty.$$

Combining the above displays gives

$$\sum_{k \in \mathbb{Z}} \int_1^\infty |u_k^x(t)| dt < \infty. \quad (2.10)$$

With (2.9) and (2.10), we conclude that

$$\sum_{k \in \mathbb{Z}} \int_0^\infty |u_k^x(t)| dt < \infty.$$

Step 3: evaluation of the integral. Combining (2.9) and (2.10) and invoking Fubini's theorem, we may interchange the summation and the integral in (2.8), which gives

$$\begin{aligned} g(x) &= \frac{c_s}{\Gamma(\frac{1-s}{2})\sqrt{4\pi}} \sum_{k \in \mathbb{Z}} \int_0^\infty u_k^x(t) dt \\ &= \frac{c_s}{\Gamma(\frac{1-s}{2})\sqrt{4\pi}} \sum_{k \in \mathbb{Z}} \left(\int_0^\infty e^{-\frac{1}{4t}(x-k)^2} \frac{dt}{t^{1+\frac{s}{2}}} - \int_0^\infty \int_{\mathbb{T}} e^{-\frac{1}{4t}(y-k)^2} \frac{dt}{t^{1+\frac{s}{2}}} dy \right) \\ &= \frac{c_s \Gamma(\frac{s}{2}) 2^s}{\Gamma(\frac{1-s}{2}) 2\sqrt{\pi}} \sum_{k \in \mathbb{Z}} \left(\frac{1}{|x-k|^s} - \int_{\mathbb{T}} \frac{dy}{|y-k|^s} \right) \\ &= \lim_{n \rightarrow \infty} \sum_{k=-n}^n \left(\frac{1}{|x-k|^s} - \int_{\mathbb{T}} \frac{dy}{|y-k|^s} \right) \\ &= \lim_{n \rightarrow \infty} \left(\sum_{k=-n}^n \frac{1}{|x+k|^s} - \frac{2}{1-s} n^{1-s} \right), \end{aligned}$$

where we have used the change of variables $t \in (0, \infty) \mapsto \frac{1}{t}$ and (2.4) in the third equality, the expression of c_s (1.3) in the fourth equality, and the fact that

$$\int_{-n}^{n+1} \frac{1}{x^s} dx = \frac{2}{1-s} n^{1-s} + o_n(1)$$

in the last equality.

Step 4: rewriting g . By the Euler-Maclaurin formula, for all $a \in \mathbb{T} \setminus \{0\}$,

$$\lim_{n \rightarrow \infty} \left(\sum_{k=0}^n \frac{1}{(k+a)^s} - \frac{1}{1-s} n^{1-s} \right) = -\frac{1}{1-s} a^{1-s} + \frac{1}{a^s} - s \int_0^\infty \frac{t - [t] - \frac{1}{2}}{(t+a)^{1+s}} dt. \quad (2.11)$$

If s is complex-valued with $\operatorname{Re}(s) > 1$, then the left-hand side of (2.11) is given $\zeta(s, a)$ (2.1). It is argued in [Ber72, Eq. (5.2)] via an analytic continuation argument that $\zeta(s, a)$ coincides with the right-hand side of (2.11) when $\operatorname{Re}(s) > -1$ and $a \in (0, 1)$. As a consequence, we find that for all $x \in \mathbb{T}$ and $s \in (0, 1)$

$$g(x) = \zeta(s, 1-x) + \zeta(s, x).$$

Step 5: proof of (2.3). Let $f \in \mathcal{C}^\infty(\mathbb{T})$. For all $\lambda > 0$, we have

$$\lambda^\alpha = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - e^{-\lambda t}) \frac{dt}{t^{1+\alpha}}.$$

Hence,

$$(-\Delta)^\alpha f(k) = (2\pi|k|^2)^\alpha \hat{f}(k) = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - e^{-4\pi^2|k|^2 t}) \hat{f}(k) \frac{dt}{t^{1+\alpha}}.$$

We deduce that

$$(-\Delta)^\alpha f = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (f - f * W_t) \frac{dt}{t^{1+\alpha}}.$$

Recalling (2.7), we have

$$f * W_t(x) = \frac{1}{\sqrt{4\pi t}} \sum_{n \in \mathbb{Z}} \int f(y) e^{-\frac{1}{4t}(x-y-n)^2}.$$

By proceeding as in Step 2, we may show that one can invert the integral and the sum, which yields

$$(-\Delta)^\alpha f = \frac{\alpha}{\Gamma(1-\alpha)\sqrt{4\pi}} \sum_{n \in \mathbb{Z}} \left(\int (f(x) - f(y)) \int_0^\infty e^{-\frac{(x-y-n)^2}{4t}} \frac{dt}{t^{\frac{3}{2}+\alpha}} dy \right).$$

We can check that

$$\int_0^\infty e^{-\frac{(x-y-n)^2}{4t}} \frac{dt}{t^{\frac{3}{2}+\alpha}} = 2^{1+2\alpha} \Gamma(\alpha + \frac{1}{2}) \frac{1}{|x-y-n|^{1+2\alpha}}.$$

Combining the above displays gives (2.3). $\square$

2.2. Inverse Riesz transform.

Lemma 2.2. *Let $\xi \in \mathcal{C}^\infty(\mathbb{T})$. Let $\psi \in \mathcal{C}^\infty(\mathbb{T})$ be given by*

$$\psi' = -\frac{1}{2c_s} (-\Delta)^{\frac{1-s}{2}} \xi \quad \text{and} \quad \int \psi = 0. \quad (2.12)$$

(1) *We have*

$$\psi(x) = -\frac{4^{-s}}{\pi \tan(\frac{\pi}{2}s)} \text{p.v.} \sum_{k \in \mathbb{Z}} \int_{\mathbb{T}} \frac{\xi(y) - \int \xi}{|x-y-k|^{1-s}} \text{sgn}(x-y-k) dy, \quad \text{for all } x \in \mathbb{T}. \quad (2.13)$$

(2) *We have*

$$\begin{aligned} \frac{(2\pi)^{1-s}}{2c_s} \|\xi\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 &= \iint g''(x-y) (\psi(x) - \psi(y))^2 dx dy - 2 \int \xi'(x) \psi(x) dx \\ &= - \int \xi'(x) \psi(x) dx. \end{aligned} \quad (2.14)$$

(3) *Assume that ξ is supported on $(-\frac{1}{2}, \frac{1}{2})$. Let $\ell_N \in (0, 1]$. Let $\xi_0 : \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$\xi_0(x) = \begin{cases} \xi(x) & \text{if } |x| \leq \frac{1}{2} \\ 0 & \text{if } |x| > \frac{1}{2} \end{cases}. \quad (2.15)$$

Then, there exists a constant $C > 0$ depending on s such that

$$\left| \|\xi(\ell_N^{-1} \cdot)\|_{H^{\frac{1-s}{2}}(\mathbb{T})} - \ell_N^s \|\xi_0\|_{H^{\frac{1-s}{2}}(\mathbb{R})}^2 \right| \leq C \ell_N^2 \|\xi_0\|_{L^2(\mathbb{T})}^2. \quad (2.16)$$

Proof. Let $\chi \in \mathcal{C}^\infty(\mathbb{T})$ such that $\xi - \int \xi = \chi'$. Since $\partial_x (-\Delta)^{\frac{1-s}{2}} = (-\Delta)^{\frac{1-s}{2}} \partial_x$ on $\mathcal{C}^\infty(\mathbb{T})$, we have

$$\psi = -\frac{1}{2c_s} (-\Delta)^{\frac{1-s}{2}} \chi. \quad (2.17)$$

Since $\chi \in \mathcal{C}^\infty(\mathbb{T})$, by (2.3),

$$\psi(x) = -\frac{c'_s}{2c_s} \text{p.v.} \int_{\mathbb{T}} \sum_{k \in \mathbb{Z}} \frac{\chi(x) - \chi(y)}{|x-y-k|^{2-s}} dy, \quad (2.18)$$

where

$$c'_s = \frac{2^{1-s} \Gamma(1 - \frac{s}{2})}{|\Gamma(-\frac{1-s}{2})| \pi^{\frac{1}{2}}} = \frac{2^{-s} (1-s) \Gamma(1 - \frac{s}{2})}{\Gamma(\frac{1+s}{2}) \pi^{\frac{1}{2}}}.$$

Using Euler's reflection lemma, we next compute

$$\frac{c'_s}{c_s} = \frac{2^{1-2s} (1-s)}{\pi} \frac{\Gamma(1 - \frac{s}{2}) \Gamma(\frac{s}{2})}{\Gamma(\frac{1-s}{2}) \Gamma(\frac{1+s}{2})} = \frac{2^{1-2s} (1-s)}{\pi} \frac{\sin(\frac{\pi(1-s)}{2})}{\sin(\frac{\pi s}{2})} = \frac{(1-s) 2^{1-2s}}{\pi \tan(\frac{\pi}{2}s)}.$$

Thus for any $x \in \mathbb{T}$

$$\psi(x) = -\frac{1-s}{4^s \pi \tan(\frac{\pi}{2}s)} \text{p.v.} \sum_{y \in (0,1)} \sum_{k \in \mathbb{Z}} \int_{\mathbb{T}} \frac{\chi(x) - \chi(y)}{|x - y - k|^{2-s}} dy.$$

By integration by parts, and using that $\int \psi = 0$ and $\psi \in \mathcal{C}^\infty(\mathbb{T})$, we get

$$\psi(x) = -\frac{4^{-s}}{\pi \tan(\frac{\pi}{2}s)} \text{p.v.} \sum_{k \in \mathbb{Z}} \int_{\mathbb{T}} \frac{\xi(y) - \int \xi}{|x - y - k|^{1-s}} \text{sgn}(x - y - k) dy.$$

By integration by parts,

$$\begin{aligned} \int g''(x-y)(\psi(x) - \psi(y))^2 dx dy &= - \int g'(x-y) \psi'(x) 2(\psi(x) - \psi(y)) dx dy \\ &= 2 \int g(x-y) \psi'(x) \psi'(y) = \int \psi'(2g * \psi') = - \int \psi' \xi = \int \psi \xi', \end{aligned}$$

proving (2.14). Moreover,

$$\int_{\mathbb{T}} \psi' \xi = \sum_{k \in \mathbb{Z}} \widehat{\psi'}(k) \hat{\xi}(k) = -\frac{(2\pi)^{1-s}}{2c_s} \sum_{k \in \mathbb{Z}} |k|^{1-s} |\hat{\xi}(k)|^2 = -\frac{(2\pi)^{1-s}}{2c_s} \|\xi\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2.$$

Assembling the two above displays shows (2.14).

Let us now prove (2.16). Assume that ξ is supported on $(-\frac{1}{2}, \frac{1}{2})$. Let $\xi_0 : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\xi_0(x) = \begin{cases} \xi(x) & \text{if } |x| \leq \frac{1}{2} \\ 0 & \text{if } |x| > \frac{1}{2}. \end{cases}$$

We have

$$\begin{aligned} \|\xi(\ell_N^{-1} \cdot)\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 &= \int_{\mathbb{T}} \mathbf{1}_{|x| < \frac{1}{2} \ell_N} \xi(\ell_N^{-1} x) \sum_{k \in \mathbb{Z}} \int_{\mathbb{T}} \frac{\xi(\ell_N^{-1} x) - \xi(\ell_N^{-1} y)}{|x - y + k|^{2-s}} \mathbf{1}_{|y| < \frac{1}{2} \ell_N} dy dx \\ &= \int_{\mathbb{R}} \xi_0(\ell_N^{-1} x) \sum_{k \in \mathbb{Z}} \int_{\mathbb{R}} \frac{\xi_0(\ell_N^{-1} x) - \xi_0(\ell_N^{-1} y)}{|x - y + k|^{2-s}} dy dx \\ &= \ell_N^s \|\xi_0\|_{H^{\frac{1-s}{2}}(\mathbb{R})}^2 + \sum_{k \in \mathbb{Z} \setminus \{0\}} \int_{\mathbb{R}} \xi_0(\ell_N^{-1} x) \frac{\xi_0(\ell_N^{-1} x) - \xi_0(\ell_N^{-1} y)}{|x - y + k|^{2-s}} dy dx \\ &= \ell_N^s \|\xi_0\|_{H^{\frac{1-s}{2}}(\mathbb{R})}^2 + O\left(\ell_N^2 \int |\xi_0(x)| |\xi_0(x) - \xi_0(y)| dx dy\right) \\ &= \ell_N^s \|\xi_0\|_{H^{\frac{1-s}{2}}(\mathbb{R})}^2 + O(\ell_N^2 \|\xi_0\|_{L^2(\mathbb{R})}^2). \end{aligned}$$

□

Next, we apply the pointwise formula (2.13) to indicator functions and inverse power functions.

Lemma 2.3 (Explicit formulas). *Let $\zeta(s, a)$ be the Hurwitz zeta function (2.1).*

(1) *Let $\xi := \mathbf{1}_{(-a, a)} - 2a$ for some $0 < a < \frac{1}{2}$ and ψ be given by (2.13). We have*

$$\psi(x) = \frac{1}{4^s \pi s \tan(\frac{\pi}{2}s)} (\zeta(-s, x+a) - \zeta(-s, x-a)), \quad \text{for all } x \in \mathbb{T}. \quad (2.19)$$

Moreover,

$$\frac{(2\pi)^{1-s}}{2c_s} \|\xi\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 = \frac{2 \cotan(\frac{\pi}{2}s)}{4^s \pi s} \zeta(-s, 2a). \quad (2.20)$$

(2) *Let $\alpha \in (0, \frac{s}{2})$ and $\xi := \zeta(-\alpha, \cdot)$. We have*

$$\frac{(2\pi)^{1-s}}{2c_s} \|\xi\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 = (2\pi)^{-1+2\alpha-s} \frac{c_\alpha^2}{c_s} \zeta(1+s-2\alpha).$$

Proof. Let $\xi := \mathbb{1}_{(-a,a)}$ for $a \in (0, \frac{1}{2})$. Formula (2.19) follows from (2.13). Moreover, by (2.14) and since $\zeta(-s, 0) = 0$,

$$\frac{(2\pi)^{1-s}}{2c_s} \|\xi\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 = \int \xi \psi' = \psi(a) - \psi(-a) = \frac{\cotan(\frac{\pi}{2}s)}{4^s \pi s} 2\zeta(-s, 2a).$$

Let $\alpha \in (0, \frac{s}{2})$ and $\xi := \zeta(-\alpha, \cdot)$. By Lemma 2.1, we have

$$(-\Delta)^{\frac{1-\alpha}{2}} \xi = c_\alpha \delta_0.$$

Therefore, for every $k \in \mathbb{Z}$,

$$\hat{\xi}(k) = c_\alpha \frac{\mathbb{1}_{k \neq 0}}{|k|^{1-\alpha} (2\pi)^{1-\alpha}}.$$

We have

$$\|\xi\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 = \sum_{k \in \mathbb{Z}} |k|^{1-s} |\hat{\xi}(k)|^2 = \frac{c_\alpha^2}{(2\pi)^{2(1-\alpha)}} \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{1}{k^{1-2\alpha+s}}.$$

Moreover,

$$\sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{1}{k^{1-2\alpha+s}} = 2\zeta(1-2\alpha+s).$$

Combining the above displays, we get

$$\frac{(2\pi)^{1-s}}{2c_s} \|\xi\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 = (2\pi)^{1-s-2(1-\alpha)} \frac{c_\alpha^2}{c_s} \zeta(1+s-2\alpha).$$

□

3. THE HELFFER-SJÖSTRAND EQUATION

In this section, we review some basic properties of the Helffer-Sjöstrand equation. We first state some existence and uniqueness results, then adapt a known comparison principle to the circular setting, and finally record various concentration estimates.

3.1. Basic properties. In this subsection we introduce the H.-S. equation and state some standard existence and uniqueness results, following in part [AW19]. Let μ be a probability measure on D_N of the form

$$d\mu(X_N) = e^{-H(X_N)} \mathbb{1}_{D_N}(X_N) dX_N,$$

with $H : D_N \rightarrow \mathbb{R}$ measurable. We make the following assumption on H :

Assumption 3.1. *The Hamiltonian $H : D_N \rightarrow \mathbb{R}$ is of the form*

$$H : X_N \mapsto \sum_{i \neq j} \chi_{ij}(|x_i - x_j|), \quad (3.1)$$

for a family of functions $\chi_{ij} : (0, \infty) \rightarrow \mathbb{R}$ such that for every $i, j \in \{1, \dots, N\}$, $i \neq j$, $\chi_{ij} = \chi_{ji}$ and

$$\chi_{ij} \in \mathcal{C}^\infty(0, \infty), \quad \chi_{ij}'' \geq c > 0 \quad \text{and} \quad \lim_{x \rightarrow 0} \chi_{ij}(x) = +\infty. \quad (3.2)$$

Let $F : D_N \rightarrow \mathbb{R}$ be a smooth enough function. We seek to rewrite the variance of F under μ in a tractable way. Define on $\mathcal{C}^\infty(D_N)$ the Langevin operator

$$\mathcal{L}^\mu = \nabla H \cdot \nabla - \Delta,$$

with ∇ and Δ denoting the gradient and Laplace operators on $\mathbb{T}^N$. Recall the integration by parts formula under μ : for all $\phi, \psi \in \mathcal{C}^\infty(D_N)$, we have

$$\mathbb{E}_\mu[(\mathcal{L}^\mu \phi) \psi] = \mathbb{E}[\nabla \phi \cdot \nabla \psi]. \quad (3.3)$$

We study the well-posedness of the Helffer-Sjöstrand equation

$$A_1^\mu \nabla \phi = \nabla F \quad \text{on } D_N. \quad (3.4)$$

Define the norm

$$\|F\|_{H^1(\mu)} = \mathbb{E}_\mu[F^2]^{\frac{1}{2}} + \mathbb{E}_\mu[|\nabla F|^2]^{\frac{1}{2}}.$$

Let $H^1(\mu)$ be the completion of $\mathcal{C}^\infty(D_N)$ with respect to the norm $\|\cdot\|_{H^1(\mu)}$. Also, define the norm

$$\|F\|_{H^{-1}(\mu)} := \sup\{|\mathbb{E}_\mu[FG]| : G \in H^1(\mu), \|G\|_{H^1(\mu)} \leq 1\}$$

and let $H^{-1}(\mu)$ be the dual of $H^1(\mu)$, defined as the completion of $\mathcal{C}^\infty(D_N)$ with respect to the norm $\|\cdot\|_{H^{-1}(\mu)}$.

Define the map

$$\text{Gap}_N : X_N \in D_N \mapsto N(x_2 - x_1, x_3 - x_1, \dots, x_N - x_{N-1}, x_1 - x_N) \in \mathbb{R}^N. \quad (3.5)$$

Proposition 3.1 (Existence and representation). *Let μ satisfy Assumption 3.1 and let $\mu' := \text{Gap}_N \# \mu$. Let $F \in H^1(\mu)$. Assume that F is of the form $F = G \circ \text{Gap}_N$ with $G \in H^1(\mu')$. Then, there exists a unique $\nabla\phi \in L^2(\{1, \dots, N\}, H^1(\mu))$ such that*

$$A_1^\mu \nabla\phi = \nabla F \quad \text{on } D_N, \quad (3.6)$$

with the equality being, for each coordinate, an equality of elements of $H^{-1}(\mu)$.

Moreover, the solution of (3.6) is the unique minimizer of the functional

$$\nabla\phi \mapsto \mathbb{E}_\mu[\nabla\phi \cdot \nabla^2 H \nabla\phi + \|\nabla^2 \phi\|_F^2 - 2\nabla F \cdot \nabla\phi], \quad (3.7)$$

on maps $\nabla\phi \in L^2(\{1, \dots, N\}, H^1(\mu))$ such that $\nabla\phi \cdot \vec{n} = 0$ a.e. on ∂D_N .

Finally, the variance of F may be represented as

$$\text{Var}_\mu[F] = \mathbb{E}_\mu[\nabla\phi \cdot \nabla F] \quad (3.8)$$

and the covariance between F and any function $G \in H^1(\mu)$ as

$$\text{Cov}_\mu[F, G] = \mathbb{E}_\mu[\nabla\phi \cdot \nabla G].$$

The identity (3.8) is called the Helffer-Sjöstrand formula. The proof of Proposition 3.1 is postponed to Appendix A.

When ∇F is replaced by a non-gradient vector field v , the solution is generally non-unique. To ensure uniqueness, we also need to assume that $\sum_{i=1}^N v_i = 0$ and that for every $i \in \{1, \dots, N\}$, v_i is a function of the gaps.

Proposition 3.2 (Well-posedness for non-gradient vector fields). *Let μ satisfy Assumption 3.1 and let $\mu' := \text{Gap}_N \# \mu$. Let $v \in L^2(\{1, \dots, N\}, H^{-1}(\mu))$ be such that $v \cdot (e_1 + \dots + e_N) = 0$ and for each $i \in \{1, \dots, N\}$, $v_i = w_i \circ \text{Gap}_N$ for some $w_i \in H^{-1}(\mu')$. Then, there exists a unique $\psi \in L^2(\{1, \dots, N\}, H^1(\mu))$ such that*

$$\begin{cases} A_1^\mu \psi = v & \text{on } D_N \\ \psi \cdot (e_1 + \dots + e_N) = 0 & \text{on } D_N. \end{cases} \quad (3.9)$$

Moreover, the solution of (3.9) is also the unique minimizer of

$$\psi \mapsto \mathbb{E}_\mu[\psi \cdot \nabla^2 H \psi + \|D\psi\|_F^2 - 2v \cdot \psi],$$

over maps $\psi \in L^2(\{1, \dots, N\}, H^1(\mu))$.

The proof of Proposition 3.2 is postponed to Appendix A. When v satisfies the assumption of Proposition 3.2, we denote $\psi = (A_1^\mu)^{-1}v$ as the solution of (3.9).

Lemma 3.3. *Let μ satisfy Assumption 3.1 and let $\mu' := \text{Gap}_N \# \mu$. Let $v, w \in L^2(\{1, \dots, N\}, H^{-1}(\mu))$ satisfy the assumption of Proposition 3.2. We have*

$$\mathbb{E}_\mu[(v + w) \cdot (A_1^\mu)^{-1}(v + w)] \leq 2\left(\mathbb{E}_\mu[v \cdot (A_1^\mu)^{-1}v] + \mathbb{E}_\mu[w \cdot (A_1^\mu)^{-1}w]\right). \quad (3.10)$$

Proof. Since $v - w$ satisfies $(e_1 + \dots + e_N) \cdot (v - w) = 0$, by Proposition 3.2, $(A_1^\mu)^{-1}(v - w)$ is well-defined. Moreover, by integration by parts (3.3),

$$\begin{aligned} \mathbb{E}_\mu[(v - w) \cdot (A_1^\mu)^{-1}(v - w)] \\ = \mathbb{E}_\mu\left[(A_1^\mu)^{-1}(v - w) \cdot \nabla^2 H (A_1^\mu)^{-1}(v - w) + \|D((A_1^\mu)^{-1}(v - w))\|_F^2\right] \geq 0, \end{aligned}$$

since $\nabla H^2 \geq 0$. By linearity, this implies (3.10). $\square$

3.2. Monotonicity and consequences. We now state some monotonicity results related to the FKG inequality. Recall that a measure ρ on $\mathbb{R}^N$ is said to satisfy the FKG inequality if for any functions $F : \mathbb{R}^N \rightarrow \mathbb{R}$, $G : \mathbb{R}^N \rightarrow \mathbb{R}$ in $L^2(\rho)$ which are increasing in each variable, we have

$$\text{Cov}_\rho[F, G] \geq 0.$$

A standard condition for ρ to satisfy the FKG inequality [BÉ85] is that ρ can be written as

$$d\rho(X_N) = e^{-H(X_N)} dX_N$$

with $H : \mathbb{R}^N \rightarrow \mathbb{R}$ is a twice continuously differentiable function satisfying

$$\partial_{ij}H \leq 0 \quad \text{for every } 1 \leq i, j \leq N, i \neq j. \quad (3.11)$$

Note that for smooth Gibbs measures, the FKG property is roughly equivalent to a maximum principle for the equation (3.6), which states that $(\mathcal{L}^\mu)^{-1}$ preserves the cone of smooth functions that are increasing in each variable. In the next lemma, we show that the FKG inequality allows us to compare the variance of two functions via their gradients.

Lemma 3.4. *Let μ be a probability measure on D_N of the form $d\mu(X_N) = e^{-H(X_N)} dX_N$ with $H \in \mathcal{C}^2(D_N)$ such that*

$$\lim_{d(x, \partial D_N) \rightarrow 0} H(x) = +\infty. \quad (3.12)$$

Assume that

$$\partial_{ij}H \leq 0 \quad \text{for every } i, j \in \{1, \dots, N\}, i \neq j. \quad (3.13)$$

Let $F, G \in H^1(\mu)$ be such that for every $i \in \{1, \dots, N\}$,

$$|\partial_i F| \leq \partial_i G. \quad (3.14)$$

Then,

$$\text{Var}_\mu[F] \leq \text{Var}_\mu[G]. \quad (3.15)$$

Proof. It is standard, see, for example [BM92, Th. 1.3] for the case of $\mathbb{R}^N$, that μ satisfies the FKG inequality. That is, for all measurable non-decreasing functions f and g , the covariance between f and g under μ is non-negative.

Let $F, G \in H^1(\mu)$ satisfy (3.14). One may write

$$\text{Var}_\mu[G] = \text{Var}_\mu[F] + \text{Cov}_\mu[G + F, G - F]$$

Since $G - F$ and $F + G$ are non-decreasing, their covariance is non-negative, which concludes the proof of (3.15). $\square$

We observe that μ which satisfies Assumption 3.1 satisfies the condition (3.13). Since particles are ordered, one may write, for all $X_N \in D_N$,

$$H(X_N) = 2 \sum_{i < j} \chi_{ij}(x_j - x_i).$$

Notice that for every $i \in \{1, \dots, N\}$ and all $X_N \in D_N$,

$$\partial_i H(X_N) = -2 \sum_{j:j>i} \chi'_{ij}(x_j - x_i) + 2 \sum_{j:j<i} \chi'_{ij}(x_i - x_j).$$

Therefore, for every $i, j \in \{1, \dots, N\}$ with $i \neq j$ and all $X_N \in D_N$, we have

$$\partial_{ij}H(X_N) = -2\chi''_{ij}(|x_j - x_i|) \leq 0.$$

Therefore, if μ which satisfies Assumption 3.1, then it satisfies the FKG inequality and, therefore, the conclusion of Lemma 3.4.

In fact, for such measures μ , one can establish a stronger maximum principle. Let us work with the push-forward μ^* of μ by the anchoring map

$$\text{Anch}_N : x \in D_N \mapsto (x_2 - x_1, x_3 - x_1, \dots, x_N - x_1) \in \mathbb{R}^{N-1}. \quad (3.16)$$

We now prove a maximum principle for the solution of $A_1^{\mu^*} \psi = v$. This is similar to the maximum principle derived in [Car74, HS94].

Lemma 3.5 (Monotonicity). *Let μ satisfy Assumption 3.1 and let $\mu^* := \text{Anch}_N \# \mu$, where Anch_N is the linear map defined in (3.16). Let $v \in L^2(\{1, \dots, N-1\}, H^{-1}(\mu^*))$, and let $\psi \in L^2(\{1, \dots, N-1\}, H^1(\mu^*))$ be the solution of*

$$A_1^{\mu^*} \psi = v. \quad (3.17)$$

If

$$v_i \geq 0 \quad \text{a.e. on } \text{Anch}_N(D_N), \quad \text{for every } i \in \{1, \dots, N-1\},$$

then,

$$\psi_i \geq 0 \quad \text{a.e. on } \text{Anch}_N(D_N), \quad \text{for every } i \in \{1, \dots, N-1\}. \quad (3.18)$$

Remark 3.2. Let H^* be such that $H = H^*(\text{Anch}_N(x))$. We can notice that for all $U_{N-1} = (u_2, \dots, u_N) \in \mathbb{R}^{N-1}$,

$$\begin{aligned} U_{N-1} \cdot \nabla^2 H^* U_{N-1} &\geq \chi''_{12}(x_2 - x_1)u_2^2 + \sum_{i=2}^{N-1} \chi''_{i,i+1}(x_{i+1} - x_i)(u_{i+1} - u_i)^2 \\ &\geq \frac{1}{C} \left(u_1^2 + \sum_{i=2}^{N-1} (u_{i+1} - u_i)^2 \right). \end{aligned}$$

Thus, there exists $C > 0$ such that $\nabla^2 H^* \geq \frac{1}{C} \text{Id}$. Since μ^* is uniformly log-concave, the existence and uniqueness of a solution to (3.17) follow from the same reasoning as in the proof of Proposition 3.2.

Proof of Lemma 3.5. Let H^* be such that $H = H^*(\text{Anch}_N(x))$. Let ψ^+ and ψ^- be the positive and negative parts of ψ . Taking the scalar product of the equation $A_1^{\mu^*} \psi = v$ with ψ^- gives

$$\psi^- \cdot \nabla^2 H^* \psi + \psi^- \cdot (\mathcal{L}^{\mu^*} \otimes I_{N-1}) \psi = \psi^- \cdot v \geq 0.$$

By integration by parts under μ^* , since for every $i, j = 1, \dots, N$, $i \neq j$, we have $\lim_{x \rightarrow 0} \chi_{ij}(x) = +\infty$,

$$\mathbb{E}_{\mu^*}[\psi^- \cdot (\mathcal{L}^{\mu^*} \otimes I_{N-1}) \psi] = \mathbb{E}_{\mu^*} \left[\sum_{i=1}^N \nabla \psi_i^- \cdot \nabla \psi_i \right].$$

Indeed, the boundary term in this integration by parts vanishes due to (3.2). Combining the two above displays, we get

$$\mathbb{E}_{\mu^*} \left[\psi^- \cdot \nabla^2 H^* \psi + \sum_{i=1}^{N-1} \nabla \psi_i^- \cdot \nabla \psi_i \right] \geq 0,$$

Note first that for every $i \in \{1, \dots, N-1\}$, $\nabla \psi_i^- \cdot \nabla \psi_i = -|\nabla \psi_i^-|^2$. Hence,

$$\begin{aligned} \mathbb{E}_{\mu^*} \left[\psi^- \cdot \nabla^2 H^* \psi + \sum_{i=1}^{N-1} \nabla \psi_i^- \cdot \nabla \psi_i \right] &= \mathbb{E}_{\mu^*} [\psi^- \cdot \nabla^2 H^* \psi - \|D\psi^-\|_F^2] \\ &= \mathbb{E}_{\mu^*} [-\psi^- \cdot \nabla^2 H^* \psi^+ + \psi^- \cdot \nabla^2 H^* \psi^+ - \|D\psi^-\|_F^2]. \end{aligned}$$

Also observe that, since $\psi_i^+ \psi_i^- = 0$,

$$\psi^- \cdot \nabla^2 H^* \psi^+ = 2 \sum_{i < j} \chi''_{ij}(x_j - x_i) (\psi_i^- \psi_i^+ - \psi_i^- \psi_j^+) = -2 \sum_{i < j} \chi''_{ij}(x_j - x_i) \psi_i^- \psi_j^+ \leq 0, \quad (3.19)$$

where we have used that $\chi''_{ij} \geq 0$ in the last inequality. We deduce from the above three displays that

$$\mathbb{E}_{\mu^*}[\psi^- \cdot \nabla^2 H^* \psi^-] \leq 0.$$

By Assumption 3.1 (see Remark 3.2), there exists $c > 0$ such that

$$\mathbb{E}_{\mu^*}[\psi^- \cdot \nabla^2 H^* \psi^-] \geq c \mathbb{E}_{\mu^*} \left[\sum_{i=1}^{N-1} (\psi_i^-)^2 \right].$$

This shows that $\psi^- = 0$, which concludes the proof of the lemma. $\square$

The maximum principle of Lemma 3.5 allows us to extend Lemma 3.4.

Lemma 3.6 (A_1^{-1} -energy comparison). *Let μ satisfy Assumption 3.1 and let $\mu^* := \text{Anch}_N \# \mu$, where Anch_N is the linear map defined in (3.16). Let $v, w \in L^2(\{1, \dots, N-1\}, H^{-1}(\mu^*))$. If*

$$|v_i| \leq w_i, \quad \text{a.e. on } \text{Anch}_N(D_N) \quad \text{for every } i \in \{1, \dots, N-1\}, \quad (3.20)$$

then

$$\mathbb{E}_{\mu^*} \left[v \cdot (A_1^{\mu^*})^{-1} v \right] \leq \mathbb{E}_{\mu^*} \left[w \cdot (A_1^{\mu^*})^{-1} w \right].$$

In particular, if $F, G \in H^1(\mu)$ satisfy

$$|\partial_i F| \leq \partial_i G, \quad \text{a.e. on } \text{Anch}_N(D_N) \quad \text{for every } i \in \{1, \dots, N-1\},$$

then

$$\text{Var}_{\mu^*}[F] \leq \text{Var}_{\mu^*}[G].$$

Proof of Lemma 3.6. For $y = (y_1, \dots, y_{N-1}) \in \mathbb{R}^{N-1}$, we use the notation $y \geq 0$ when for every $i \in \{1, \dots, N-1\}$, $y_i \geq 0$.

Let v^+ and v^- be the positive and negative parts of v . Using the fact that $A_1^{\mu^*}$ is self-adjoint on $L^2(\{1, \dots, N-1\}, H^{-1}(\mu^*))$, we have

$$\mathbb{E}_{\mu^*} [w \cdot (A_1^{\mu^*})^{-1} w] - \mathbb{E}_{\mu^*} [v \cdot (A_1^{\mu^*})^{-1} v] = \mathbb{E}_{\mu^*} [(v+w) \cdot (A_1^{\mu^*})^{-1} (w-v)].$$

Note that since $w-v \geq 0$, by Lemma 3.5, we have $(A_1^{\mu^*})^{-1}(w-v) \geq 0$. Moreover, since $w+v \geq 0$, we get

$$\mathbb{E}_{\mu^*} \left[(v+w) \cdot (A_1^{\mu^*})^{-1} (w-v) \right] \geq 0,$$

which is the desired result. The in particular assertion is immediate. $\square$

3.3. Brascamp-Lieb inequality.

Lemma 3.7 (Brascamp-Lieb inequality [BL02]). *Let μ satisfy Assumption 3.1. Let $\mu' := \text{Gap}_N \# \mu$. Let $F \in L^2(\mu)$ be of the form $F = G \circ \text{Gap}_N$ with $\nabla G \in L^2(\{1, \dots, N\}, H^{-1}(\mu'))$. We have*

$$\text{Var}_{\mu}[F] \leq \mathbb{E}_{\mu} [\nabla F \cdot (\nabla^2 H)^{-1} \nabla F], \quad (3.21)$$

where $(\nabla^2 H)^{-1}$ is taken in the sense of the inverse on $\{x \in \mathbb{R}^N : x \cdot (e_1 + \dots + e_N) = 0\}$.

This result is standard, but for completeness, the proof is sketched below.

Proof. By Proposition 3.1, the variance of F may be written as

$$\text{Var}_{\mu}[F] = - \min_{\phi \in H^1(\mu)} \mathbb{E}_{\mu} \left[\nabla \phi \cdot \nabla^2 H \nabla \phi + \|\nabla^2 \phi\|_F^2 - 2 \nabla F \cdot \nabla \phi \right].$$

Discarding the non-negative term $\mathbb{E}_{\mu} [\|\nabla^2 \phi\|_F^2]$ yields

$$\text{Var}_{\mu}[F] \leq - \min_{\phi \in H^1(\mu)} \mathbb{E}_{\mu} \left[\nabla \phi \cdot \nabla^2 H \nabla \phi - 2 \nabla F \cdot \nabla \phi \right].$$

Notice that for any $\phi \in H^1(\mu)$,

$$\nabla \phi \cdot \nabla^2 H \nabla \phi - 2 \nabla F \cdot \nabla \phi \geq \min_{U_N \in \mathbb{R}^N} (U_N \cdot \nabla^2 H U_N - 2 \nabla F \cdot U_N).$$

It follows that

$$\text{Var}_{\mu}[F] \leq - \mathbb{E}_{\mu} \left[\min_{U_N \in \mathbb{R}^N} (U_N \cdot \nabla^2 H U_N - 2 \nabla F \cdot U_N) \right].$$

We have $\nabla F \cdot (e_1 + \dots + e_N) = 0$. Therefore, for all $U_N \in \mathbb{R}^N$,

$$U_N \cdot \nabla^2 H U_N - 2 \nabla F \cdot U_N = U'_N \cdot \nabla^2 H U'_N - 2 \nabla F \cdot U'_N,$$

where

$$U'_N = U_N - \frac{1}{N} U_N \cdot (e_1 + \dots + e_N) (e_1 + \dots + e_N).$$

Hence, we may restrict the minimization to $U_N \in \mathbb{R}^N$ satisfying $U_N \cdot (e_1 + \dots + e_N) = 0$. Recall that $\nabla^2 H$ preserves this subspace and is uniformly positive definite on it, and thus invertible. In particular,

$$\min_{U_N : U_N \cdot (e_1 + \dots + e_N) = 0} (U_N \cdot \nabla^2 H U_N - 2 \nabla F \cdot U_N) = - \nabla F \cdot (\nabla^2 H)^{-1} \nabla F.$$

Therefore,

$$\mathrm{Var}_\mu[F] \leq \mathbb{E}_\mu \left[\nabla F \cdot (\nabla^2 H)^{-1} \nabla F \right],$$

which is the desired inequality. $\square$

Note that Lemma 3.7 is only valid for functions of the gaps since $\nabla^2 H$ is only strictly positive definite on the subspace of vectors of mean 0, i.e. on the subspace orthogonal to $e_1 + \dots + e_N$.

3.4. Log-Sobolev inequalities and Gaussian concentration. We now review some standard results on log-Sobolev inequalities and Gaussian concentration for log-concave measures on $\mathbb{R}^N$ and derive some stronger estimates valid for measures satisfying Assumption 3.1, following [BEY⁺14b].

Let us first recall a crucial convexity result proved in [BL02].

Lemma 3.8. *Let μ satisfy Assumption 3.1. Let $I \subset \{1, \dots, N\}$ be of cardinality m and denote by π_I the projection on the coordinates $(x_i)_{i \in I}$. Split H into $H = H_1 \circ \pi_I + H_2$ with*

$$H_1 : (x_i)_{i \in I} \in D_m \mapsto \sum_{i \neq j \in I} \chi(|x_i - x_j|),$$

$$H_2 : X_N \in D_N \mapsto \sum_{i \in I^c} \sum_{j \in I^c: j \neq i} \chi(|x_i - x_j|) + 2 \sum_{i \in I^c} \sum_{j \in I} \chi(|x_i - x_j|).$$

Define

$$\tilde{H} : z \in D_m \mapsto -\log \int e^{-H_2(z, y)} \mathbf{1}_{D_N}(z, y) dy, \quad (3.22)$$

where we use the coordinates $z = (x_i)_{i \in I}$, $y = (x_i)_{i \in I^c}$.

(1) The density of $\nu := \pi_I \# \mu$ is given by

$$d\nu(x) \propto e^{-(H_1 + \tilde{H})(x)} \mathbf{1}_{D_m}(x) dx. \quad (3.23)$$

(2) The Hessian of $\tilde{H}$ is positive-definite,

$$\nabla^2 \tilde{H} \geq 0.$$

Proof. The statement (3.23) is immediate. We prove that the Hessian of (3.22) is non-negative.

For every smooth map $f : D_N \rightarrow \mathbb{R}$ define

$$\nabla_1 f := (\partial_i f)_{i \in I}, \quad \nabla_2 f := (\partial_i f)_{i \in I^c}.$$

Accordingly, split the Hessian into its four block components:

$$\nabla^2 f = \begin{pmatrix} \nabla_{11}^2 f & \nabla_{12}^2 f \\ \nabla_{21}^2 f & \nabla_{22}^2 f \end{pmatrix},$$

where for example $\nabla_{11}^2 f = (\partial_{ij} f)_{i \in I, j \in I}$.

Fix $z \in D_m$, $v \in \mathbb{R}^m$ and let

$$h : t \mapsto \tilde{H}(z + tv).$$

Since for every $i, j \in \{1, \dots, N\}$, $i \neq j$,

$$\lim_{x \rightarrow 0} \chi_{ij}(x) = +\infty,$$

one can check that for all $t \in \mathbb{R}$,

$$h''(t) = \mathbb{E}_{\mathbb{P}(x+tv)}[v \cdot (\nabla_{11}^2 H_2)v] - \mathrm{Var}_{\mathbb{P}(x+tv)}[v \cdot \nabla_1 H_2], \quad (3.24)$$

where for any $z \in D_m$, we define $\mathbb{P}(z)$ to be the probability measure

$$d\mathbb{P}(z) = \frac{1}{Z(z)} e^{-H_2(z, y)} \mathbf{1}_{(z, y) \in D_N} dy.$$

Since the Hessian of $y \mapsto H_2(x, y)$ is non-negative, the Brascamp-Lieb inequality (Lemma 3.7) implies

$$\mathrm{Var}_{\mathbb{P}(x+tv)}[v \cdot \nabla_1 H_2] \leq \mathbb{E}_{\mathbb{P}(x+tv)}[v \cdot (\nabla_{12}^2 H_2)(\nabla_{22}^2 H_2)^{-1}(\nabla_{12}^2 H_2)v].$$

Furthermore, since $\nabla^2 H_2$ is non-negative, its Schur complement is non-negative, which gives

$$\nabla_{11}^2 H_2 - \nabla_{12}^2 H_2 (\nabla_{22}^2 H_2)^{-1} \nabla_{12}^2 H_2 \geq 0.$$

Inserting this into (3.24) shows that $\nabla^2 \tilde{H} \geq 0$. $\square$

Let us recall the standard log-Sobolev inequality for uniformly log-concave measures on $\mathbb{R}^N$, which is a special case of the Bakry-Emery criterion [BÉ85]. Let μ and ν be two probability measures on D_N . Recall that the relative entropy of μ with respect to ν is defined by

$$\text{Ent}(\mu \mid \nu) = \int \log\left(\frac{d\mu}{d\nu}\right) d\mu \in [0, +\infty], \quad (3.25)$$

if μ is absolutely continuous with respect to ν and $\text{Ent}(\mu \mid \nu) = +\infty$ otherwise. Also recall the Fisher information of μ with respect to ν ,

$$\text{Fisher}(\mu \mid \nu) = \int \left| \nabla \log \frac{d\mu}{d\nu} \right|^2 d\mu, \quad (3.26)$$

if μ is absolutely continuous with respect to ν and $\text{Fisher}(\mu \mid \nu) = +\infty$ otherwise.

Lemma 3.9 (Bakry-Emery [BÉ85]). *Let K be a convex domain of $\mathbb{R}^N$. Let $w > 0$ and let γ^w be a centered Gaussian distribution on $\mathbb{R}^N$ with covariance matrix $\frac{1}{w}I_N$. Let γ_K^w be defined by conditioning γ^w on K . Assume that ν is a probability measure on K of the form $d\nu = f d\gamma_K^w$ with $f : K \rightarrow \mathbb{R}$ Borel and log-concave. Then ν satisfies a log-Sobolev inequality with constant $2w$, meaning for all probability measures μ on K ,*

$$\text{Ent}(\mu \mid \nu) \leq 2w \text{Fisher}(\mu \mid \nu).$$

Moreover ν satisfies Gaussian concentration: for every $F \in H^1$, we have

$$\log \mathbb{E}_\nu[e^{tF}] \leq t \mathbb{E}_\nu[F] + \frac{w}{2} t^2 \sup_K |\nabla F|^2, \quad \text{for all } t \in \mathbb{R}. \quad (3.27)$$

We now state a key concentration result due to [BEY⁺14b]. Recall that if μ satisfies Assumption 3.1, then there exists $c > 0$ such that

$$U_N \cdot \nabla^2 H U_N \geq c \sum_{i,j} (u_i - u_j)^2, \quad \text{for all } U_N \in \mathbb{R}^N.$$

The crucial observation is that when $\sum_{i=1}^N u_i = 0$, then

$$\sum_{i,j} (u_i - u_j)^2 = \sum_{i,j} (u_i^2 + u_j^2) = 2N \sum_{i=1}^N u_i^2.$$

Therefore, for all $U_N \in \mathbb{R}^N$ such that $\sum_{i=1}^N u_i = 0$, we have

$$U_N \cdot \nabla^2 H U_N \geq 2Nc \sum_{i=1}^N u_i^2. \quad (3.28)$$

Furthermore, one can observe that $\nabla \phi := (A_1^\mu)^{-1} \nabla F$ satisfies $\partial_1 \phi + \dots + \partial_N \phi = 0$ when $\partial_1 F + \dots + \partial_N F = 0$, since ϕ is a function of the gaps. As we will see below, combining this with (3.28) gives the following Gaussian concentration estimate:

Lemma 3.10. *Let $I \subset \{1, \dots, N\}$ be an interval of integers and π_I be the projection on the coordinates $(x_i)_{i \in I}$. Let μ satisfy Assumption 3.1. Suppose that for every $i, j \in \{1, \dots, N\}$, $i \neq j$,*

$$\lim_{x \rightarrow 0} \chi_{ij}(x) = +\infty,$$

and that there exists a constant $c_1 > 0$ such that for every $i, j \in I$ with $i \neq j$ and $X_N \in D_N$,

$$\chi_{ij}''(x_j - x_i) \geq \frac{c_1}{2} > 0. \quad (3.29)$$

Let $F = G \circ \pi_I$, for some $G \in H^1(\pi_I \# \mu)$. If F is independent of $\sum_{i \in I} x_i$, i.e. $\sum_{i \in I} \partial_i F = 0$, then for all $t \in \mathbb{R}$, we have

$$\log \mathbb{E}_\mu[e^{tF}] \leq t \mathbb{E}_\mu[F] + \frac{t^2}{2c_1 |I|} \sup |\nabla F|^2.$$

The proof of Lemma 3.10 can be found in [BEY⁺14b, Le. 3.9] and can be readily adapted to our circular setting. For completeness, we give the proof, following essentially line by line the proof of [BEY⁺14b].

Proof. Without loss of generality, suppose that $I = \{1, \dots, m\}$.

On D_N introduce the coordinates (x, y) with $x = (x_i)_{i \in I} \in D_m$ and $y = (x_i)_{i \in I^c} \in D_{N-m}$. The energy H can be split into $H(x, y) = H_1(x) + H_2(x, y)$ with

$$H_1(x) = \sum_{i, j \in I, i \neq j} \chi_{ij}(|x_i - x_j|)$$

and

$$H_2(x, y) = \sum_{i, j \in I^c, i \neq j} \chi_{ij}(|x_i - x_j|) + 2 \sum_{i \in I, j \in I^c} \chi_{ij}(|x_i - x_j|).$$

Observe that $\nabla^2 H_1 > 0$ and $\nabla^2 H_2 \geq 0$, and that H_1 is independent of $\sum_{i \in I} x_i$, i.e., $\sum_{i \in I} \partial_i H_1 = 0$.

Now we introduce on D_m the coordinates $x = (z, \omega)$ with $z = (x_1, \dots, x_{m-1}) \in D_{m-1}$ and $\omega = m^{-\frac{1}{2}} \sum_{i=1}^m x_i$. Observe that this change of variables can be written $(z, \omega) = M(x_1, \dots, x_m)$, with M orthogonal. Since H_1 is independent of ω , one can write it as $H_1 = \tilde{H}_1(z)$ for some map $\tilde{H}_1$. Similarly, F can be written $F = \tilde{F}(z)$ for some map $\tilde{F}$.

Set $q = (\omega, y)$. Let ν be the push-forward of μ by the map $(z, q) \mapsto z$. The density of ν is given by $d\nu(z) \propto e^{-\tilde{H}(z)} dz$ where

$$\tilde{H}(z) = -\log \int e^{-H(z, q)} dq. \quad (3.30)$$

Fix $z \in D_{m-1}$ and $v \in \mathbb{R}^{m-1}$. Consider $h : t \mapsto \tilde{H}(z + tv)$. As in Lemma 3.8, denote by $\nabla_{qq}^2, \nabla_{qz}^2, \nabla_{zq}^2, \nabla_{zz}^2$ block matrices of the Hessian. We have

$$h''(t) = \mathbb{E}_{\mathbb{P}(z+tv)}[v \cdot \nabla_{zz}^2 H(z, \cdot) v] - \text{Var}_{\mathbb{P}(z+tv)}[v \cdot (\nabla_z H)(z, \cdot)],$$

where for any $z \in D_{m-1}$, $\mathbb{P}(z)$ stands for the probability measure

$$d\mathbb{P}(z) = \frac{1}{Z(z)} e^{-H(z, q)} dq.$$

Using the Brascamp-Lieb inequality (see Lemma 3.7), one can write

$$\text{Var}_{\mathbb{P}(z+tv)}[v \cdot (\nabla_z H)(z, \cdot)] \leq \mathbb{E}_{\mathbb{P}(z+tv)}[v \cdot (\nabla_{zq}^2 H (\nabla_{qq}^2 H)^{-1} \nabla_{zq}^2 H)(z, \cdot) v].$$

Hence,

$$h''(t) \geq \mathbb{E}_{\mathbb{P}(z+tv)}[v \cdot (\nabla_{zz}^2 H)(z, \cdot) v - v \cdot (\nabla_{zq}^2 H (\nabla_{qq}^2 H)^{-1} \nabla_{zq}^2 H)(z, \cdot) v].$$

Thus, by taking $t = 0$,

$$\nabla_{zz}^2 \tilde{H}(z) \geq \mathbb{E}_{\mathbb{P}(z)}[\nabla_{zz}^2 H(z, \cdot) - (\nabla_{zq}^2 H (\nabla_{qq}^2 H)^{-1} \nabla_{zq}^2 H)(z, \cdot)]. \quad (3.31)$$

Since H_1 is independent of q , we have

$$\nabla_{zq}^2 H (\nabla_{qq}^2 H)^{-1} \nabla_{zq}^2 H = \nabla_{zq}^2 H_2 (\nabla_{qq}^2 H_2)^{-1} \nabla_{zq}^2 H_2.$$

Hence,

$$\nabla_{zz}^2 H - \nabla_{zq}^2 H (\nabla_{qq}^2 H)^{-1} \nabla_{zq}^2 H = \nabla_{zz}^2 H_1 + \nabla_{zz}^2 H_2 - \nabla_{zq}^2 H_2 (\nabla_{qq}^2 H_2)^{-1} \nabla_{zq}^2 H_2. \quad (3.32)$$

Since $\nabla_{zz}^2 H_2 \geq 0$, the Schur complement is also non-negative:

$$\nabla_{zz}^2 H_2 - \nabla_{zq}^2 H_2 (\nabla_{qq}^2 H_2)^{-1} \nabla_{zq}^2 H_2 \geq 0.$$

Therefore, by (3.32),

$$\nabla_{zz}^2 H - \nabla_{zq}^2 H (\nabla_{qq}^2 H)^{-1} \nabla_{zq}^2 H \geq \nabla_{zz}^2 H_1.$$

Inserting this into (3.31), we deduce from (3.29) that for all $u \in \mathbb{R}^{m-1}$,

$$u \cdot (\nabla_{zz}^2 \tilde{H})(z) u \geq \mathbb{E}_{\mathbb{P}(z)}[u \cdot \nabla_{zz}^2 H_1(z, \cdot) u].$$

Next notice that, in view of the assumption (3.29),

$$u \cdot \nabla_{zz}^2 H_1 u = \tilde{M} u \cdot \nabla_{xx}^2 H_1(x) (\tilde{M} u) \geq \frac{c_1}{2} \sum_{i \neq j} ((\tilde{M} u)_i - (\tilde{M} u)_j)^2,$$

where $\tilde{M}$ denotes the first $m-1$ columns of M . Moreover,

$$\sum_{i \neq j} ((\tilde{M} u)_i - (\tilde{M} u)_j)^2 = 2m \sum_{i=1}^m u_i^2.$$

Combining the last three displays, we conclude that

$$\nabla_{zz}^2 \tilde{H} \geq c_1 m \text{Id}.$$

Therefore, by the Bakry-Emery criterion stated in Lemma 3.9, for all $t \in \mathbb{R}$, we have

$$\mathbb{E}_\mu[e^{tF}] = \mathbb{E}_\nu[e^{t\tilde{F}}] \leq e^{t\mathbb{E}_\nu[\tilde{F}] + \frac{t^2}{2c_1 m} \sup |\nabla_z \tilde{F}|^2}.$$

Since M is orthogonal, $|\nabla_z \tilde{F}|^2 = |\nabla F|^2$. This concludes the proof. $\square$

4. NEAR-OPTIMAL RIGIDITY

This section is dedicated to the proof of the rigidity result stated in Theorem 1. The approach draws on several techniques introduced in the seminal work [BEY⁺14b, Th. 3.1]. Since our setting is the circle, the expected values of the gaps can be computed explicitly. As a result, we avoid one of the main technical challenges of [BEY⁺14b], namely estimating the accuracy of the standard positions. Our first objective is to establish the following local law for the gaps: we show that for every $i \in \{1, \dots, N\}$ and $1 \leq k \leq \lfloor \frac{N}{2} \rfloor$, the quantity $N(x_{i+k} - x_i)$ is typically of order k . To achieve this, we carry out a multiscale analysis, inspired by [BEY⁺14b], which allows us to bootstrap the local law down to microscale. The argument relies on a convexification procedure, which we describe first.

4.1. Comparison to a constrained Gibbs measures. Since the Hessian of the energy degenerates when particles are far apart, one cannot directly obtain Gaussian concentration estimates for $\mathbb{P}_{N,\beta}$. Following [BEY⁺14b], we add a convexifying term to the Hamiltonian that penalizes configurations with large gaps. Let $\theta : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be a smooth function such that $\theta(x) = x^2$ for $x > 2$, $\theta(x) = 0$ for $x \in [0, 1]$ and $\theta'' \geq 0$ on $(0, \infty)$. Let $i_0 \in \{1, \dots, N\}$, $L \in \{1, \dots, \lfloor \frac{N}{2} \rfloor\}$ and $R > 0$. Let us define

$$I := \{i, \dots, i + L\},$$

with the convention that this discrete interval is taken modulo N . Then, we define the forcing term

$$F := 2 \sum_{i,j \in I: i < j} \theta\left(\frac{N}{R}(x_j - x_i)\right) \quad (4.1)$$

and the locally constrained Gibbs measure

$$d\mathbb{Q}_{N,\beta}(X_N) = \frac{1}{K_{N,\beta}} e^{-\beta(\mathcal{H}_N + F)(X_N)} \mathbb{1}_{D_N}(X_N) dX_N, \quad (4.2)$$

where

$$K_{N,\beta} := \int_{D_N} e^{-\beta(\mathcal{H}_N + F)(X_N)} \mathbb{1}_{D_N}(X_N) dX_N.$$

In what follows, we often choose $R = L^{1+\varepsilon}$ for some $\varepsilon > 0$.

Recall that the total variation distance between two probability measures μ and ν on D_N is defined by

$$\text{TV}(\mu, \nu) = \sup_{\mathcal{A} \in \mathcal{B}(D_N)} |\mu(\mathcal{A}) - \nu(\mathcal{A})|,$$

where $\mathcal{B}(D_N)$ is the set of measurable subsets of D_N . By the Pinsker inequality, see [ABC⁺00, Ch. 5], we have

$$\text{TV}(\mu, \nu)^2 \leq 2 \text{Ent}(\mu \mid \nu), \quad (4.3)$$

where $\text{Ent}(\cdot \mid \nu)$ is the relative entropy with respect to ν . Using (4.3) and the uniform log-concavity of the constrained measure (4.2) in the window I , one may derive the following control:

Lemma 4.1. *Let $i_0 \in \{1, \dots, N\}$, $L \in \{1, \dots, \lfloor \frac{N}{2} \rfloor\}$ and $R > 0$. Let*

$$I := \{i, \dots, i + L\}.$$

Denote by π_I the projection $\pi_I : X_N \in D_N \mapsto (x_i)_{i \in I} \in D_{L+1}$. Let $\mathbb{Q}_{N,\beta}$ be the probability measure (4.2). There exists a constant $C > 0$ depending only on β and s such that

$$\text{TV}(\pi_I \# \mathbb{P}_{N,\beta}, \pi_I \# \mathbb{Q}_{N,\beta})^2 \leq CL^5 R^s \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{N}{R} (x_{i_0+L} - x_{i_0}) \right)^2 \mathbb{1}_{x_{i_0+L} - x_{i_0} \geq \frac{R}{N}} \right].$$

Proof. For every $k = 1, \dots, \lfloor N/2 \rfloor$, let us denote

$$\overline{\text{Gap}}_k : X_k \in D_k \mapsto (N(x_2 - x_1), \dots, N(x_k - x_{k-1})) \in \mathbb{R}^{k-1}.$$

Recall the relative entropy defined in (3.25). Applying Pinsker's inequality (4.3) to $\mu = \pi_I \# \mathbb{P}_{N,\beta}$ and $\nu = \pi_I \# \mathbb{Q}_{N,\beta}$ gives

$$\text{TV}(\pi_I \# \mathbb{P}_{N,\beta}, \pi_I \# \mathbb{Q}_{N,\beta})^2 \leq 2 \text{Ent}(\pi_I \# \mathbb{P}_{N,\beta} \mid \pi_I \# \mathbb{Q}_{N,\beta}). \quad (4.4)$$

Let us recall that for any measurable map $T : \pi_I(D_N) \rightarrow \mathbb{R}^{L+1}$, we have

$$\text{Ent}((T \circ \pi_I) \# \mathbb{P}_{N,\beta} \mid (T \circ \pi_I) \# \mathbb{Q}_{N,\beta}) \leq \text{Ent}(\pi_I \# \mathbb{P}_{N,\beta} \mid \pi_I \# \mathbb{Q}_{N,\beta}).$$

This can be proven by using, for instance, the variational representation of the entropy. Applying this to the map $T : (x_i)_{i \in I} \mapsto (x_{i_0}, \overline{\text{Gap}}_{L+1}((x_i)_{i \in I}))$ and to its inverse, we obtain that

$$\text{Ent}(\pi_I \# \mathbb{P}_{N,\beta} \mid \pi_I \# \mathbb{Q}_{N,\beta}) = \text{Ent}(\text{Law}_{\mathbb{P}_{N,\beta}}(x_{i_0}, \overline{\text{Gap}}_{L+1} \circ \pi_I(X_N)) \mid \text{Law}_{\mathbb{P}_{N,\beta}}(x_{i_0}, \overline{\text{Gap}}_{L+1} \circ \pi_I(X_N))). \quad (4.5)$$

Recall that the law of x_{i_0} under $\mathbb{P}_{N,\beta}$ is uniform, as is the law of x_{i_0} under $\mathbb{Q}_{N,\beta}$. Moreover, under $\mathbb{P}_{N,\beta}$ (similarly $\mathbb{Q}_{N,\beta}$), x_{i_0} is independent from $\overline{\text{Gap}}_{L+1}(x_{i_0})$. We thus get from the tensorization property of the entropy and (4.5) that

$$\begin{aligned} \text{Ent}(\pi_I \# \mathbb{P}_{N,\beta} \mid \pi_I \# \mathbb{Q}_{N,\beta}) &= \text{Ent}((\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{P}_{N,\beta} \mid (\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{Q}_{N,\beta}) + \text{Ent}(dx|dx) \\ &= \text{Ent}((\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{P}_{N,\beta} \mid (\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{Q}_{N,\beta}). \end{aligned}$$

By Lemma 3.8, the push-forward of $\mathbb{Q}_{N,\beta}$ by the map π_I has density proportional to $\exp(-\beta(H + \tilde{H}))$ with $\nabla^2 \tilde{H} \geq 0$ and H defined by

$$H : (x_i)_{i \in I} \in D_{L+1} \mapsto N^{-s} \sum_{i,j \in D_{L+1}: i \neq j} g(x_i - x_j) + F \circ \pi_I.$$

By definition of F (4.1) and θ , for all $u \in \mathbb{R}^{L+1}$,

$$u \cdot \nabla^2 H u \geq C_0 \sum_{i,j \in I: i < j} \frac{(N(u_j - u_i))^2}{R^{s+2}} \geq C_0 \sum_{i \in I \setminus \max I} \frac{(N(u_{i+1} - u_i))^2}{R^{s+2}},$$

for some constant $C_0 > 0$ independent of N, L and R . We deduce that the push-forward of $\mathbb{P}_{N,\beta}$ by the map $\overline{\text{Gap}}_{L+1} \circ \pi_I$ has density proportional to $\exp(-\beta(H_0 + \tilde{H}_0))$ with $\nabla^2 H_0 \geq \frac{C_0}{R^{s+2}} I$ and $\nabla^2 \tilde{H}_0 \geq 0$.

Therefore, by Lemma 3.9, $(\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{Q}_{N,\beta}$ satisfies a log-Sobolev inequality with constant $2c^{-1}$ where $c := \frac{C\beta}{R^{s+2}}$. It follows that

$$\begin{aligned} \text{Ent}((\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{P}_{N,\beta} \mid (\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{Q}_{N,\beta}) \\ \leq 2c^{-1} \text{Fisher}((\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{P}_{N,\beta} \mid (\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{Q}_{N,\beta}), \end{aligned}$$

where the Fisher information has been defined in (3.26). One can write $F = G \circ \overline{\text{Gap}}_{L+1} \circ \pi_I$ for some $G : \mathbb{R}^L \rightarrow \mathbb{R}$. This gives the bound

$$\text{Ent}((\overline{\text{Gap}}_{L+1} \circ \pi_I) \# \mathbb{P}_{N,\beta} \mid \nu) \leq CR^{s+2} \mathbb{E}_{\mathbb{P}_{N,\beta}} [|\nabla(G \circ \overline{\text{Gap}}_{L+1} \circ \pi_I)|^2]. \quad (4.6)$$

Let us bound the right-hand side of the last display. Note that we have the explicit expression

$$G(y_{i_0}, \dots, y_{i_0+L-1}) = 2 \sum_{i,j \in I: i < j} \theta\left(\frac{1}{R} \sum_{k=i}^{j-1} y_k\right).$$

Therefore, for every $l \in I \setminus \{\max(I)\}$,

$$\partial_l G(y_{i_0}, \dots, y_{i_0+L-1}) = 2 \sum_{i,j \in I: i < j} \frac{1}{R} \theta'\left(\frac{1}{R} \sum_{k=i}^{j-1} y_k\right) \mathbf{1}_{i \leq l < j}.$$

Hence, since θ' is non-negative and increasing,

$$\begin{aligned} |\partial_l G(y_{i_0}, \dots, y_{i_0+L-1})| &\leq 2 \sum_{i,j \in I: i < j} \frac{1}{R} \theta' \left(\frac{1}{R} \sum_{k=i_0}^{i_0+L-1} y_k \right) \mathbb{1}_{i \leq l < j} \\ &\leq \frac{2}{R} \sum_{i,j \in I: i < j} \theta' \left(\frac{N}{R} (x_{i_0+L} - x_{i_0}) \right) \mathbb{1}_{i \leq l < j} \\ &\leq C \frac{L^2}{R} \theta' \left(\frac{N}{R} (x_{i_0+L} - x_{i_0}) \right). \end{aligned}$$

Hence,

$$\sum_{l \in I \setminus \{\max I\}} |\partial_l G(y_{i_0}, \dots, y_{i_0+L-1})|^2 \leq C \frac{L^5}{R^2} (\theta')^2 \left(\frac{N}{R} (x_{i_0+L} - x_{i_0}) \right).$$

Since θ is smooth and $\theta'(0) = 0$, we have $|\theta'(x) - \theta'(0)| = \theta'(x) \leq C|x|$ on $(0, 2]$. Since $\theta(x) = x^2$ for $x > 2$, there exists a constant $C > 0$ such that for all $x \in (0, \infty)$, $|\theta'(x)| \leq Cx$. We deduce from this and the last display that the Fisher information is bounded by

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{N,\beta}} [|\nabla(G \circ \overline{\text{Gap}}_{L+1} \circ \pi_I)|^2] &\leq C \frac{L^5}{R^2} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\theta' \left(\frac{N}{R} (x_{i_0+L} - x_{i_0}) \right) \right)^2 \right] \\ &\leq C \frac{L^5}{R^2} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\theta' \left(\frac{N}{R} (x_{i_0+L} - x_{i_0}) \right) \right)^2 \mathbb{1}_{|x_{i_0+L} - x_{i_0}| \geq \frac{R}{N}} \right] \\ &\leq C \frac{L^5}{R^2} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{N}{R} (x_{i_0+L} - x_{i_0-L}) \right)^2 \mathbb{1}_{x_{i_0+L} - x_{i_0} \geq \frac{R}{N}} \right], \end{aligned}$$

where we have used that $\theta'(x) = 0$ if $x \in (0, 1)$. Inserting this into (4.6) and using (4.4) concludes the proof of the lemma. $\square$

4.2. First local law. We now prove that each gap $N(x_{i+k} - x_i)$ is typically of order k with an exponentially small probability of deviation.

Lemma 4.2. *Set*

$$\delta_0 := \frac{1-s}{2(s+2)}. \quad (4.7)$$

Let $\delta > 0$. There exist $c > 0$ and $C > 0$ such that for every $i \in \{1, \dots, N\}$ and $1 \leq k \leq \lfloor \frac{N}{2} \rfloor$,

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq C \exp\left(-ck^{2 \min(\delta, \delta_0 \frac{1+\delta}{1+\delta_0})}\right). \quad (4.8)$$

The proof of Lemma 4.2 is inspired by the multiscale analysis of [BEY+14b]. We proceed by a bootstrap on scales: assuming that the local law (4.8) holds for some $k \in \{1, \dots, \lfloor \frac{N}{2} \rfloor\}$, Lemma 4.1 allows us to convexify the measure within a window of size k without significantly perturbing it. Moreover, the convexified measure satisfies better concentration estimates, allowing one to prove via Lemma 3.10 that (4.8) holds at a slightly smaller scale.

Proof of Lemma 4.2.

Step 1: setting the bootstrap. Define

$$\delta_0 := \frac{1-s}{2(s+2)} \in (0, 1).$$

We wish to prove that there exist $c_0 > 0$ and $C_0 > 0$ independent of N such that for every $i \in \{1, \dots, N\}$, $1 \leq k \leq \lfloor \frac{N}{2} \rfloor$ and all $\delta > 0$,

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq \begin{cases} C_0 e^{-c_0 k^{2\delta}} & \text{if } \delta \in (0, \delta_0] \\ C_0 e^{-c_0 k^{2\delta_0 \frac{1+\delta}{1+\delta_0}}} & \text{if } \delta > \delta_0. \end{cases} \quad (4.9)$$

Observe that (4.9) trivially holds for $k = \lfloor N/2 \rfloor$ since $0 \leq N(x_{i+\lfloor N/2 \rfloor} - x_i) \leq N$.

Let $K \in \{1, \dots, \lfloor N/2 \rfloor\}$. Assume that (4.9) holds for every $k \geq K$. Fix $\delta \in (0, \delta_0]$ and fix

$$\alpha_0 \in \left(0, 1 - \frac{1+s}{2-\delta(s+2)}\right), \quad (4.10)$$

to be determined below. Note that since $\delta \leq \delta_0 < \frac{1-s}{s+2}$, we have $1 - \frac{1+s}{2-\delta(s+2)} > 0$. We will show, for a particular choice of α_0 , that (4.9) holds for every $k \geq K^{1-\alpha_0}$.

Let $i \in \{1, \dots, N\}$, $k \in \{K^{1-\alpha_0}, \dots, \lfloor \frac{N}{2} \rfloor\}$ and fix

$$\gamma \in (\delta(1-\alpha_0), \delta_0), \quad (4.11)$$

to be determined below. Define

$$I := \{i, \dots, i+K\}.$$

Let $\theta : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be a smooth cutoff function with $\theta(x) = x^2$ for $x > 2$, $\theta = 0$ on $[0, 1]$ and $\theta'' \geq 0$ on $(0, \infty)$. Define, as in (4.1)

$$F := 2 \sum_{l, j \in I: l < j} \theta\left(\frac{N}{K^{1+\gamma}}(x_l - x_j)\right), \quad (4.12)$$

and, as in (4.2),

$$d\mathbb{Q}_{N,\beta}(X_N) = \frac{1}{K_{N,\beta}} e^{-\beta(\mathcal{H}_N + F)(X_N)} \mathbb{1}_{D_N}(X_N) dX_N.$$

Since $x_{i+k} - x_i$ is a function of $(x_i)_{i \in I}$, we have that

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq \mathbb{Q}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) + \text{TV}(\pi_I \# \mathbb{P}_{N,\beta}, \pi_I \# \mathbb{Q}_{N,\beta}). \quad (4.13)$$

Step 2: upper bound on the total variation distance. Applying Lemma 4.1 with $L := K$ and $R := K^{1+\gamma} \leq K^2$, we obtain

$$\begin{aligned} \text{TV}(\pi_I \# \mathbb{P}_{N,\beta}, \pi_I \# \mathbb{Q}_{N,\beta})^2 &\leq CK^7 \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{1}{K^{1+\gamma}} N(x_{i+K} - x_i) \right)^2 \mathbb{1}_{N(x_{i+K} - x_i) \geq K^{1+\gamma}} \right] \\ &\leq CK^7 \sum_{j \geq K^{1+\gamma}} \frac{j^2}{K^{2(1+\gamma)}} \mathbb{P}_{N,\beta}(N(x_{i+K} - x_i) \geq j). \end{aligned} \quad (4.14)$$

Notice that

$$\delta_0 \frac{1+\delta}{1+\delta_0} \leq \delta \iff \delta \geq \delta_0.$$

It follows that

$$C_0 e^{-c_0 k^{2 \min(\delta, \delta_0 \frac{1+\delta}{1+\delta_0})}} = \begin{cases} C_0 e^{-c_0 k^{2\delta}} & \text{if } \delta \in (0, \delta_0] \\ C_0 e^{-c_0 k^{2\delta_0 \frac{1+\delta}{1+\delta_0}}} & \text{if } \delta > \delta_0. \end{cases}$$

Let $n \geq 1$. By (4.9) and the above display, for every $j \in \{K^{1+n\gamma}, \dots, K^{1+(n+1)\gamma} - 1\}$,

$$\mathbb{P}_{N,\beta}(N(x_{i+K} - x_i) \geq j) \leq C_0 \exp\left(-c_0 K^{2 \min(n\gamma, \delta_0 \frac{1+n\gamma}{1+\delta_0})}\right). \quad (4.15)$$

Hence, using (4.15),

$$\begin{aligned} &\sum_{j \geq K^{1+\gamma}} \frac{j^2}{K^{2(1+\gamma)}} \mathbb{P}_{N,\beta}(N(x_{i+K} - x_i) \geq j) \\ &\leq C \sum_{n=1}^{\infty} \sum_{j: K^{1+n\gamma} \leq j < K^{1+(n+1)\gamma}} \frac{j^2}{K^{2(1+\gamma)}} \mathbb{P}_{N,\beta}(N(x_{i+K} - x_i) \geq j) \\ &\leq C \sum_{n=1}^{\infty} \frac{K^{3(1+(n+1)\gamma)}}{K^{2(1+\gamma)}} \exp\left(-c_0 K^{2 \min(n\gamma, \delta_0 \frac{1+n\gamma}{1+\delta_0})}\right). \end{aligned} \quad (4.16)$$

Therefore, using that $k \geq K^{1-\alpha_0}$, we have that for all $\kappa < \frac{\gamma}{1-\alpha_0}$, there exists $C > 0$ independent of N and K such that

$$\text{TV}(\pi_I \# \mathbb{P}_{N,\beta}, \pi_I \# \mathbb{Q}_{N,\beta}) \leq C e^{-c_0 k^{2\kappa}}. \quad (4.17)$$

Step 3: accuracy under $\mathbb{Q}_{N,\beta}$. Since $N(x_{i+k} - x_i)$ is not uniformly bounded from above with respect to N , one cannot directly apply the total variation bound (4.17) to approximate its expectation under $\mathbb{Q}_{N,\beta}$. It is therefore necessary to first establish a tightness result for the variable $N(x_{i+k} - x_i)$ under $\mathbb{Q}_{N,\beta}$. Let $\varepsilon > 0$. One can write

$$\log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{(N(x_{i+k}-x_i))^\varepsilon}] = \log \mathbb{E}_{\mathbb{P}_{N,\beta}}[e^{(N(x_{i+k}-x_i))^\varepsilon - \beta F}] - \log \mathbb{E}_{\mathbb{P}_{N,\beta}}[e^{-\beta F}].$$

By Jensen's inequality and the fact $F \geq 0$,

$$\log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{(N(x_{i+k}-x_i))^\varepsilon}] \leq \log \mathbb{E}_{\mathbb{P}_{N,\beta}}[e^{(N(x_{i+k}-x_i))^\varepsilon}] + \beta \mathbb{E}_{\mathbb{P}_{N,\beta}}[F]. \quad (4.18)$$

By the definition of θ ,

$$\begin{aligned} F &= 2 \sum_{i,j \in I: i < j} \theta\left(\frac{N(x_j - x_i)}{K^{1+\gamma}}\right) \\ &= 2 \sum_{i,j \in I: i < j} \theta\left(\frac{N(x_j - x_i)}{K^{1+\gamma}}\right) \mathbb{1}_{N(x_j - x_i) \geq K^{1+\gamma}} \\ &\leq 2 \sum_{i,j \in I: i < j} \theta\left(\frac{N(x_j - x_i)}{K^{1+\gamma}}\right) \mathbb{1}_{N(x_{i+K} - x_i) \geq K^{1+\gamma}} \\ &\leq C \sum_{i,j \in I: i < j} \left(\frac{N(x_j - x_i)}{K^{1+\gamma}}\right)^2 \mathbb{1}_{N(x_{i+K} - x_i) \geq K^{1+\gamma}} \\ &\leq CK^2 \left(\frac{N(x_{i+K} - x_i)}{K^{1+\gamma}}\right)^2 \mathbb{1}_{N(x_{i+K} - x_i) \geq K^{1+\gamma}}. \end{aligned}$$

Therefore,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[F] \leq CK^2 \sum_{j \geq K^{1+\gamma}} \left(\frac{j}{K^{1+\gamma}}\right)^2 \mathbb{P}_{N,\beta}(N(x_{i+K} - x_i) \geq j).$$

Splitting the sum over j as in (4.16), we deduce that there exists $C > 0$ independent of N and K such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[F] \leq C. \quad (4.19)$$

We then turn to bounding the first term on the right of (4.18),

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[e^{(N(x_{i+k}-x_i))^\varepsilon}] \leq e^{K^\varepsilon(1+\gamma)} + \sum_{j > K^{1+\gamma}} e^{j^\varepsilon} \mathbb{P}_{N,\beta}(N(x_{i+K} - x_i) \geq j).$$

Using (4.15), we find that for $\varepsilon > 0$ small enough,

$$\log \mathbb{E}_{\mathbb{P}_{N,\beta}}[e^{(N(x_{i+k}-x_i))^\varepsilon}] \leq CK^{\varepsilon(1+\gamma)},$$

for some constant $C > 0$ independent of N and K . Fix $\varepsilon > 0$ accordingly. Plugging in the above display and (4.19) into (4.18) yields

$$\log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{(N(x_{i+k}-x_i))^\varepsilon}] \leq CK^{\varepsilon(1+\gamma)}, \quad (4.20)$$

for some constant $C > 0$ independent of N and K . One can next write

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i) \mathbb{1}_{N(x_{i+k}-x_i) \geq K^2}] &\leq \sum_{j \geq K^2} j \mathbb{Q}_{N,\beta}(N(x_{i+k} - x_i) \geq j) \\ &= \sum_{j \geq K^2} j \mathbb{Q}_{N,\beta}(e^{(N(x_{i+k}-x_i))^\varepsilon} \geq e^{j^\varepsilon}) \\ &\leq \sum_{j \geq K^2} j e^{-j^\varepsilon} e^{CK^{\varepsilon(1+\gamma)}}, \end{aligned}$$

where we have used Markov's inequality and (4.20) in the last inequality. We conclude that since $\gamma \in (0, 1)$,

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i) \mathbb{1}_{N(x_{i+k}-x_i) \geq K^2}] \leq C, \quad (4.21)$$

for some positive constant C independent of N and K . On the other hand, using (4.15), we get that by proceeding as in (4.16) that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_{i+k} - x_i)\mathbb{1}_{N(x_{i+k}-x_i)>K^2}] \leq C, \quad (4.22)$$

for some positive constant C independent of N and K .

Recall that for every function f bounded and measurable, we have

$$|\mathbb{E}_{\mathbb{Q}_{N,\beta}}[f] - \mathbb{E}_{\mathbb{P}_{N,\beta}}[f]| \leq \|f\|_{L^\infty} \text{TV}(\mathbb{Q}_{N,\beta}, \mathbb{P}_{N,\beta}).$$

It follows that

$$\begin{aligned} & \left| \mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_{i+k} - x_i)\mathbb{1}_{N(x_{i+k}-x_i)\leq K^2}] - \mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)\mathbb{1}_{N(x_{i+k}-x_i)\leq K^2}] \right| \\ & \leq K^2 \text{TV}(\pi_I \# \mathbb{P}_{N,\beta}, \pi_I \# \mathbb{Q}_{N,\beta}). \end{aligned}$$

Using (4.17), we get that for all $\kappa < \frac{\gamma}{1-\alpha_0}$, there exists a constant $C > 0$ such that

$$\left| \mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_{i+k} - x_i)\mathbb{1}_{N(x_{i+k}-x_i)\leq K^2}] - \mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)\mathbb{1}_{N(x_{i+k}-x_i)\leq K^2}] \right| \leq CK^2 e^{-c_0 k^{2\kappa}}.$$

Furthermore, combining (4.21) and (4.22), one gets

$$|\mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_{i+k} - x_i)\mathbb{1}_{N(x_{i+k}-x_i)>K^2}] - \mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)\mathbb{1}_{N(x_{i+k}-x_i)>K^2}]| \leq C,$$

for some positive constant C independent of N and K . Let us observe that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_{i+k} - x_i)] = k \mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_2 - x_1)],$$

since under $\mathbb{P}_{N,\beta}$, particles are identically distributed. Since $\sum_{i=1}^N N(x_{i+1} - x_i) = N$, we therefore get that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_2 - x_1)] = 1.$$

Thus,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_{i+k} - x_i)] = k.$$

We conclude that

$$|\mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)] - \mathbb{E}_{\mathbb{P}_{N,\beta}}[N(x_{i+k} - x_i)]| = |\mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)] - k| \leq C, \quad (4.23)$$

for some positive constant C independent of N and K . Now, we observe that

Step 4: fluctuations under $\mathbb{Q}_{N,\beta}$. We now study the fluctuations of $N(x_{i+k} - x_i)$ under the probability measure $\mathbb{Q}_{N,\beta}$. Denote

$$G : X_N \in D_N \mapsto N(x_{i+k} - x_i).$$

Observe that $\sum_{j \in I} \partial_j G = 0$, $\partial_j G = 0$ for every $j \in I^c$ and $\sup |\nabla G|^2 = 2N^2$. Moreover, $\mathbb{Q}_{N,\beta}$ satisfies Assumption 3.1 with, for every $i, j \in \{1, \dots, N\}$ with $i \neq j$,

$$\chi_{ij} : x \in \mathbb{T} \mapsto \beta \left(N^{-s} g(x) + \theta \left(\frac{N}{K^{1+\gamma}} x \right) \mathbb{1}_{i,j \in I} \right).$$

For every $i, j \in I$ with $i \neq j$ and $x \in \mathbb{T}$, we have

$$\chi_{ij}''(x) = \beta \left(N^{-s} g''(x) + \theta'' \left(\frac{N}{K^{1+\gamma}} x \right) \left(\frac{N}{K^{1+\gamma}} \right)^2 \right).$$

If $N|x| \leq 2K^{1+\gamma}$, then one can write

$$N^{-s} g''(x) + \theta'' \left(\frac{N}{K^{1+\gamma}} x \right) \left(\frac{N}{(2K)^{1+\gamma}} \right)^2 \geq N^{-s} g''(x) \geq c_0 \frac{N^2}{K^{(s+2)(1+\gamma)}},$$

for some constant $c_0 \in (0, 2)$. If $N|x| \geq 2K^{1+\gamma}$, then we use

$$N^{-s} g''(x) + \theta'' \left(\frac{N}{K^{1+\gamma}} x \right) \left(\frac{N}{K^{1+\gamma}} \right)^2 \geq \theta'' \left(\frac{N}{K^{1+\gamma}} x \right) \left(\frac{N}{K^{1+\gamma}} \right)^2 = 2 \left(\frac{N}{K^{1+\gamma}} \right)^2 \geq c_0 \frac{N^2}{K^{(s+2)(1+\gamma)}},$$

since by assumption $\theta(x) = x^2$ for all $x \geq 2$. Combining the two last displays, we deduce that for every $i, j \in I$ with $i \neq j$,

$$\chi_{ij}''(x) \geq \frac{c_1}{2} \quad \text{where } c_1 := \frac{4c_0 \beta N^2}{K^{(1+\gamma)(s+2)}}.$$

Therefore, by applying Lemma 3.10, we obtain that for all $t \in \mathbb{R}$,

$$0 \leq \log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{tG}] - t\mathbb{E}_{\mathbb{Q}_{N,\beta}}[G] \leq \frac{t^2}{2c_1|I|} \sup |\nabla G|^2 \leq Ct^2 K^{(1+\gamma)(s+2)-1},$$

for some constant $C > 0$ independent of N and K . By Markov's inequality, recalling that $k \geq K^{1-\alpha_0}$, this implies that

$$\mathbb{Q}_{N,\beta}(|N(x_{i+k} - x_i) - \mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)]| \geq k^{1+\delta}) \leq C \exp\left(-c \frac{k^{2(1+\delta)}}{K^{(1+\gamma)(s+2)-1}}\right),$$

for some constants $c > 0, C > 0$ independent of N and K . Since $k \geq K^{1-\alpha_0}$, we get

$$\mathbb{Q}_{N,\beta}(|N(x_{i+k} - x_i) - \mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)]| \geq k^{1+\delta}) \leq C \exp\left(-ck^{2(1+\delta)-(1-\alpha_0)^{-1}((1+\gamma)(s+2)-1)}\right).$$

Using (4.23), this gives

$$\mathbb{Q}_{N,\beta}(|N(x_{i+k} - x_i) - k| \geq k^{1+\delta}) \leq C \exp\left(-ck^{2(1+\delta)-(1-\alpha_0)^{-1}((1+\gamma)(s+2)-1)}\right), \quad (4.24)$$

for some constants $c > 0, C > 0$ independent of N and K .

Step 5: conclusion. We have

$$2(1+\delta) - \frac{1}{1-\alpha_0}((1+\gamma)(s+2)-1) > 2\delta \iff \gamma < \frac{2(1-\alpha_0) - (1+s)}{2+s}. \quad (4.25)$$

Observe that the conditions (4.11) and (4.25) can be satisfied if and only if

$$(1-\alpha_0)\delta < \frac{2(1-\alpha_0) - (1+s)}{2+s} \iff 1-\alpha_0 > \frac{1+s}{2-\delta(s+2)}. \quad (4.26)$$

Therefore, by (4.10) we can choose $\gamma \in (\delta(1-\alpha_0), \delta_0)$ satisfying (4.25). Therefore, by (4.24), there exist $\delta' > \delta$ and $C > 0, c > 0$ independent of N and K such that

$$\mathbb{Q}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq Ce^{-ck^{2\delta'}}.$$

In combination with (4.17) and (4.13), since $\frac{\gamma}{1-\alpha_0} > \delta$, we deduce that there exist $\delta' > \delta$ and $C > 0, c > 0$ independent of N and K such that

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq Ce^{-ck^{2\delta'}}. \quad (4.27)$$

Notice that if k is large enough, then

$$Ce^{-ck^{2\delta'}} \leq C_0 e^{-c_0 k^{2\delta}},$$

where C_0 and c_0 are the constants in the bootstrap assumption (4.9). We deduce that there exists an integer k_0 independent of N and K such that for every k with $k \geq k_0$ and $k \geq K^{1-\alpha_0}$, and for all $\delta \in (0, \delta_0]$,

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq C_0 e^{-c_0 k^{2\delta}}. \quad (4.28)$$

Let us now prove that (4.9) holds for $\delta > \delta_0$. Let $\delta > \delta_0$ and $L := \lfloor k^{\frac{1+\delta}{1+\delta_0}} \rfloor \geq k$. One can write

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq \mathbb{P}_{N,\beta}(N(x_{i+L} - x_i) \geq k^{1+\delta}) \leq \mathbb{P}_{N,\beta}(N(x_{i+L} - x_i) \geq L^{1+\delta_0}).$$

Therefore, by (4.27), there exists $\delta'_0 > \delta_0$ such that

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq Ce^{-cL^{2\delta'_0}} \leq Ce^{-ck^{\frac{2\delta'_0}{1+\delta_0} \frac{1+\delta}{1+\delta_0}}}.$$

Therefore, there exists k_0 independent of K and N such that for every k with $k \geq k_0$ and $k \geq K^{1-\alpha_0}$, we have

$$\mathbb{P}_{N,\beta}(N(x_{i+k} - x_i) \geq k^{1+\delta}) \leq C_0 e^{-c_0 k^{\frac{2\delta'_0}{1+\delta_0} \frac{1+\delta}{1+\delta_0}}}.$$

We deduce with (4.28) that there exists k_0 independent of N and K such that for every k with $k \geq k_0$ and $k \geq K^{1-\alpha_0}$, and every $\delta > 0$, (4.9) holds.

We conclude by induction that there exists a constant k_0 , independent of N and K , such that (4.9) holds for all $k \geq k_0$. By adjusting the constant C_0 if necessary, we may extend the validity of (4.9) to all $k \geq 1$. $\square$

From the proof of Lemma 4.2, we deduce the following estimate on the expectation of gaps under the locally constrained measure:

Lemma 4.3. *Let $L \in \{1, \dots, \lfloor N/2 \rfloor - 1\}$ and $i \in \{1, \dots, N\}$. Let*

$$I := \{i, \dots, i + L\}.$$

Let $j \in I$ and $k \in \{1, \dots, \frac{N}{2}\}$ such that $j + k \in I$. Let $\gamma \in (0, 1)$. Let $\mathbb{Q}_{N,\beta}$ be the locally constrained measure (4.2) with $R = L^{1+\gamma}$. There exists $C > 0$ depending on γ, s and β such that if γ is small enough,

$$|\mathbb{E}_{\mathbb{Q}_{N,\beta}}[N(x_{i+k} - x_i)] - k| \leq C.$$

Proof. We have proved in Lemma 4.2 that $\mathbb{P}_{N,\beta}$ satisfies (4.9). We thus conclude by (4.23). $\square$

4.3. Reduction to a block average. In this subsection, we implement a method developed in [BEY⁺14b] to study the fluctuations of particle positions. The strategy consists of replacing a single point x_i with the average of the x_j 's over a block centered at x_i .

For every $i \in \{1, \dots, N\}$ and $1 \leq k \leq \lfloor \frac{N}{2} \rfloor$, let $I_k(i)$ stand for the interval of indices

$$I_k(i) := \{j \in \{1, \dots, N\} : d(i, j) \leq k\}.$$

Define the block average

$$x_i^{[k]} := \frac{1}{|I_k(i)|} \sum_{j \in I_k(i)} x_j. \quad (4.29)$$

Lemma 4.4 (Comparison to a block average). *Let $\varepsilon > 0$ be small enough. There exist $C > 0$ and $c > 0$ independent of N such that for every $i \in \{1, \dots, N\}$ and $1 \leq k \leq \lfloor \frac{N}{2} \rfloor$,*

$$\mathbb{P}_{N,\beta}(|N(x_i - x_i^{[k]})| \geq k^{\frac{s}{2} + \varepsilon}) \leq Ce^{-ck^{\frac{\varepsilon}{s+2}}}.$$

Proof. Let $i \in \{1, \dots, N\}$ and $1 \leq k \leq \lfloor \frac{N}{2} \rfloor$. Fix $\varepsilon > 0$. Let $p \geq 1$ be a large number and $\alpha := \frac{1}{p}$. Since $x_i^{[0]} = x_i$, one can break $N(x_i - x_i^{[k]})$ into

$$N(x_i - x_i^{[k]}) = \sum_{m=0}^{p-1} N(x_i^{[k^{m\alpha}]} - x_i^{[k^{(m+1)\alpha}]}).$$

For each $m \in \{0, \dots, p-1\}$, denote

$$G_m := N(x_i^{[k^{m\alpha}]} - x_i^{[k^{(m+1)\alpha}]}), \quad \text{and} \quad I_m := I_{\lfloor k^{(m+1)\alpha} \rfloor}(i).$$

Observe that G_m only depends on the variables $(x_j)_{j \in I_m}$ and $\sum_{j \in I_m} \partial_j G_m = 0$.

Fix $m \in \{0, \dots, p-1\}$. Let $\varepsilon' > 0$. Let $\mathbb{Q}_{N,\beta}$ be the constrained Gibbs measure (4.2) with $I = I_m$ and $R := K^{1+\varepsilon'}$ where $K := k^{(m+1)\alpha}$.

We begin by bounding from above the total variation distance between $\mathbb{P}_{N,\beta}$ and $\mathbb{Q}_{N,\beta}$. By Lemma 4.1, we have

$$\text{TV}(\pi_{I_m} \# \mathbb{P}_{N,\beta}, \pi_{I_m} \# \mathbb{Q}_{N,\beta})^2 \leq C |I_m|^7 \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{N}{R} (x_{\max I_m} - x_{\min I_m}) \right)^2 \mathbf{1}_{x_{\max I_m} - x_{\min I_m} \geq \frac{R}{N}} \right].$$

Moreover, by Lemma 4.2 applied with $\delta = n\varepsilon'$

$$\begin{aligned} & \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{N}{R} (x_{\max I_m} - x_{\min I_m}) \right)^2 \mathbf{1}_{x_{\max I_m} - x_{\min I_m} \geq \frac{R}{N}} \right] \\ & \leq \sum_{j \geq R} \left(\frac{j}{R} \right)^2 \mathbb{P}_{N,\beta}(N(x_{\max I_m} - x_{\min I_m}) \geq j) \\ & \leq \sum_{n=1}^{\infty} \sum_{j: K^{1+n\varepsilon'} \leq j < K^{1+(n+1)\varepsilon'}} \left(\frac{j}{R} \right)^2 e^{-c_0 K^{2 \min(n\varepsilon', \delta_0 \frac{1+n\varepsilon'}{1+\delta_0})}}, \end{aligned}$$

where δ_0 is as in (4.7). Therefore, if $\varepsilon' > 0$ is small enough, namely $\varepsilon' \leq \delta_0$, there exists $C > 0$ such that

$$\text{TV}(\pi_{I_m} \# \mathbb{P}_{N,\beta}, \pi_{I_m} \# \mathbb{Q}_{N,\beta}) \leq Ce^{-cK^{\varepsilon'}} = Ce^{-ck^{(m+1)\alpha\varepsilon'}}.$$

We fix ε' so that $(m+1)\alpha\varepsilon' = \frac{\varepsilon}{s+2}$. This gives

$$\text{TV}(\pi_{I_m} \# \mathbb{P}_{N,\beta}, \pi_{I_m} \# \mathbb{Q}_{N,\beta}) \leq C e^{-cK^{\varepsilon'}} = C e^{-ck^{\frac{\varepsilon}{s+2}}}. \quad (4.30)$$

We now bound the fluctuations of G_m under $\mathbb{Q}_{N,\beta}$. One can check that the measure $\mathbb{Q}_{N,\beta}$ satisfies the assumption of Lemma 3.10 with $c_1 := \beta c_0 N^2 R^{-(s+2)}$, for some constant $c_0 > 0$. Moreover, we can compute that

$$\sup |\nabla G_m|^2 \leq C N^2 k^{-\alpha m},$$

for some constant C independent of N, α, k , and m .

Thus, by Lemma 3.10, for all $t \in \mathbb{R}$, we have

$$\begin{aligned} \left| \log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{tG_m}] - t \mathbb{E}_{\mathbb{Q}_{N,\beta}}[G_m] \right| &\leq C t^2 \frac{\sup |\nabla G_m|^2}{c_1 |I|} \\ &\leq C t^2 \frac{N^2 k^{-\alpha m}}{N^2 R^{-(s+2)} k^{(m+1)\alpha}} \\ &\leq C t^2 k^{-\alpha m - (m+1)\alpha + ((m+1)\alpha)(1+\varepsilon')(s+2)} \\ &= C t^2 k^{\alpha s(m+1) + \alpha + \alpha(m+1)\varepsilon'(s+2)} \\ &\leq C t^2 k^{\alpha s(m+1) + \alpha + \varepsilon'(s+2)}, \end{aligned}$$

since $\alpha(m+1) \leq 1$. Inserting $\varepsilon' = \frac{\varepsilon}{(m+1)(s+2)\alpha}$, we get that for all $t \geq \mathbb{R}$,

$$\left| \log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{tG_m}] - t \mathbb{E}_{\mathbb{Q}_{N,\beta}}[G_m] \right| \leq C t^2 k^{\alpha s(m+1) + \alpha + \frac{\varepsilon}{(m+1)\alpha}}.$$

Let us optimize this over $\alpha(m+1) \in (0, 1)$. The maximum is attained at the boundary, therefore either for $m = 0$ or for $m+1 = p = \frac{1}{\alpha}$. For $m = 0$, we have

$$\alpha(m+1)s + \alpha + \frac{\varepsilon}{(m+1)\alpha} = \alpha(s+1) + \frac{\varepsilon}{\alpha}.$$

For $m = p - 1$, we have

$$\alpha(m+1)s + \alpha + \frac{\varepsilon}{(m+1)\alpha} = s + \alpha + \varepsilon.$$

If p is large enough and if $\varepsilon \leq \frac{s}{2}\alpha$, then

$$\alpha(s+1) + \frac{\varepsilon}{\alpha} \leq s + \alpha + \varepsilon.$$

Therefore, fixing p and ε accordingly, we get that for every $t \in \mathbb{R}$,

$$\left| \log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{tG_m}] - t \mathbb{E}_{\mathbb{Q}_{N,\beta}}[G_m] \right| \leq C t^2 k^{s + \alpha + \varepsilon}.$$

Hence, using the standard Markov's inequality argument, we obtain

$$\mathbb{Q}_{N,\beta}(|G_m - \mathbb{E}_{\mathbb{Q}_{N,\beta}}[G_m]| \geq k^{\frac{s}{2} + \varepsilon}) \leq C e^{-ck^{s+2\varepsilon - (s+\alpha+\varepsilon)}} = C e^{-ck^{\varepsilon - \alpha}} \leq C e^{-ck^{\frac{\varepsilon}{2}}}, \quad (4.31)$$

provided $\alpha \leq \frac{\varepsilon}{2}$. Using the result of Lemma 4.3 and (4.15), we get

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}}[G_m] \leq C, \quad (4.32)$$

for $C > 0$ depending only on ε, β , and s . Combining (4.32) with (4.31) and (4.30), we deduce that

$$\mathbb{P}_{N,\beta}(|G_m| \geq k^{\frac{s}{2} + \varepsilon}) \leq C e^{-ck^{\frac{\varepsilon}{s+2}}}.$$

We conclude that there exist constants $C > 0, c > 0$ depending on ε, β and s such that

$$\mathbb{P}_{N,\beta} \left(\sum_{m=0}^{p-1} |G_m| \geq p k^{\frac{s}{2} + \varepsilon} \right) \leq C e^{-ck^{\frac{\varepsilon}{s+2}}},$$

which concludes the proof of Lemma 4.4. $\square$

4.4. Proof of Theorem 1. We now turn to the analysis of the fluctuations of $N(x_{i+k} - x_i)$ and prove Theorem 1, which follows from Lemma 4.4. By this lemma, x_i and x_{i+k} can be replaced by their block averages at scale k , up to a well-controlled error. Moreover, the fluctuations of the difference between these block averages can be bounded using Lemma 3.10.

Proof of Theorem 1. Let $i \in \{1, \dots, N\}$, $1 \leq k \leq \lfloor \frac{N}{2} \rfloor$ and $\varepsilon \in (0, 1)$ small enough as in Lemma 4.4. Let us split the gap $N(x_{i+k} - x_i)$ into

$$N(x_{i+k} - x_i) = N(x_{i+k} - x_{i+k}^{[k]}) - N(x_i - x_i^{[k]}) + N(x_{i+k}^{[k]} - x_i^{[k]}). \quad (4.33)$$

By Lemma 4.4, letting $\delta := \frac{\varepsilon}{s+2}$, we have

$$\mathbb{P}_{N,\beta}(|N(x_i^{[k]} - x_i)| \geq k^{\frac{s}{2}+\varepsilon}) \leq Ce^{-ck^\delta}, \quad (4.34)$$

and

$$\mathbb{P}_{N,\beta}(|N(x_{i+k}^{[k]} - x_{i+k})| \geq k^{\frac{s}{2}+\varepsilon}) \leq Ce^{-ck^\delta}. \quad (4.35)$$

Thus it remains to control the third term on the right in (4.33). Define

$$G : X_N \in D_N \mapsto N(x_{i+k}^{[k]} - x_i^{[k]}).$$

Let $\mathbb{Q}_{N,\beta}$ be the constrained Gibbs measure (4.1) with $I := \{j : d(i, j) \leq k\}$ and $R := k^{(1+\varepsilon')}$ for some $\varepsilon' \in (0, 1)$ to be fixed later. We have

$$\sup |\nabla G|^2 \leq C \frac{N^2}{k},$$

for some constant C independent of N and k . Moreover, $\mathbb{Q}_{N,\beta}$ satisfies the assumption of Lemma 3.10 with $c_1 := \frac{c_0 \beta N^2}{R^{s+2}}$, for some constant $c_0 > 0$ depending only on s . Consequently, by Lemma 3.10, for all $t \in \mathbb{R}$,

$$\begin{aligned} 0 \leq \log \mathbb{E}_{\mathbb{Q}_{N,\beta}}[e^{tG}] - t \mathbb{E}_{\mathbb{Q}_{N,\beta}}[G] &\leq Ct^2 \frac{\sup |\nabla G|^2}{N^2 R^{-(s+2)} k} \\ &\leq Ct^2 k^{(1+\varepsilon')(s+2)-2} = Ct^2 k^{s+\varepsilon'(s+2)}. \end{aligned}$$

Therefore, by Markov's inequality,

$$\begin{aligned} \mathbb{Q}_{N,\beta}(|G - \mathbb{E}_{\mathbb{Q}_{N,\beta}}[G]| \geq k^{\frac{s}{2}+\varepsilon}) &\leq C \exp\left(-c\left(k^{s+2\varepsilon-(s+\varepsilon'(s+2))}\right)\right) \\ &= C \exp\left(-ck^{2\varepsilon-\varepsilon'(s+2)}\right). \end{aligned} \quad (4.36)$$

Next, inserting the accuracy estimate of Lemma 4.3 we find

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}}[G] \leq C, \quad (4.37)$$

for some constant C depending only on ε, β, s . Fix $\varepsilon' = \frac{\varepsilon}{s+2}$. By (4.36) and (4.37) one finds

$$\mathbb{Q}_{N,\beta}(|G| \geq k^{\frac{s}{2}+\varepsilon}) \leq Ce^{-ck^\varepsilon}. \quad (4.38)$$

Set

$$\delta_0 := \frac{1-s}{2(s+2)}.$$

Suppose that ε is small enough, so that $\varepsilon' \leq \delta_0$. Then, by Lemma 4.1 and Lemma 4.2, one has

$$\begin{aligned} \text{TV}(\pi_I \# \mathbb{P}_{N,\beta}, \pi_I \# \mathbb{Q}_{N,\beta}) &\leq k^7 \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{N(x_{i+k} - x_i)}{R} \right)^2 \mathbf{1}_{N(x_{i+k} - x_i) \geq k^{1+\varepsilon'}} \right] \\ &\leq Ck^7 \sum_{n=1}^{\infty} \sum_{j: k^{1+\varepsilon'n} \leq j < k^{1+\varepsilon'(n+1)}} \left(\frac{j}{R} \right)^2 e^{-ck^{2 \min(n\varepsilon', \delta_0) \frac{1+n\varepsilon'}{1+\delta_0}}} \\ &\leq Ce^{-ck^{\frac{\varepsilon}{s+2}}}. \end{aligned} \quad (4.39)$$

Combining (4.38) and (4.39) one deduces that

$$\mathbb{P}_{N,\beta}(|G| \geq k^{\frac{s}{2}+\varepsilon}) \leq Ce^{-ck^{\frac{\varepsilon}{s+2}}}.$$

Together with (4.34) and (4.35), this proves (1.6). The proof of (1.7) follows from this and the fact that for each $i \in \{1, \dots, N\}$, x_i is uniformly distributed on $\mathbb{T}$. This concludes the proof of Theorem 1. $\square$

4.5. Controlling close by particles. In this subsection, we prove the following bound:

Lemma 4.5. *For all $\alpha > 0$, there exists a constant $C > 0$ depending on β , s and α such that*

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\frac{1}{(N(x_2 - x_1))^\alpha} \right] \leq C.$$

We begin by proving a weaker estimate.

Lemma 4.6. *Let $q > 1$. There exists a constant $C > 0$ depending on β , s and q such that*

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i=1}^N \frac{1}{(N(x_{i+1} - x_i))^s} \right)^q \right]^{\frac{1}{q}} \leq CN.$$

Proof. Let $q > 1$. By Lemma B.6, there exists a constant $C > 0$ depending on β , s and q such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\max \left\{ \left(\sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} N^{-s} (g(x_{i+k} - x_i) - g(\frac{k}{N})) \right), 0 \right\}^q \right]^{\frac{1}{q}} \leq CN.$$

Next, we can write

$$\begin{aligned} \sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} (g(x_{i+k} - x_i) - g(k)) &= \sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} (g(x_{i+k} - x_i) - g(k)) \mathbf{1}_{N(x_{i+k} - x_i) < \frac{k}{2}} \\ &\quad + \sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} (g(x_{i+k} - x_i) - g(k)) \mathbf{1}_{N(x_{i+k} - x_i) \geq \frac{k}{2}} := X_1 + X_2. \end{aligned}$$

Notice that there exists a constant $c > 0$ depending only on s such that for every i and k ,

$$(g(x_{i+k} - x_i) - g(k)) \mathbf{1}_{N(x_{i+k} - x_i) < \frac{k}{2}} \geq \frac{1}{c} g(x_{i+k} - x_i) \mathbf{1}_{N(x_{i+k} - x_i) < \frac{k}{2}} \geq 0. \quad (4.40)$$

Observe that $\max(X_1 + X_2, 0) \geq \max(X_1, 0) - |X_2|$. Thus,

$$\max(X_1, 0)^q \leq (\max(X_1 + X_2, 0) + |X_2|)^q \leq 2^{q-1} (\max(X_1 + X_2, 0)^q + |X_2|^q).$$

Thus, since $X_1 \geq 0$, we get that there exists $C > 0$ depending on β , s and q such that

$$\mathbb{E}[\max(X_1, 0)^q] = \mathbb{E}[X_1^q] \leq C(N^q + \mathbb{E}[X_2^q]).$$

By Theorem 1, there exists $C > 0$ depending on β , s and q such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} [X_2^q]^{\frac{1}{q}} \leq CN.$$

Combining the two last displays, we deduce that there exists $C > 0$ depending on β , s and q such that

$$\mathbb{E}[X_1^q]^{\frac{1}{q}} \leq CN. \quad (4.41)$$

By (4.40), we have

$$X_1 \geq \frac{1}{c} \sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} g(x_{i+k} - x_i) \mathbf{1}_{N(x_{i+k} - x_i) < \frac{k}{2}} \geq \frac{1}{c} \sum_{i=1}^N g(x_{i+1} - x_i) \mathbf{1}_{N(x_{i+1} - x_i) < \frac{1}{2}}.$$

We conclude by combining this with (4.41) that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i=1}^N \frac{1}{(N(x_{i+1} - x_i))^s} \mathbf{1}_{N(x_{i+1} - x_i) \leq \frac{1}{2}} \right)^q \right]^{\frac{1}{q}} \leq CN,$$

which clearly proves the lemma. $\square$

Proof of Lemma 4.5. We prove the statement by induction on the parameter α . The inductive hypothesis is that there exists $\alpha \geq s$ and a constant $C > 0$ depending on s , β and α such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\frac{1}{(N(x_2 - x_1))^\alpha} \right] \leq C. \quad (4.42)$$

The base case $\alpha = s$ follows from Lemma 4.6. Assuming that (4.42) holds for some $\alpha \geq s$, we will show that (4.42) holds for some $\alpha' > \alpha$.

Step 1: integration by parts under the gap measure. Recall

$$\text{Gap}_N : X_N \in D_N \mapsto (N(x_2 - x_1), N(x_3 - x_2), \dots, N(x_N - x_{N-1}), N(x_1 - x_N)).$$

Let us denote $\mathbb{P}'_{N,\beta} := \text{Gap}_N \# \mathbb{P}_{N,\beta}$ and set

$$\Sigma_N := \text{Gap}_N(D_N) = \{(y_1, \dots, y_N) \in (0, \infty)^N : y_1 + \dots + y_N = N\}.$$

The gap measure $\mathbb{P}'_{N,\beta}$ can be written

$$d\mathbb{P}'_{N,\beta} = \frac{1}{Z_\beta} e^{-\beta \tilde{H}(Y_N)} d\sigma_N(Y_N),$$

for some $\tilde{H} : \Sigma_N \rightarrow \mathbb{R}$ and where $d\sigma_N$ is the $(N-1)$ -dimensional surface measure on Σ_N .

Let $u \in \mathcal{C}^1(\Sigma_N, T\Sigma_N)$ be tangent to Σ_N , i.e. $\sum_{i=1}^N u_i = 0$. The intrinsic divergence theorem gives

$$\int_{\Sigma_N} \text{div}_{\Sigma_N}(u e^{-\beta \tilde{H}}) d\sigma_N = \int_{\partial \Sigma_N} (u e^{-\beta \tilde{H}}) \cdot \nu_{\Sigma_N} dS_N,$$

with ν_{Σ_N} the outward unit normal on Σ_N and dS_N the surface measure on $\partial \Sigma_N$.

Because $H(y) \rightarrow +\infty$ as $d(y, \partial \Sigma_N) \rightarrow 0$, the factor $e^{-\beta H(y)}$ decays faster than any power of the distance to $\partial \Sigma_N$. Assuming that $|u|(y)$ grows at most like $d(y, \partial \Sigma_N)^{-\alpha}$ for some $\alpha > 0$, we obtain

$$\int_{\partial \Sigma_N} (u e^{-\beta \tilde{H}}) \cdot \nu_{\Sigma_N} dS_N = 0,$$

so that

$$\int_{\Sigma_N} \text{div}_{\Sigma_N}(u e^{-\beta \tilde{H}}) d\sigma_N = 0.$$

Expanding the divergence and dividing by Z_β yields

$$\int_{\Sigma_N} (\text{div}_{\Sigma_N} u - \beta \nabla_{\Sigma_N} \tilde{H} \cdot u) e^{-\beta \tilde{H}} d\sigma_N = 0 \implies \beta \mathbb{E}_{\mathbb{P}'_{N,\beta}}[\nabla_{\Sigma_N} \tilde{H} \cdot u] = \mathbb{E}_{\mathbb{P}'_{N,\beta}}[\text{div}_{\Sigma_N} u].$$

Since $\text{div}_{\Sigma_N} u = \text{div}(u)$ whenever u is tangent ($\sum_{i=1}^N u_i = 0$), we arrive at

$$\beta \mathbb{E}_{\mathbb{P}'_{N,\beta}}[\nabla \tilde{H} \cdot u] = \mathbb{E}_{\mathbb{P}'_{N,\beta}}[\text{div}(u)]. \quad (4.43)$$

Step 2: construction of the tangent vector field. We claim that there exist $\varepsilon \in (0, 1)$ and $M > 1$ depending only on s, β , and α such that

$$\mathbb{P}'_{N,\beta}(y_i \in (2\varepsilon, M)) \geq \frac{1}{2}. \quad (4.44)$$

Indeed, we have $\mathbb{E}_{\mathbb{P}'_{N,\beta}}[y_i] = 1$ since $y_1 + \dots + y_N = N$ and since the y_i are identically distributed. Therefore, by Markov's inequality,

$$\mathbb{P}'_{N,\beta}(y_i > M) \leq \frac{1}{M}.$$

Now, by Markov's inequality again,

$$\mathbb{P}'_{N,\beta}(y_i < 2\varepsilon) = \mathbb{P}'_{N,\beta}\left(\frac{1}{y_i^\alpha} > \frac{1}{(2\varepsilon)^\alpha}\right) \leq (2\varepsilon)^\alpha \mathbb{E}_{\mathbb{P}'_{N,\beta}}\left[\frac{1}{y_1^\alpha}\right].$$

By using the induction hypothesis (4.42), we get by combining the two above displays that for $\varepsilon > 0$ small enough and for $M > 1$ large enough, (4.44) holds.

We now show that there exists a smooth function $\chi : [0, N] \rightarrow \mathbb{R}$ with $\chi(x) = 1$ for $x \in (0, \varepsilon)$, $\chi(x) = 0$ for $x \geq M$, $|\chi| \leq C$ with C depending only on β, s, ε and α and

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}}\left[\frac{1}{y_1^\alpha} \chi(y_1)\right] = 0. \quad (4.45)$$

Consider a smooth non-positive bump function ϕ such that $\phi \in \mathcal{C}^\infty(0, N)$, $\phi(x) = 0$ for $x \leq \varepsilon$, $-1 \leq \phi \leq 0$, $\phi(x) = 0$ for $x \geq M$, $\phi(x) \leq -\frac{1}{2}$ for $x \in (2\varepsilon, M-1)$. We then set, for $t \geq 0$, $\chi_t(x) := 1 + t\phi(x)$. We have $\chi_t \geq 0$, χ_t supported on $[0, M]$, $\chi_t(x) = 1$ for $x \in [0, \varepsilon]$. We also have

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}}\left[\frac{1}{y_1^\alpha} \chi_t(y_1)\right] = 0 \implies t = \frac{\mathbb{E}_{\mathbb{P}'_{N,\beta}}\left[\frac{1}{y_1^\alpha}\right]}{\mathbb{E}_{\mathbb{P}'_{N,\beta}}\left[\frac{-\phi(y_1)}{y_1^\alpha}\right]}.$$

We now observe that the parameter t in the above display is upper bounded by a constant depending only on β, s and α . Indeed, by the induction hypothesis (4.42), there exists a constant $C > 0$ depending only on s, β and α such that

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{y_1^\alpha} \right] \leq C.$$

Moreover, since $\phi \leq 0$, we have

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{-\phi(y_1)}{y_1^\alpha} \right] \geq \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{-\phi(y_1)}{y_1^\alpha} \mathbf{1}_{y_1 \in (2\varepsilon, M-1)} \right] \geq \frac{1}{2M^\alpha} \mathbb{P}'_{N,\beta}(y_1 \in (2\varepsilon, M-1)).$$

By (4.44) and the above display, we deduce that there exists $c > 0$ depending only on α such that

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{-\phi(y_1)}{y_1^\alpha} \right] \geq c.$$

Thus, $t \in (0, C)$ for some C depending only on β, s and α , as claimed. We finally set $\chi := \chi_t$ for this chosen t , which satisfies the desired assumption.

Consider $u \in \mathcal{C}^1(\Sigma_N, T\Sigma_N)$ the vector field defined for every $i = 1, \dots, N$ and $Y_N \in \Sigma_N$ by

$$u_i(Y_N) = -\frac{1}{y_i^\alpha} \chi(y_i) + \frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \chi(y_j).$$

Notice that $\sum_{i=1}^N u_i = 0$ and that $|u|(y)$ grows at most like $d(y, \partial\Sigma_N)^{-\alpha}$ as $d(y, \partial\Sigma_N) \rightarrow 0$. Thus, by (4.43),

$$\beta \mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H} \cdot u] = \mathbb{E}_{\mathbb{P}'_{N,\beta}} [\operatorname{div}(u)]. \quad (4.46)$$

One can check that the energy in gap coordinates is given, for all $Y_N \in \Sigma_N$, by

$$\tilde{H}(Y_N) = 2N^{-s} \sum_{k=1}^{\lfloor \frac{N-1}{2} \rfloor} \sum_{i=1}^N g\left(\frac{y_i + \dots + y_{i+k-1}}{N}\right) + \mathbf{1}_{N \text{ even}} N^{-s} \sum_{i=1}^N g\left(\frac{y_i + \dots + y_{i+\frac{N}{2}-1}}{N}\right).$$

Therefore,

$$\begin{aligned} \nabla \tilde{H} \cdot u &= 2N^{-(s+1)} \sum_{k=1}^{\lfloor \frac{N-1}{2} \rfloor} \sum_{i=1}^N g'\left(\frac{y_i + \dots + y_{i+k-1}}{N}\right) (u_i + \dots + u_{i+k-1}) \\ &\quad + \mathbf{1}_{\{N \text{ even}\}} N^{-(s+1)} \sum_{i=1}^N g'\left(\frac{y_i + \dots + y_{i+\frac{N}{2}-1}}{N}\right) (u_i + \dots + u_{i+\frac{N}{2}-1}). \end{aligned}$$

Let us decompose u into $u^1 + u^2$ where

$$u_i^1 := -\frac{1}{y_i^\alpha} \chi(y_i) \quad \text{for every } i = 1, \dots, N$$

and

$$u^2 := \left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \chi(y_j) \right) (e_1 + \dots + e_N),$$

where we recall that $(e_1, \dots, e_N)$ stands for the canonical basis on $\mathbb{R}^N$. By (4.46), we have

$$\beta \mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H} \cdot u^1] = \mathbb{E}_{\mathbb{P}'_{N,\beta}} [\operatorname{div}(u)] - \beta \mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H} \cdot u^2]. \quad (4.47)$$

Step 3: reduction to nearest neighbor gaps for $\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H} \cdot u^2]$. Let $Y_N^0 = (1, \dots, 1)$, which corresponds to all gaps equal to 1, i.e. to points regularly spaced. By (4.45), we have

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H}(Y_N^0) \cdot u^2] = \nabla \tilde{H}(Y_N^0) \cdot (e_1 + \dots + e_N) \mathbb{E} \left[\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \chi(y_j) \right] = 0.$$

Thus,

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H}(Y_N) \cdot u^2] = \mathbb{E}_{\mathbb{P}'_{N,\beta}} [(\nabla \tilde{H}(Y_N) - \nabla \tilde{H}(Y_N^0)) \cdot u^2].$$

By Taylor expansion, there exists a constant $C > 0$ depending on s such that

$$\begin{aligned} N^{-(s+1)} & \left| g' \left(\frac{y_i + \dots + y_{i+k-1}}{N} \right) - g' \left(\frac{k}{N} \right) \right| \\ & \leq C \left(\frac{1}{k^{s+2}} |y_i + \dots + y_{i+k-1} - k| + \frac{1}{(y_i + \dots + y_{i+k-1})^{s+1}} \mathbb{1}_{y_i + \dots + y_{i+k-1} \leq \frac{k}{2}} \right) \\ & \leq C \left(\frac{1}{k^{s+2}} |y_i + \dots + y_{i+k-1} - k| + \frac{1}{y_i^{s+1}} \mathbb{1}_{y_i + \dots + y_{i+k-1} \leq \frac{k}{2}} \right). \end{aligned}$$

Therefore, there exists a constant $C > 0$ depending on s such that

$$\begin{aligned} |(\nabla \tilde{H}(Y_N) - \nabla \tilde{H}(Y_N^0)) \cdot u^2| & \leq C \left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right) \\ & \times \left(\sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} \frac{1}{k^{s+1}} |y_i + \dots + y_{i+k-1} - k| + \sum_{i=1}^N \frac{1}{y_i^{s+1}} \left(\sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} k \mathbb{1}_{y_i + \dots + y_{i+k-1} \leq \frac{k}{2}} \right) \right). \quad (4.48) \end{aligned}$$

Let $p, q > 1$ such that $\frac{1}{p} + \frac{1}{q} = 1$ with q large to be selected at the end of the proof below. By Hölder's inequality,

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right) |y_i + \dots + y_{i+k-1} - k| \right] \leq \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right)^p \right]^{\frac{1}{p}} \mathbb{E}_{\mathbb{P}'_{N,\beta}} [|y_i + \dots + y_{i+k-1} - k|^q]^{\frac{1}{q}}.$$

By Theorem 1, there exists a constant $C > 0$ depending on β, s and p such that

$$\sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} \frac{1}{k^{s+1}} \mathbb{E}_{\mathbb{P}'_{N,\beta}} [|y_i + \dots + y_{i+k-1} - k|^q]^{\frac{1}{q}} \leq CN. \quad (4.49)$$

Let us now turn to the second term in the right-hand side of (4.48). By Hölder's inequality again, for every $i \in \{1, \dots, N\}$,

$$\begin{aligned} \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right) \frac{1}{y_i^{1+s}} k \mathbb{1}_{y_i + \dots + y_{i+k-1} \leq \frac{k}{2}} \right] \\ \leq \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right)^p \frac{1}{y_i^{(1+s)p}} \right]^{\frac{1}{p}} \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} k \mathbb{P}'_{N,\beta} \left(y_i + \dots + y_{i+k-1} \leq \frac{k}{2} \right)^{\frac{1}{q}} \end{aligned}$$

Thus, by Theorem 1, there exists a constant $C > 0$ depending on β, s, α and p such that

$$\sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right) \frac{1}{y_i^{1+s}} k \mathbb{1}_{y_i + \dots + y_{i+k-1} \leq \frac{k}{2}} \right] \leq C \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right)^p \frac{1}{y_i^{(1+s)p}} \right]^{\frac{1}{p}}. \quad (4.50)$$

We conclude by assembling (4.48), (4.49) and (4.50): there exists a constant $C > 0$ depending on β, s, α and p such that

$$|\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H}(Y_N) \cdot u^2]| \leq C \left(\sum_{i=1}^N \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{j=1}^N \frac{1}{y_j^\alpha} \right)^p \left(1 + \frac{1}{y_i^{(1+s)p}} \right) \right]^{\frac{1}{p}} \right). \quad (4.51)$$

Step 4: rearrangement inequality and control on $\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla H \cdot u^2]$. Since $\alpha \geq s$, we get from a convexity argument that

$$\left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^\alpha} \right) \left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{1+s}} \right) \leq \left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^s} \right) \left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{1+\alpha}} \right). \quad (4.52)$$

Indeed, let us define

$$M(\gamma) := \frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^\gamma}.$$

It is well known that $\gamma \mapsto \log M(\gamma)$ is convex on $(0, \infty)$. Observe that the intervals $[\alpha, s]$ and $[1+s, 1+\alpha]$ have the same length. Since $\log M$ is convex, the slope is non-decreasing and therefore

$$\frac{\log M(s) - \log M(\alpha)}{\alpha - s} \leq \frac{\log M(1+s) - \log M(1+\alpha)}{\alpha - s}.$$

It follows that

$$\frac{M(\alpha)}{M(s)} \leq \frac{M(1+\alpha)}{M(1+s)},$$

which is the desired inequality (4.52).

By Hölder's inequality,

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^s} \right)^p \left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{1+\alpha}} \right)^p \right]^{\frac{1}{p}} \leq \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^s} \right)^{pq} \right]^{\frac{1}{pq}} \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{1+\alpha}} \right)^{p^2} \right]^{\frac{1}{p^2}}.$$

On the one hand, by the estimate of Lemma 4.6, there exists a constant $C > 0$ depending on β, s, α and p such that

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^s} \right)^{pq} \right]^{\frac{1}{pq}} \leq C.$$

On the other hand, by the convexity of $x \in (0, \infty) \mapsto x^{p^2}$ (since $p > 1$), we have

$$\left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{1+\alpha}} \right)^{p^2} \leq \frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}}.$$

Thus,

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^s} \right)^p \left(\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{1+\alpha}} \right)^p \right]^{\frac{1}{p}} \leq C \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}} \right]^{\frac{1}{p^2}}.$$

Combining this with (4.51) and (4.52), we conclude that there exists a constant $C > 0$ depending on β, s, α and p such that

$$|\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H}(Y_N) \cdot u^2]| \leq C \mathbb{E} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}} \right]^{\frac{1}{p^2}}. \quad (4.53)$$

Hence, by (4.47), there exists a constant $C > 0$ depending on β, s, α and p such that

$$\beta |\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H} \cdot \nabla u^2]| \leq C N \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}} \right]^{\frac{1}{p^2}} + \mathbb{E}_{\mathbb{P}'_{N,\beta}} [\operatorname{div}(u)]. \quad (4.54)$$

Step 5: positivity argument for $\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H} \cdot \nabla u^1]$. Recall that $g' \leq 0$ on $(0, \frac{N}{2}]$ and $g' \geq 0$ on $[\frac{N}{2}, N)$. Moreover,

$$\begin{aligned} \nabla \tilde{H} \cdot u^1 &= 2 N^{-(s+1)} \sum_{k=2}^{\lfloor \frac{N-1}{2} \rfloor} \sum_{i=1}^N g' \left(\frac{y_i + \dots + y_{i+k-1}}{N} \right) (u_i^1 + \dots + u_{i+k-1}^1) \\ &\quad + \mathbf{1}_{\{N \text{ even}\}} N^{-(s+1)} \sum_{i=1}^N g' \left(\frac{y_i + \dots + y_{i+\frac{N}{2}-1}}{N} \right) (u_i^1 + \dots + u_{i+\frac{N}{2}-1}^1). \end{aligned}$$

Recall that $u_i^1 = -\frac{1}{y_i^\alpha} \leq 0$. One can write

$$\begin{aligned} \nabla \tilde{H} \cdot u^1 &\geq 2N^{-(s+1)} \sum_{i=1}^N g' \left(\frac{y_i}{N} \right) \left(-\frac{1}{y_i^\alpha} \right) \\ &\quad + 2N^{-(s+1)} \sum_{k=1}^{\lfloor \frac{N-1}{2} \rfloor} \sum_{i=1}^N g' \left(\frac{y_i + \dots + y_{i+k-1}}{N} \right) (u_i^1 + \dots + u_{i+k-1}^1) \mathbb{1}_{y_i + \dots + y_{i+k-1} \geq \frac{N}{2}} \\ &\quad + \mathbb{1}_{\{N \text{ even}\}} N^{-(s+1)} \sum_{i=1}^N g' \left(\frac{y_i + \dots + y_{i+\frac{N}{2}-1}}{N} \right) (u_i^1 + \dots + u_{i+\frac{N}{2}-1}^1) \mathbb{1}_{y_i + \dots + y_{i+\frac{N}{2}-1} \geq \frac{N}{2}}. \end{aligned}$$

By the rigidity estimate of Theorem 1, we therefore get that there exists a constant $c > 0$ depending on s and a constant $C > 0$ depending on β, α and s such that

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} [\nabla \tilde{H} \cdot u^1] \geq c \sum_{i=1}^N \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{y_i^{1+s+\alpha}} \right] - CN. \quad (4.55)$$

Assembling (4.54) and (4.54) therefore gives

$$\sum_{i=1}^N \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{y_i^{1+s+\alpha}} \right] \leq CN \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}} \right]^{\frac{1}{p^2}} + \mathbb{E}_{\mathbb{P}'_{N,\beta}} [\text{div}(u)] + CN,$$

for some constant $C > 0$ depending on β, s, α and p . Notice that there exists $C > 0$ depending only on α such that

$$|\text{div}(u)| \leq C \sum_{i=1}^N \frac{1}{y_i^{1+\alpha}}.$$

Thus, combining the two last displays, we obtain

$$\begin{aligned} \beta \sum_{i=1}^N \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{y_i^{1+s+\alpha}} \right] &\leq CN + CN \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}} \right]^{\frac{1}{p^2}} \\ &\leq CN + CN \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{N} \sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}} \right]. \end{aligned} \quad (4.56)$$

for some constant $C > 0$ depending on β, s, α and p .

Step 6: conclusion. By taking $p > 1$ small enough so that $(1+\alpha)p^2 < 1+s+\alpha$, we deduce that from (4.56) that there exists a constant $C > 0$ depending on β, s, α such that

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\sum_{i=1}^N \frac{1}{y_i^{1+\alpha}} \right] \leq \mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\sum_{i=1}^N \frac{1}{y_i^{(1+\alpha)p^2}} \right] \leq CN.$$

Hence, since the y_i are identically distributed,

$$\mathbb{E}_{\mathbb{P}'_{N,\beta}} \left[\frac{1}{y_1^{1+\alpha}} \right] \leq C,$$

which shows that (4.42) holds for $\alpha' := 1 + \alpha$.

Since (4.42) holds for $\alpha = s$, we conclude by induction that it holds for every $\alpha > 0$, which concludes the proof of the lemma. $\square$

5. OPTIMAL RIGIDITY FOR SINGULAR LINEAR STATISTICS

In this section, we prove the optimal variance estimate stated in Theorem 2. This will give the optimal scaling of the fluctuations of gaps and discrepancies.

5.1. Regularization estimates. In this subsection, we record various regularization estimates that will be used later in the proof.

Let $K \in \mathcal{C}^\infty(\mathbb{T})$ be an even, non-negative mollifier with $\int_{\mathbb{T}} K(x) = 1$ and K supported on $(-\frac{1}{2}, \frac{1}{2})$. For all $\ell \in (0, 1)$ define

$$K_\ell := \ell^{-1} K(\ell^{-1} \cdot). \quad (5.1)$$

Lemma 5.1. *Suppose $\xi \in H^{\frac{1-s}{2}+\delta}(\mathbb{T})$ for some $\delta > 0$. Then, there exists a constant $C > 0$ depending on K such that*

$$\|\xi - \xi * K_\ell\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 \leq C \ell^{2 \min(\delta, \frac{1}{2})} \|\xi\|_{H^{\min(\delta, \frac{1}{2})}(\mathbb{T})}^2. \quad (5.2)$$

Let $\phi \in L^2(\mathbb{T}) \cap \mathcal{C}^1(\mathbb{T} \setminus \{0\})$ be such that for all $x \in \mathbb{T} \setminus \{0\}$,

$$|\phi'(x)| \leq \frac{1}{|x|^{1+\gamma}}, \quad (5.3)$$

for some $\gamma \in [0, 1)$. Then, there exists a constant $C > 0$ depending on K and γ such that for all $x \in \mathbb{T}$,

$$|(\phi * K_\ell)'(x)| \leq \frac{C}{\min(|x|, \ell)^{1+\gamma}}. \quad (5.4)$$

Proof. We prove the (5.2) with Fourier series. One can suppose without loss of generality that $\delta \leq \frac{1}{2}$.

First, by the Parseval identity,

$$\begin{aligned} \|\xi - \xi * K_\ell\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 &= \sum_{k \in \mathbb{Z}} |\hat{\xi}(k)|^2 |k|^{1-s} |1 - \hat{K}(\ell k)|^2 \\ &= \sum_{k \in \mathbb{Z}: |k| \leq \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s} |1 - \hat{K}(\ell k)|^2 + \sum_{k \in \mathbb{Z}: |k| > \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s} |1 - \hat{K}(\ell k)|^2. \end{aligned}$$

Since K is smooth, there exists a constant $C > 0$ such that

$$|1 - \hat{K}(t)| \leq C|t| \quad \text{for every } |t| \leq 1.$$

Moreover, for every $n \geq 1$, there exists $C_n > 0$ such that

$$|\hat{K}(t)| \leq \frac{C_n}{|t|^n} \quad \text{for every } |t| > 1.$$

Hence, for low frequencies,

$$\sum_{k \in \mathbb{Z}: |k| \leq \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s} |1 - \hat{K}(\ell k)|^2 \leq C \sum_{k \in \mathbb{Z}: |k| \leq \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s} |k| \ell.$$

Since by assumption $\xi \in H^{\frac{1-s}{2}+\delta}(\mathbb{T})$ with $\delta \leq \frac{1}{2}$, we have

$$\sum_{k \in \mathbb{Z}: |k| \leq \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s} |1 - \hat{K}(\ell k)|^2 \leq C \sum_{k \in \mathbb{Z}} |\hat{\xi}(k)|^2 |k| \ell^{1-s+2\delta} = C \ell^{2\delta} \|\xi\|_{H^{\frac{1-s}{2}+\delta}(\mathbb{T})}^2.$$

On the other hand, for high frequencies, there exists a constant $C > 0$ depending on K such that

$$\begin{aligned} \sum_{k \in \mathbb{Z}: |k| > \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s} |1 - \hat{K}(\ell k)|^2 &\leq C \sum_{k \in \mathbb{Z}: |k| > \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s} \\ &\leq C \ell^\delta \sum_{k \in \mathbb{Z}: |k| > \ell^{-1}} |\hat{\xi}(k)|^2 |k|^{1-s+2\delta} \leq C \ell^{2\delta} \|\xi\|_{H^{\frac{1-s}{2}+\delta}(\mathbb{T})}^2. \end{aligned}$$

Combining the two above displays proves (5.2).

Let us now prove (5.4). Let $r := |x|$ and recall $\text{supp } K_\ell \subset [-\ell, \ell]$. Suppose that $r \geq 2\ell$. Then $|x - y| \geq \frac{r}{2}$. Recall

$$(\phi * K_\ell)'(x) = \int_{-\ell}^{\ell} \phi'(x - y) K_\ell(y) dy.$$

Using $|\phi'(z)| \leq |z|^{-1-\gamma}$, we therefore get

$$|(\phi * K_\ell)'(x)| \leq C \frac{1}{r^{1+\gamma}}.$$

Suppose now that $r \leq 2\ell$. Recall

$$(\phi * K_\ell)'(x) = - \int_{\mathbb{T}} \phi(y) \partial_y K_\ell(x-y) dy.$$

Integrating by parts gives

$$|\partial_y K_\ell(x-y)| \leq C\ell^{-2} \mathbf{1}_{|x-y| \leq \ell}.$$

Since $|\phi(y)| \leq C|y|^{-\gamma}$, we have

$$|(\phi * K'_\ell)(x)| \leq C\ell^{-2} \int_{|x-y| \leq \ell} |\phi(y)| dy \leq C\ell^{-2} \ell^{1-\gamma} = C \frac{1}{\ell^{1+\gamma}}.$$

Both bounds combine to

$$|(\phi * K'_\ell)(x)| \leq \frac{C}{(|x| + \ell)^{1+\gamma}} = \frac{C}{\min(|x|, \ell)^{1+\gamma}},$$

as desired. $\square$

5.2. A priori estimate on the fluctuations. In the following lemma, we prove a near-optimal variance estimate for linear statistics with smooth but N -dependent test-functions that are regularized at the microscopic scale. Our proof is based on the comparison principle of Lemma 3.6 and on the gaps fluctuation estimate of Theorem 1.

Lemma 5.2. *Let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$ and $\ell_N \in (0, 1]$. Suppose that ξ is supported on $(-\frac{1}{2}, \frac{1}{2})$ if $\ell_N \neq 1$. Let $\alpha \in \mathbb{R}$. Suppose that*

$$|\xi'(x)| \leq \frac{1}{\max(|x|, \frac{1}{N\ell_N})^{1+\alpha}} \mathbf{1}_{|x| \leq 2} + \frac{1}{|x|^{3-s}} \mathbf{1}_{|x| \geq 2}.$$

Then, for all $\varepsilon > 0$, there exists $C > 0$ depending on β, s, α and ε such that

$$\text{Var}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi(\ell_N^{-1}(x_i - x_1)) \right] \leq C(N\ell_N)^\varepsilon (N\ell_N)^{\max(2\alpha, s)} \quad (5.5)$$

and

$$\text{Var}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi(\ell_N^{-1}x_i) \right] \leq C(N\ell_N)^\varepsilon (N\ell_N)^{\max(2\alpha, s)}. \quad (5.6)$$

Proof. We work with the push-forward of $\mathbb{P}_{N,\beta}$ by the map Anch_N defined in (3.16). Fix $\gamma \geq 1$ and let

$$t_\gamma := \min\left(\frac{(N\ell_N)^\gamma}{N}, \frac{1}{2}\right)$$

and

$$I_\gamma := \{i \in \{1, \dots, N\} : d(i, 1) \leq Nt_\gamma\}.$$

Define

$$F_N := \sum_{i \in I_\gamma} \xi(\ell_N^{-1}(x_i - x_1)).$$

Step 1: reduction to a good event. Let $\varepsilon > 0$. We let $\chi_{1,2} : \mathbb{R}^+ \rightarrow \mathbb{R}$ be given for all $x \in \mathbb{R}^+$ by

$$\chi_{1,2}(x) = \begin{cases} 1, & 0 \leq x \leq 1, \\ 2-x, & 1 < x < 2, \\ 0, & x \geq 2. \end{cases}$$

Define the cutoff function

$$\eta_{\varepsilon,\gamma} := \prod_{i \in I_\gamma - \frac{N}{2} \leq k \leq \frac{N}{2} : i+k \in I_\gamma} \chi_{1,2}\left(2 \frac{N|(x_{i+k} - x_i) - k|}{|k|^{\frac{s}{2}}(N\ell_N)^\varepsilon}\right) \prod_{i \in I_\gamma : i+1 \in I_\gamma} \chi_{1,2}\left(\frac{1}{N(x_{i+1} - x_i)(N\ell_N)^\varepsilon}\right). \quad (5.7)$$

First, by sub-additivity

$$\text{Var}_{\mathbb{P}_{N,\beta}}[F_N] \leq 2\text{Var}_{\mathbb{P}_{N,\beta}}[\eta_{\varepsilon,\gamma}F_N] + 2\text{Var}_{\mathbb{P}_{N,\beta}}[(1 - \eta_{\varepsilon,\gamma})F_N].$$

By Theorem 1 and Lemma 4.5, there exists a constant $C > 0$ depending on β, s, α and γ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[(1 - \eta_{\varepsilon,\gamma})F_N] \leq \frac{C}{(N\ell_N)^{10}}. \quad (5.8)$$

Step 2: convexification. As in Section 4, we convexify the measure $\mathbb{P}_{N,\beta}$ through a penalty for configurations with unusually large nearest-neighbor gaps in the window I_γ .

Let $\theta : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be a smooth cutoff function such that $\theta(x) = |x|^2$ for $x > 2$, $\theta(x) = 0$ for $x \in [0, 1]$ and $\theta'' \geq 0$ on $\mathbb{R}^+$. Let

$$F := \sum_{i \in I_\gamma : i+1 \in I_\gamma} \theta\left(\frac{N(x_{i+1} - x_i)}{(N\ell_N)^\varepsilon}\right)$$

and $\mathbb{Q}_{N,\beta}$ be the locally constrained measure

$$d\mathbb{Q}_{N,\beta}(X_N) = \frac{1}{K_{N,\beta}} e^{-\beta(\mathcal{H}_N + F)(X_N)} \mathbf{1}_{D_N}(X_N) dX_N. \quad (5.9)$$

Observe that under $\mathbb{P}_{N,\beta}$ (resp. $\mathbb{Q}_{N,\beta}$), x_1 is independent of $\text{Gap}_N(X_N)$ and is distributed according to the Lebesgue measure on $\mathbb{T}$. Therefore, arguing as in Lemma 4.1,

$$\text{Ent}(\mathbb{P}_{N,\beta} \mid \mathbb{Q}_{N,\beta}) = \text{Ent}(\text{Gap}_N \# \mathbb{P}_{N,\beta} \mid \text{Gap}_N \# \mathbb{Q}_{N,\beta}).$$

By Lemma 4.1 and Theorem 1, there exists a constant $\delta > 0$ depending on β, s, ε and γ such that

$$\text{Ent}(\text{Gap}_N \# \mathbb{P}_{N,\beta} \mid \text{Gap}_N \# \mathbb{Q}_{N,\beta}) \leq e^{-(N\ell_N)^\delta}.$$

Combining the two above displays, we deduce that there exists a constant $\delta > 0$ depending on β, s, ε and γ such that

$$\text{Ent}(\mathbb{P}_{N,\beta} \mid \mathbb{Q}_{N,\beta}) \leq e^{-(N\ell_N)^\delta}.$$

Thus, by the Pinsker inequality (4.3), there exists a constant $\delta > 0$ depending on β, s, ε and γ such that

$$\text{TV}(\mathbb{P}_{N,\beta}, \mathbb{Q}_{N,\beta}) \leq e^{-(N\ell_N)^\delta}. \quad (5.10)$$

Observe that on the support of $\eta_{\varepsilon,\gamma}$, there exists a constant $\kappa > 0$ depending on α such that

$$|F_N| \leq (N\ell_N)^\kappa.$$

Therefore, by (5.10), there exists $\delta > 0$ depending on $\beta, s, \alpha, \varepsilon$ and γ such that

$$\left| \text{Var}_{\mathbb{P}_{N,\beta}}[\eta_{\varepsilon,\gamma} F_N] - \text{Var}_{\mathbb{Q}_{N,\beta}}[\eta_{\varepsilon,\gamma} F_N] \right| \leq e^{-(N\ell_N)^\delta}. \quad (5.11)$$

Let F_N^* be such that $F_N = F_N^* \circ \text{Anch}_N$ and $\eta_{\varepsilon,\gamma}^*$ be such that $\eta_{\varepsilon,\gamma} = \eta_{\varepsilon,\gamma}^* \circ \text{Anch}_N$. We let $\mathbb{P}_{N,\beta}^* := \text{Anch}_N \# \mathbb{P}_{N,\beta}$ and $\mathbb{Q}_{N,\beta}^* := \text{Anch}_N \# \mathbb{Q}_{N,\beta}$. Clearly,

$$\text{Var}_{\mathbb{Q}_{N,\beta}}[\eta_{\varepsilon,\gamma} F_N] = \text{Var}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,\gamma}^* F_N^*].$$

Our goal is now to estimate the variance of $\eta_{\varepsilon,\gamma}^* F_N^*$ under $\mathbb{Q}_{N,\beta}^*$.

Set $A_1 := A_1^{\mathbb{Q}_{N,\beta}^*}$. By sub-additivity,

$$\text{Var}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,\gamma}^* F_N^*] \leq 2\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,\gamma}^* \nabla F_N^* \cdot A_1^{-1}(\eta_{\varepsilon,\gamma}^* \nabla F_N^*)] + 2\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[F_N^* \nabla \eta_{\varepsilon,\gamma}^* \cdot A_1^{-1}(F_N^* \nabla \eta_{\varepsilon,\gamma}^*)]. \quad (5.12)$$

Step 3: estimating the A_1^{-1} -energy of $F_N^* \nabla \eta_{\varepsilon,\gamma}^*$. Let $\mathcal{H}_N^*$ and F^* be such that $\mathcal{H}_N = \mathcal{H}_N^* \circ \text{Anch}_N$ and $F = F^* \circ \text{Anch}_N$. By Proposition 3.2,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[F_N^* \nabla \eta_{\varepsilon,\gamma}^* \cdot A_1^{-1}(F_N^* \nabla \eta_{\varepsilon,\gamma}^*)] \\ \leq - \min_{\phi \in H^1(\mathbb{Q}_{N,\beta}^*)} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[\|\nabla^2 \phi\|_F^2 + \beta \nabla \phi \cdot \nabla^2(\mathcal{H}_N^* + F^*) \nabla \phi - \nabla \phi \cdot (F_N^* \nabla \eta_{\varepsilon,\gamma}^*)]. \end{aligned}$$

Proceeding as in the proof of Lemma 3.7, we get

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[F_N^* \nabla \eta_{\varepsilon,\gamma}^* \cdot A_1^{-1}(F_N^* \nabla \eta_{\varepsilon,\gamma}^*)] \\ \leq -\beta^{-1} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*} \left[\min_{U_{N-1} \in \mathbb{R}^{N-1}} U_{N-1} \cdot \nabla^2(\mathcal{H}_N^* + F^*) U_{N-1} - 2(F_N^* \nabla \eta_{\varepsilon,\gamma}^*) \cdot U_{N-1} \right]. \end{aligned}$$

Moreover, by the definition of F , for all $U_{N-1} = (u_2, \dots, u_N) \in \mathbb{R}^{N-1}$, there exists a constant $C > 0$ depending on s such that

$$U_{N-1} \cdot \nabla^2(\mathcal{H}_N^* + F^*) U_{N-1} \geq \frac{1}{C} (N\ell_N)^{-(s+2)\varepsilon} \sum_{i \in I_\gamma} (N(u_{i+1} - u_i))^2,$$

where we set $u_1 = 0$. Let Gap_N^* be such that $\text{Gap}_N = \text{Gap}_N^* \circ \text{Anch}_N$.

Observe that $\eta_{\varepsilon,\gamma}^*$ is a function of the gaps: it can be written $\eta_{\varepsilon,\gamma}^* = \tilde{\eta}_\varepsilon \circ \text{Gap}_N^*$ for some smooth function $\tilde{\eta}_\varepsilon : \mathbb{R}^N \rightarrow \mathbb{R}$. Therefore,

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [F_N^* \nabla \eta_{\varepsilon,\gamma}^* \cdot A_1^{-1} (F_N^* \nabla \eta_{\varepsilon,\gamma}^*)] \leq (N\ell_N)^{\kappa\varepsilon} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [(F_N^*)^2 |\nabla \tilde{\eta}_\varepsilon|^2 \circ \text{Gap}_N^*]. \quad (5.13)$$

Notice that on the support of $\eta_{\varepsilon,\gamma}^*$, there exists a constant $\kappa > 0$ depending on α such that $|F_N^*| \leq (N\ell_N)^\kappa$. Thus, using Theorem 1, Lemma 4.5 and (5.11), we deduce from the above display that there exists $C > 0$ depending on β , s and α such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [F_N^* \nabla \eta_{\varepsilon,\gamma}^* \cdot A_1^{-1} (F_N^* \nabla \eta_{\varepsilon,\gamma}^*)] \leq \frac{C}{(N\ell_N)^{10}}. \quad (5.14)$$

Step 4: estimating the A_1^{-1} -energy of $\eta_{\varepsilon,\gamma}^* \nabla F_N^*$. By the assumption on ξ , there exists a constant $C > 0$ such that for every $i \in I_\gamma$,

$$\begin{aligned} \eta_{\varepsilon,\gamma}^* |\partial_i F_N^*| &\leq C \eta_{\varepsilon,\gamma}^* \ell_N^{-1} \left(\frac{1}{|\max(\ell_N^{-1} x_i, \frac{1}{N\ell_N})|^{1+\alpha}} \mathbb{1}_{|\ell_N^{-1} x_i| \leq 2} + \frac{1}{|\ell_N^{-1} x_i|^{3-s}} \mathbb{1}_{|\ell_N^{-1} x_i| \geq 2} \right) \\ &= C \eta_{\varepsilon,\gamma}^* N \left(\frac{(N\ell_N)^\alpha}{|\max(Nx_i, 1)|^{1+\alpha}} \mathbb{1}_{|\ell_N^{-1} x_i| \leq 2} + \frac{(N\ell_N)^{3-s}}{|Nx_i|^{2-s}} \mathbb{1}_{|\ell_N^{-1} x_i| \geq 2} \right). \end{aligned}$$

Therefore, by the definition of $\eta_{\varepsilon,\gamma}$, there exists $\kappa > 0$ such that for every $i \in I_\gamma$,

$$\eta_{\varepsilon,\gamma}^* \partial_i F_N^* \leq C (N\ell_N)^{\kappa\varepsilon} \eta_{\varepsilon,\gamma}^* N \left(\frac{(N\ell_N)^\alpha}{1 + d(i, 1)^{1+\alpha}} \mathbb{1}_{d(i, 1) \leq (N\ell_N)^{1+\varepsilon}} + \frac{(N\ell_N)^{2-s}}{1 + d(i, 1)^{3-s}} \mathbb{1}_{d(i, 1) \geq (N\ell_N)^{1-\varepsilon}} \right). \quad (5.15)$$

Set

$$G_N^* := (N\ell_N)^\alpha \sum_{i \in I_\gamma} \frac{Nx_i}{1 + d(i, 1)^{1+\alpha}} \mathbb{1}_{d(i, 1) \leq (N\ell_N)^{1+\varepsilon}} + (N\ell_N)^{2-s} \sum_{i \in I_\gamma} \frac{Nx_i}{1 + d(i, 1)^{3-s}} \mathbb{1}_{d(i, 1) \geq (N\ell_N)^{1-\varepsilon}}.$$

By (5.15),

$$\eta_{\varepsilon,\gamma}^* |\nabla F_N^*| \leq C (N\ell_N)^{\kappa\varepsilon} \eta_{\varepsilon,\gamma}^* \nabla G_N^*,$$

where the above inequality is understood coordinate-wise. Thus, applying Lemma 3.6, we get

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* \nabla F_N^* \cdot A_1^{-1} (\eta_{\varepsilon,\gamma}^* \nabla F_N^*)] \leq (N\ell_N)^{2\kappa\varepsilon} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* \nabla G_N^* \cdot A_1^{-1} (\eta_{\varepsilon,\gamma}^* \nabla G_N^*)]. \quad (5.16)$$

By sub-additivity,

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* \nabla G_N^* \cdot A_1^{-1} (\eta_{\varepsilon,\gamma}^* \nabla G_N^*)] \leq 2 \text{Var}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* G_N^*] + 2 \mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* \nabla G_N^* \cdot A_1^{-1} (\eta_{\varepsilon,\gamma}^* \nabla G_N^*)]. \quad (5.17)$$

Proceeding as in (5.14), we get that there exists $C > 0$ depending on β , s and α such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* \nabla G_N^* \cdot A_1^{-1} (\eta_{\varepsilon,\gamma}^* \nabla G_N^*)] \leq \frac{C}{(N\ell_N)^{10}}. \quad (5.18)$$

Now, by the definition of $\eta_{\varepsilon,\gamma}$, there exists $\kappa > 0$ depending on α such that

$$|\eta_{\varepsilon,\gamma}^* G_N^*| \leq (N\ell_N)^{\kappa\varepsilon+\alpha} \sum_{i \in I_\gamma} \frac{1}{1 + d(i, 1)^{1+\alpha-\frac{s}{2}}}.$$

Therefore, there exist $C > 0$ and $\kappa > 0$ depending on α such that

$$\begin{aligned} |\eta_{\varepsilon,\gamma}^* G_N^*| &\leq C (N\ell_N)^{\kappa\varepsilon} \left(\sum_{i \in I_\gamma} \frac{(N\ell_N)^\alpha}{1 + d(i, 1)^{1+\alpha-\frac{s}{2}}} \mathbb{1}_{d(i, 1) \leq (N\ell_N)^{1+\varepsilon}} + \sum_{i \in I_\gamma} \frac{(N\ell_N)^{2-s}}{1 + d(i, 1)^{3-\frac{3s}{2}}} \mathbb{1}_{d(i, 1) \geq (N\ell_N)^{1-\varepsilon}} \right) \\ &\leq C (N\ell_N)^{\kappa\varepsilon} (N\ell_N)^{\max(2\alpha, s)}. \end{aligned}$$

Thus, there exists $\kappa > 0$ such that

$$\text{Var}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* G_N^*] \leq C (N\ell_N)^{\kappa\varepsilon} (N\ell_N)^{\max(2\alpha, s)}. \quad (5.19)$$

Combining (5.16), (5.17), (5.18) and (5.19), we conclude that there exist $C > 0$ and $\kappa > 0$ depending on β , s and α such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,\gamma}^* \nabla F_N^* \cdot A_1^{-1} (\eta_{\varepsilon,\gamma}^* \nabla F_N^*)] \leq C (N\ell_N)^{\kappa\varepsilon} (N\ell_N)^{\max(2\alpha, s)}. \quad (5.20)$$

Step 5: proof of (5.5). Putting (5.12), (5.14) and (5.20) together, we get that for all $\varepsilon > 0$, there exists $C > 0$ depending on β, s, α and ε such that

$$\text{Var}_{\mathbb{Q}_{N,\beta}}[\eta_{\varepsilon,\gamma} F_N] = \text{Var}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,\gamma}^* F_N^*] \leq C(N\ell_N)^\varepsilon (N\ell_N)^{\max(2\alpha,s)}. \quad (5.21)$$

Combining this with (5.8) and (5.11) gives

$$\text{Var}_{\mathbb{P}_{N,\beta}}[F_N] \leq C(N\ell_N)^\varepsilon (N\ell_N)^{\max(2\alpha,s)}.$$

On the other hand, using Theorem 1 and Lemma 4.5, we get that for γ large enough, there exists $C > 0$ depending on β, s, γ and α such that

$$\text{Var}_{\mathbb{P}_{N,\beta}} \left[\sum_{i \notin I_\gamma} \xi(\ell_N^{-1}(x_i - x_1)) \right] \leq C.$$

Combining the two above displays gives (5.5).

Step 6: proof of (5.6). Fix an index $i_0 \in \{1, \dots, N\}$. One can check that

$$\begin{aligned} \sum_{i=1}^N \xi(\ell_N^{-1} x_i) - \sum_{i=1}^N \xi(\ell_N^{-1}(x_i - x_1)) &= \left(\sum_{i=1}^N \xi(\ell_N^{-1}(x_{i_0-1+i})) - \sum_{i=1}^N \xi(\ell_N^{-1}(x_{i_0-1+i} - x_{i_0})) \right) \\ &\quad + \left(\sum_{i=1}^N \xi(\ell_N^{-1}(x_{i_0-1+i} - x_{i_0})) - \sum_{i=1}^N \xi(\ell_N^{-1}(x_i - x_1)) \right). \end{aligned}$$

We can notice that

$$\sum_{i=1}^N \xi(\ell_N^{-1}(x_{i_0-1+i} - x_{i_0})) - \sum_{i=1}^N \xi(\ell_N^{-1}(x_i - x_1)) = 0.$$

Hence, by Taylor expansion,

$$\left| \sum_{i=1}^N \xi(\ell_N^{-1} x_i) - \sum_{i=1}^N \xi(\ell_N^{-1}(x_i - x_1)) \right| \leq \ell_N^{-1} |x_{i_0-1}| \sum_{i=1}^N \left(\max_{t \in [x_{i_0-1+i} - x_{i_0}, x_{i_0-1+i}]} |\xi'(\ell_N^{-1} t)| \right).$$

By the assumption on ξ ,

$$\begin{aligned} &\left| \sum_{i=1}^N \xi(\ell_N^{-1} x_i) - \sum_{i=1}^N \xi(\ell_N^{-1}(x_i - x_1)) \right| \\ &\leq \ell_N^{-1} |x_{i_0-1}| \sum_{i=1}^N |\ell_N^{-1} \max(x_{i_0-1+i} - x_{i_0}, \frac{1}{N})|^{-1-\alpha} \mathbf{1}_{|\ell_N^{-1} x_{i_0-1+i}| \leq 2} \\ &\quad + \ell_N^{-1} |x_{i_0-1}| \sum_{i=1}^N |\ell_N^{-1}(x_{i_0-1+i} - x_{i_0})|^{-(3-s)} \mathbf{1}_{|\ell_N^{-1} x_{i_0-1+i}| \geq 2}. \end{aligned}$$

We then take i_0 to be the index in $\{1, \dots, N\}$ such that x_i is closest to 0. By using the local law of Theorem 1, we deduce that for all $\varepsilon > 0$, there exists $C > 0$ depending on β, s, α and ε such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i=1}^N \xi(\ell_N^{-1} x_i) - \sum_{i=1}^N \xi(\ell_N^{-1}(x_i - x_1)) \right)^2 \right] \leq C(N\ell_N)^\varepsilon ((N\ell_N)^{\max(2\alpha,0)} + 1). \quad (5.22)$$

Assembling this with (5.5) proves the estimate (5.6). $\square$

5.3. Reduction to a smooth test function. In the following lemma, we consider a singular test function ξ and control the fluctuations of $\xi - \xi_{\text{reg}}$, where ξ_{reg} is a mollification of ξ at microscopic scale.

Lemma 5.3. *Let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$ and $\ell_N \in (0, 1]$. Assume either that $\ell_N = 1$ or that ξ is supported on $(-\frac{1}{2}, \frac{1}{2})$. Let $\alpha \in [0, \frac{s}{2})$. Suppose that for all $x \in \ell_N^{-1}\mathbb{T}$,*

$$|\xi'(x)| \leq \frac{1}{|x|^{1+\alpha}}.$$

Let $h_N := (N\ell_N)^{-1}$ and $\xi_{\text{reg}} := \xi * K_{h_N}$, where K_ℓ is as in (5.1). Then, for all $\varepsilon > 0$, there exists $C > 0$ depending on β, s and ε such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\xi - \xi_{\text{reg}})(\ell_N^{-1}\cdot)]] \leq C(N\ell_N)^{\varepsilon+2\alpha}.$$

Proof. Let η be a smooth cutoff function supported on $[-\frac{2}{N\ell_N}, \frac{2}{N\ell_N}]$ and taking values in $\mathbb{R}^+$. We have

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\xi - \xi_{\text{reg}})(\ell_N^{-1}\cdot)]] &\leq 3\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(1 - \eta)(\xi - \xi_{\text{reg}})(\ell_N^{-1}\cdot)]] \\ &\quad + 3\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\eta\xi)(\ell_N^{-1}\cdot)]] + 3\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\eta\xi_{\text{reg}})(\ell_N^{-1}\cdot)]]. \end{aligned} \quad (5.23)$$

We first estimate the variance of $\text{Fluct}_N[(\xi\eta)(\ell_N^{-1}\cdot)]$. We have

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\xi\eta)(\ell_N^{-1}\cdot)]] &\leq \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i=1}^N |\xi|(\ell_N^{-1}x_i)\eta(\ell_N^{-1}x_i) \right)^2 \right] \\ &\leq C\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i=1}^N \frac{1}{|\ell_N^{-1}x_i|^\alpha} \mathbf{1}_{|x_i| \leq \frac{2}{N}} \right)^2 \right]. \end{aligned}$$

Let i_0^- (resp. i_0^+) be the index i of the point $x_i \leq 0$ (resp. $x_i \geq 0$) closest to 0. Let $i_0 \in \{i_0^-, i_0^+\}$ be the index i of the point x_i closest to 0.

For every $i \in \{1, \dots, N\}$, we have $|x_i| \geq \min(|x_i - x_{i_0^-}|, |x_i - x_{i_0^+}|)$. Therefore,

$$\begin{aligned} \sum_{i=1}^N \frac{1}{|\ell_N^{-1}x_i|^\alpha} \mathbf{1}_{|x_i| \leq \frac{2}{N}} &\leq \frac{2}{|\ell_N^{-1}x_{i_0}|^\alpha} \mathbf{1}_{|x_{i_0}| \leq \frac{2}{N}} + \sum_{i \neq i_0} \frac{1}{|\ell_N^{-1}(x_i - x_{i_0^-})|^\alpha} \mathbf{1}_{|x_i - x_{i_0^-}| \leq \frac{2}{N}} \\ &\quad + \sum_{i \neq i_0^+} \frac{1}{|\ell_N^{-1}(x_i - x_{i_0^+})|^\alpha} \mathbf{1}_{|x_i - x_{i_0^+}| \leq \frac{2}{N}}. \end{aligned}$$

Therefore, since $(x_i - x_{i_0^-})_i$ and $(x_i - x_{i_0^+})_i$ are identically distributed,

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\xi\eta)(\ell_N^{-1}\cdot)]] &\leq C\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{1}{|\ell_N^{-1}x_{i_0}|^\alpha} \mathbf{1}_{|x_{i_0}| \leq \frac{2}{N}} \right)^2 \right] \\ &\quad + C\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i \neq i_0^+} \frac{1}{|\ell_N^{-1}(x_i - x_{i_0^+})|^\alpha} \mathbf{1}_{|x_i - x_{i_0^+}| \leq \frac{2}{N}} \right)^2 \right]. \end{aligned} \quad (5.24)$$

First, by a union bound,

$$\mathbb{P}_{N,\beta}(x_{i_0^+} \leq \frac{x}{N}) \leq N\mathbb{P}_{N,\beta}(x_1 \in [0, \frac{x}{N}]) = x.$$

Hence, we get that there exists $C > 0$ such that

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\frac{1}{|\ell_N^{-1}x_{i_0^+}|^\alpha} \mathbf{1}_{|x_{i_0^+}| \leq \frac{2}{N}} \right)^2 \right] &\leq \ell_N^{2\alpha} \int_0^2 \mathbb{P}_{N,\beta}(|Nx_{i_0^+}|^{-\alpha} \geq t) dt \\ &\leq (N\ell_N)^{2\alpha} \int_0^2 t^{-\frac{1}{\alpha}} dt \\ &\leq C(N\ell_N)^{2\alpha}. \end{aligned} \quad (5.25)$$

For the second term in (5.24), by the Cauchy-Schwarz inequality,

$$\left(\sum_{i \neq i_0^+} \frac{1}{|\ell_N^{-1}(x_i - x_{i_0^+})|^\alpha} \mathbf{1}_{|x_i - x_{i_0^+}| \leq \frac{2}{N}} \right)^2 \leq C \sum_{i \neq i_0^+} \frac{1 + d(i, i_0^+)^2}{|\ell_N^{-1}(x_i - x_{i_0^+})|^{2\alpha}} \mathbf{1}_{|x_i - x_{i_0^+}| \leq \frac{2}{N}}.$$

Moreover, since $(x_i - x_{i_0^+})_{i \neq i_0^+}$ is independent from i_0 and since the x_i are undistinguishable,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i \neq i_0^+} \frac{1}{|\ell_N^{-1}(x_i - x_{i_0^+})|^\alpha} \mathbb{1}_{|x_i - x_{i_0^+}| \leq 2\frac{\ell_N}{N}} \right)^2 \right] = \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i \neq 1} \frac{1}{|\ell_N^{-1}(x_i - x_1)|^\alpha} \mathbb{1}_{|x_i - x_1| \leq 2\frac{\ell_N}{N}} \right)^2 \right].$$

Thus,

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i \neq i_0^+} \frac{1}{|\ell_N^{-1}(x_i - x_{i_0^+})|^\alpha} \mathbb{1}_{|x_i - x_{i_0^+}| \leq 2\frac{\ell_N}{N}} \right)^2 \right] \\ \leq \sum_{i=2}^N (1 + d(i, 1)^2) \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\frac{1}{|\ell_N^{-1}(x_i - x_1)|^{4\alpha}} \right]^{\frac{1}{2}} \mathbb{P}_{N,\beta}(N(x_i - x_1) \leq 2\frac{\ell_N}{N}). \end{aligned}$$

Using Theorem 1 and Lemma 4.5, we conclude that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i \neq i_0^+} \frac{1}{|\ell_N^{-1}(x_i - x_{i_0^+})|^\alpha} \mathbb{1}_{|x_i - x_{i_0^+}| \leq \frac{2}{N}} \right)^2 \right] \leq C(N\ell_N)^{2\alpha}.$$

Inserting this into (5.24) and using (5.25), we get that there exists $C > 0$ depending on β , s and α such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\xi\eta)(\ell_N^{-1}\cdot)]] \leq C(N\ell_N)^{2\alpha}. \quad (5.26)$$

Similarly, we can show that there exists $C > 0$ depending on β , s and α such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\xi_{\text{reg}}\eta)(\ell_N^{-1}\cdot)]] \leq C(N\ell_N)^{2\alpha}. \quad (5.27)$$

By (5.23), it therefore remains to control $\text{Fluct}_N[(1 - \eta)(\xi - \xi_{\text{reg}})(\ell_N^{-1}\cdot)]$. By (5.4), we have

$$|(\xi - \xi_{\text{reg}})'| \leq \frac{C}{N\ell_N} \frac{1}{|\max(x, \frac{1}{N\ell_N})|^{2+\alpha}} \mathbb{1}_{|x| \leq 2}.$$

Therefore, by Lemma 5.2, we have

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[(\eta\xi_{\text{reg}})(\ell_N^{-1}\cdot)]] \leq C(N\ell_N)^{\kappa\varepsilon} \frac{1}{(N\ell_N)^2} ((N\ell_N)^s + (N\ell_N)^{2+2\alpha}) \leq C(N\ell_N)^{\kappa\varepsilon+2\alpha}. \quad (5.28)$$

Combining (5.23), (5.26), (5.27) and (5.28), this concludes the proof of the lemma. $\square$

5.4. Mean-field transport. We first introduce the transportation argument of [Joh98, Shc13], which serves as the starting point for numerous CLTs on β -ensembles and Coulomb gases, including those in [LS18, BLS⁺18, Leb18, Ser20]. Given a linear statistic F , the method involves constructing a mean-field approximation of the solution $\nabla\Phi$ of the equation

$$-\Delta\Phi + \beta\nabla\mathcal{H}_N \cdot \nabla\Phi = F - \mathbb{E}_{\mathbb{P}_{N,\beta}}[F]$$

and then using this approximation to derive an expansion for the variance of F . This mean-field transport creates a non-local error term, sometimes called the “loop equation term”, which is defined for all measurable maps $\psi : \ell_N^{-1}\mathbb{T} \rightarrow \mathbb{R}$ by

$$\text{Loop}_{\ell_N}[\psi] := \iint_{\text{Diag}_{\mathbb{T}^2}^c} N\ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))N^{-(1+s)}g'(x - y)d\text{fluct}_N(x)d\text{fluct}_N(y), \quad (5.29)$$

where we recall that $\text{Diag}_{\mathbb{T}^2} = \{(x, y) \in \mathbb{T}^2 : x = y\}$,

$$\text{fluct}_N = \sum_{i=1}^N \delta_{x_i} - Ndx$$

and dx is the Lebesgue measure on $\mathbb{T}$.

Remark 5.1. For two-dimensional Coulomb gases, the loop equation term (5.29) is replaced by an angle contribution; see [LS18, BBNY16]. In β -ensembles, the quantity in (5.29) is smooth, and one can control it directly by bounding the fluctuation measure fluct_N via local laws. In our setting, however, the term (5.29) is as singular as the energy itself, so a more delicate analysis is required.

Proposition 5.4. *Let $\xi \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$ and $\ell_N \in (0, 1]$. Assume either that ξ is supported on $(-\frac{1}{2}, \frac{1}{2})$ or that $\ell_N \equiv 1$. Let $\psi_0 \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$ be given by*

$$\psi'_0 = -\frac{1}{2\beta c_s} \ell_N^{1-s} (-\Delta)^{\frac{1-s}{2}} (\xi(\ell_N^{-1}\cdot))(\ell_N\cdot) \quad \text{and} \quad \int \psi_0 = 0. \quad (5.30)$$

We have

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[\xi(\ell_N^{-1}\cdot)]] &= -\frac{1}{(N\ell_N)^{2(1-s)}} \mathbb{E}_{\mathbb{P}_{N,\beta}}[\beta\Psi_0 \cdot \nabla^2 \mathcal{H}_N \Psi_0 + \|D\Psi_0\|_F^2] \\ &\quad + \frac{2}{(N\ell_N)^{1-s}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi'(\ell_N^{-1}x_i) \psi_0(\ell_N^{-1}x_i) \right] \\ &\quad + \frac{1}{(N\ell_N)^{2(1-s)}} \text{Var}_{\mathbb{P}_{N,\beta}} \left[\beta \text{Loop}_{\ell_N}[\psi_0] - \sum_{i=1}^N \psi'_0(\ell_N^{-1}x_i) \right], \end{aligned} \quad (5.31)$$

where

$$\Psi_0 : X_N \in D_N \mapsto \ell_N(\psi_0(\ell_N^{-1}x_1), \dots, \psi_0(\ell_N^{-1}x_N)).$$

Remark 5.2 (Mean-field approximation). *Proposition 5.4 can be viewed as a mean-field approximation for the solution $\nabla\phi$ of*

$$\begin{cases} \mathcal{L}\phi = \text{Fluct}_N[\xi] & \text{in } D_N, \\ \nabla\phi \cdot \vec{n} = 0 & \text{on } \partial D_N, \end{cases} \quad (5.32)$$

where $\mathcal{L} := \mathcal{L}^{\mathbb{P}_{N,\beta}}$. Because of the energy scaling and the long-range character of the interaction, one can seek an approximate solution within the class of diagonal transports

$$\Psi : X_N = (x_1, \dots, x_N) \in D_N \mapsto (\psi(x_1), \dots, \psi(x_N)), \quad \psi : \mathbb{T} \rightarrow \mathbb{R}.$$

For the hypersingular Riesz gas—namely, the case $g(x) \sim |x|^{-s}$ with $s > 1$ —such a diagonal transport is no longer adequate to approximate the solution of (5.32). Here the energy is dominated by local interactions, so the term $-\text{div}(\Psi) + \beta \nabla \mathcal{H}_N \cdot \Psi$ fails to be, to leading order, a linear statistic.

Proof of Proposition 5.4. Let $\mathcal{L} := \mathcal{L}^{\mathbb{P}_{N,\beta}}$ be the operator

$$\mathcal{L} = -\Delta + \beta \nabla \mathcal{H}_N \cdot \nabla,$$

acting on $H^1(\mathbb{P}_{N,\beta})$. Let $\xi \in \mathcal{C}^\infty(\mathbb{T})$ and $F := \text{Fluct}_N[\xi]$.

Let $\phi \in \mathcal{C}^\infty(\mathbb{T})$ and $\psi = \phi'$. Define

$$\Phi : X_N \in D_N \mapsto \phi(x_1) + \dots + \phi(x_N).$$

We wish to find ϕ such that $\mathcal{L}\Phi \simeq F$. Let us expand $\nabla \mathcal{H}_N \cdot \nabla \Phi$. Letting $\mu_N := \frac{1}{N} \sum_{i=1}^N \delta_{x_i}$, we have

$$\nabla \mathcal{H}_N \cdot \nabla \Phi = \iint_{\text{Diag}_{\mathbb{T}^2}^c} N^{-(1+s)} g'(x-y) N(\psi(x) - \psi(y)) d(N\mu_N)(x) d(N\mu_N)(y),$$

a.e., where $\text{Diag}_{\mathbb{T}^2} = \{(x, y) \in \mathbb{T}^2 : x = y\}$. By decomposing μ_N into $\mu_N = dx + \frac{1}{N} \text{fluct}_N$, where dx is the Lebesgue measure on the circle, one can break $\nabla \mathcal{H}_N \cdot \nabla \Phi$ into

$$\begin{aligned} \nabla \mathcal{H}_N \cdot \nabla \Phi &= N^2 \iint N(\psi(x) - \psi(y)) N^{-(1+s)} g'(x-y) dx dy \\ &\quad + 2N \int \left(\int N(\psi(x) - \psi(y)) N^{-(1+s)} g'(x-y) dy \right) d\text{fluct}_N(x) + \text{Loop}_1[\psi], \end{aligned} \quad (5.33)$$

where $\text{Loop}_1[\psi]$ is as in (5.29) for $\ell_N = 1$. For the cross term, we can write

$$N \int N(\psi(x) - \psi(y)) N^{-(1+s)} g'(x-y) dy = -N^{1-s} g' * \psi,$$

where $g' * \psi = g * \psi'$ is well-defined since $\psi \in \mathcal{C}^\infty(\mathbb{T})$. Setting

$$C_N := N^2 \iint N(\psi(x) - \psi(y)) N^{-(1+s)} g'(x-y) dx dy,$$

we thus get

$$\mathcal{L}\Phi - F = \beta C_N + \int \left(-2\beta N^{1-s} g' * \psi - \xi \right) d\text{fluct}_N - \text{Fluct}_N[\psi'] + \beta \text{Loop}_1[\psi]. \quad (5.34)$$

Let $\psi_0 \in \mathcal{C}^\infty(\mathbb{T})$ be the solution of the master equation

$$-2\beta g' * \psi_0 = \xi - \int \xi \quad \text{with} \quad \int \psi_0 = 0.$$

Since g is the fundamental solution of the fractional Laplace equation (1.2), ψ_0 is the unique solution of

$$\psi'_0 = -\frac{1}{2\beta c_s} (-\Delta)^{\frac{1-s}{2}} \xi \quad \text{with} \quad \int \psi_0 = 0. \quad (5.35)$$

For this map, one can observe that the constant term in the splitting (5.33) vanishes:

$$C_N = -N^{2-s} \int g' * \psi_0 = 0,$$

since $\xi - \int \xi$ has zero mean. Let us choose $\psi = \frac{1}{N^{1-s}} \psi_0$ so that by (5.34),

$$\mathcal{L}\Phi - F = -\frac{1}{N^{1-s}} \text{Fluct}_N[\psi'_0] + \frac{\beta}{N^{1-s}} \text{Loop}_1[\psi_0]. \quad (5.36)$$

One can now write

$$\text{Var}_{\mathbb{P}_{N,\beta}}[F] = \text{Var}_{\mathbb{P}_{N,\beta}}[F - \mathcal{L}\Phi] - \text{Var}_{\mathbb{P}_{N,\beta}}[\mathcal{L}\Phi] + 2\text{Cov}_{\mathbb{P}_{N,\beta}}[F, \mathcal{L}\Phi].$$

By integration by parts under $\mathbb{P}_{N,\beta}$,

$$\text{Cov}_{\mathbb{P}_{N,\beta}}[F, \mathcal{L}\Phi] = \mathbb{E}_{\mathbb{P}_{N,\beta}}[\nabla F \cdot \nabla \Phi]$$

and

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\mathcal{L}\Phi] = \mathbb{E}_{\mathbb{P}_{N,\beta}}[\beta \nabla \Phi \cdot \nabla^2 \mathcal{H}_N \nabla \Phi + \|\nabla^2 \Phi\|_F^2].$$

Assembling the above gives

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[F] = & -\frac{1}{N^{2(1-s)}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\beta \sum_{i \neq j} N^{-s} g''(x_i - x_j) (\psi_0(x_i) - \psi_0(x_j))^2 + \sum_{i=1}^N \psi'_0(x_i)^2 \right] \\ & + \frac{2}{N^{1-s}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi'(x_i) \psi_0(x_i) \right] + \frac{1}{N^{2(1-s)}} \text{Var}_{\mathbb{P}_{N,\beta}} \left[\beta \text{Loop}_1[\psi_0] - \sum_{i=1}^N \psi'_0(x_i) \right]. \end{aligned} \quad (5.37)$$

This establishes the lemma when $\ell_N = 1$. We now prove the lemma when $\ell_N \in (0, 1)$ and ξ is supported on $(-\frac{1}{2}, \frac{1}{2})$ by rescaling. Let $\psi_0 \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$ be given by

$$\psi'_0 = -\frac{1}{2\beta c_s} \ell_N^{1-s} (-\Delta)^{\frac{1-s}{2}} (\xi(\ell_N^{-1} \cdot))(\ell_N \cdot) \quad \text{and} \quad \int_{\mathbb{T}} \psi_0 = 0.$$

Let $\psi_1 := \ell_N^s \psi_0(\ell_N^{-1} \cdot)$. We can check that $\psi_1 \in \mathcal{C}^\infty(\mathbb{T})$ solves

$$\psi'_1 = -\frac{1}{2\beta c_s} (-\Delta)^{\frac{1-s}{2}} (\xi(\ell_N^{-1} \cdot)) \quad \text{and} \quad \int_{\mathbb{T}} \psi_1 = 0.$$

Indeed, by the definition of ψ_1 , for all $x \in \mathbb{T}$,

$$\psi'_1(x) = \ell_N^{s-1} \psi'_0(\ell_N^{-1} x) = -\frac{1}{2\beta c_s} (-\Delta)^{\frac{1-s}{2}} (\xi(\ell_N^{-1} \cdot))(x).$$

Therefore, by (5.37),

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[F] = & -\frac{1}{N^{2(1-s)}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\beta \sum_{i \neq j} N^{-s} g''(x_i - x_j) (\psi_1(x_i) - \psi_1(x_j))^2 + \sum_{i=1}^N \psi'_1(x_i)^2 \right] \\ & + \frac{2}{N^{1-s}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \ell_N^{-1} \xi'(\ell_N^{-1} x_i) \psi_1(x_i) \right] + \frac{1}{N^{2(1-s)}} \text{Var}_{\mathbb{P}_{N,\beta}} \left[\beta \text{Loop}_1[\psi_1] - \sum_{i=1}^N \psi'_1(x_i) \right]. \end{aligned} \quad (5.38)$$

By the definition of ψ_1 ,

$$\begin{aligned} & \frac{1}{N^{2(1-s)}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\beta \sum_{i \neq j} N^{-s} g''(x_i - x_j) (\psi_1(x_i) - \psi_1(x_j))^2 + \sum_{i=1}^N \psi_1'(x_i)^2 \right] \\ &= \frac{1}{(N\ell_N)^{2(1-s)}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\beta \sum_{i \neq j} N^{-s} g''(x_i - x_j) \ell_N^2 (\psi_0(\ell_N^{-1}x_i) - \psi_0(\ell_N^{-1}x_j))^2 + \sum_{i=1}^N \psi_0'(\ell_N^{-1}x_i)^2 \right]. \end{aligned}$$

Moreover,

$$\frac{1}{N^{1-s}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \ell_N^{-1} \xi'(\ell_N^{-1}x_i) \psi_1(x_i) \right] = \frac{1}{(N\ell_N)^{1-s}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi'(\ell_N^{-1}x_i) \psi_0(\ell_N^{-1}x_i) \right].$$

Finally, recalling the definition of Loop_{ℓ_N} from (5.29),

$$\frac{1}{N^{2(1-s)}} \text{Var}_{\mathbb{P}_{N,\beta}} \left[\beta \text{Loop}_1[\psi_1] - \sum_{i=1}^N \psi_1'(x_i) \right] = \frac{1}{(N\ell_N)^{2(1-s)}} \text{Var}_{\mathbb{P}_{N,\beta}} \left[\beta \text{Loop}_{\ell_N}[\psi_0] - \sum_{i=1}^N \psi_0'(\ell_N^{-1}x_i) \right].$$

Thus, inserting the three last displays into (5.38), we obtain

$$\begin{aligned} & \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Fluct}_N[\xi(\ell_N^{-1}\cdot)]] \\ &= -\frac{1}{(N\ell_N)^{2(1-s)}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\beta \sum_{i \neq j} N^{-s} g''(x_i - x_j) \ell_N^2 (\psi(\ell_N^{-1}x_i) - \psi(\ell_N^{-1}x_j))^2 + \sum_{i=1}^N \psi'(\ell_N^{-1}x_i)^2 \right] \\ &+ \frac{2}{(N\ell_N)^{1-s}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi'(\ell_N^{-1}x_i) \psi(\ell_N^{-1}x_i) \right] + \frac{1}{(N\ell_N)^{2(1-s)}} \text{Var}_{\mathbb{P}_{N,\beta}} \left[\beta \text{Loop}_{\ell_N}[\psi] - \sum_{i=1}^N \psi'(\ell_N^{-1}x_i) \right], \end{aligned}$$

which concludes the proof of the lemma. $\square$

5.5. Variance expansion up to the control on the loop equation term. Building on Proposition 5.4, we first establish Theorem 2, provided we can bound the variance of the loop equation term (5.29).

Proposition 5.5. *Let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$ and (ℓ_N) satisfy Assumption 1.1. Let ϕ be an integral of $(-\Delta)^{\frac{1-s}{2}} \xi$. Suppose that there exists $\alpha \in [s-1, \frac{s}{2})$ and $C > 0$ such that*

$$|\phi''(x)| \leq \frac{C}{|x|^{2-s+\alpha}}. \quad (5.39)$$

Let $h_N := (N\ell_N)^{-1}$ and $\xi_{\text{reg}} := \xi * K_{h_N}$ where K_{h_N} is the kernel (5.1). Consider the map $\psi \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$ given by

$$\psi' = -\frac{1}{2c_s} \ell_N^{1-s} (-\Delta)^{\frac{1-s}{2}} (\xi_{\text{reg}}(\ell_N^{-1}\cdot))(\ell_N\cdot) \quad \text{and} \quad \int \psi = 0. \quad (5.40)$$

Let $\sigma_{\ell_N}^2(\xi)$ be as in (1.10). For all $\varepsilon > 0$, there exists a constant $C > 0$ depending on $\beta, s, \xi, \varepsilon$ and the kernel K such that

$$\begin{aligned} & \left| \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Fluct}_N[\xi(\ell_N^{-1}\cdot)]] - (N\ell_N)^s \sigma_{\ell_N}^2(\xi) \right| \\ & \leq C(N\ell_N)^{2(s-1)} \left(\text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N}[\psi]] + (N\ell_N)^{\max(2s-1, 2\alpha)+\varepsilon} \right). \end{aligned} \quad (5.41)$$

Proof. Without loss of generality, we can suppose that $\int \xi = 0$.

By Lemma 5.3, for all $\varepsilon > 0$, there exists a constant $C > 0$ depending on $\beta, s, \varepsilon, \xi$ and the kernel K such that

$$\text{Var}_{\mathbb{P}_{N,\beta}} [\text{Fluct}_N[\xi_{\text{reg}}(\ell_N^{-1}\cdot) - \xi(\ell_N^{-1}\cdot)]] \leq C(1 + (N\ell_N)^{2\alpha+2\varepsilon}). \quad (5.42)$$

We now study the fluctuations of $\text{Fluct}_N[\xi_{\text{reg}}(\ell_N^{-1}\cdot)]$. Let ψ be as in (5.40). Define

$$\Psi : X_N \in D_N \mapsto \ell_N(\psi(\ell_N^{-1}x_1), \dots, \psi(\ell_N^{-1}x_N)).$$

By Proposition 5.4,

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[\xi_{\text{reg}}(\ell_N^{-1}\cdot)]] &= -\frac{1}{(N\ell_N)^{2(1-s)}}\mathbb{E}_{\mathbb{P}_{N,\beta}}[\beta\Psi\cdot\nabla^2\mathcal{H}_N\Psi + \|D\Psi\|_F^2] \\ &\quad + \frac{2}{(N\ell_N)^{1-s}}\mathbb{E}_{\mathbb{P}_{N,\beta}}\left[\sum_{i=1}^N\xi'_{\text{reg}}(\ell_N^{-1}x_i)\psi(\ell_N^{-1}x_i)\right] \\ &\quad + \frac{1}{(N\ell_N)^{2(1-s)}}\text{Var}_{\mathbb{P}_{N,\beta}}\left[\beta\text{Loop}_{\ell_N}[\psi] - \sum_{i=1}^N\psi'(\ell_N^{-1}x_i)\right]. \end{aligned} \quad (5.43)$$

By (5.4), observe that there exists $C > 0$ depending on ξ and s such that

$$|\psi''(x)| \leq C\left(\frac{1}{\max(|x|, \frac{1}{N\ell_N})^{2-s+\alpha}}\mathbb{1}_{|x|\leq 2} + \frac{1}{|x|^{3-s}}\mathbb{1}_{|x|\geq 2}\right).$$

Therefore, by Lemma 5.2, there exists a constant $C > 0$ depending on β , s , ξ and the kernel K such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Fluct}_N[\psi'(\ell_N^{-1}\cdot)]] \leq C((N\ell_N)^s + (N\ell_N)^{2(1-s+\alpha)}) \leq C(N\ell_N + (N\ell_N)^{2(1-s+\alpha)}). \quad (5.44)$$

Splitting μ_N into $\text{d}x + \frac{1}{N}\text{fluct}_N$, one can write

$$\begin{aligned} &\mathbb{E}_{\mathbb{P}_{N,\beta}}[\Psi\cdot\nabla^2\mathcal{H}_N\Psi] \\ &= \mathbb{E}_{\mathbb{P}_{N,\beta}}\left[\iint_{\text{Diag}_{\mathbb{T}^2}^c} N^{-(s+2)}g''(x-y)(N\ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))^2\text{d}(N\mu_N)(x)\text{d}(N\mu_N)(y)\right] \\ &= \iint N^{-(s+2)}g''(x-y)(N\ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))^2(N\text{d}x)(N\text{d}y) \\ &\quad + 2\iint N^{-(s+2)}g''(x-y)(N\ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))^2(N\text{d}x)\text{dfluct}_N(y) + \text{B}_{\ell_N}[\psi], \end{aligned}$$

where

$$\text{B}_{\ell_N}[\psi] := \iint_{\text{Diag}_{\mathbb{T}^2}^c} N^{-(s+2)}g''(x-y)(N\ell_N)^2(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))^2\text{dfluct}_N(x)\text{dfluct}_N(y). \quad (5.45)$$

Since the first marginal x_1 of $\mathbb{P}_{N,\beta}$ is uniformly distributed on $\mathbb{T}$, we have

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\Psi\cdot\nabla^2\mathcal{H}_N\Psi] = \iint N^{-(s+2)}g''(x-y)(N\ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))^2(N\text{d}x)(N\text{d}y) + \mathbb{E}_{\mathbb{P}_{N,\beta}}[\text{B}_{\ell_N}[\psi]].$$

Therefore,

$$\begin{aligned} &-\frac{1}{(N\ell_N)^{2(1-s)}}\mathbb{E}_{\mathbb{P}_{N,\beta}}[\beta\Psi\cdot\nabla^2\mathcal{H}_N\Psi + \|D\Psi\|_F^2] + \frac{2}{(N\ell_N)^{1-s}}\mathbb{E}_{\mathbb{P}_{N,\beta}}\left[\sum_{i=1}^N\xi'_{\text{reg}}(\ell_N^{-1}x_i)\psi(\ell_N^{-1}x_i)\right] \\ &= N^s C_N - (N\ell_N)^{2s-1}\|\psi'\|_{L^2(\ell_N^{-1}\mathbb{T})}^2 - \beta\frac{1}{(N\ell_N)^{2(1-s)}}\mathbb{E}_{\mathbb{P}_{N,\beta}}[\text{B}_{\ell_N}[\psi]], \end{aligned} \quad (5.46)$$

where

$$C_N := -\beta\int g''(x-y)(\ell_N^{-(1-s)}\psi(\ell_N^{-1}x) - \ell_N^{-(1-s)}\psi(\ell_N^{-1}y))^2\text{d}x\text{d}y + 2\int \xi'_{\text{reg}}(\ell_N^{-1}\cdot)\ell_N^{-(1-s)}\psi(\ell_N^{-1}\cdot).$$

By (2.14),

$$C_N = \frac{1}{2\beta c_s}\|\xi_{\text{reg}}(\ell_N^{-1}\cdot)\|_{H^{\frac{1-s}{2}}(\mathbb{T})}^2 = \ell_N^s \sigma_{\ell_N}^2(\xi_{\text{reg}}).$$

We will prove in Lemma B.5 that for all $\varepsilon > 0$, there exists $C > 0$ depending on β , s , ξ and ε and the kernel K such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\text{B}_{\ell_N}[\psi]] \leq C(N\ell_N)^\varepsilon((N\ell_N + (N\ell_N)^{2(1-s+\alpha)}). \quad (5.47)$$

Recall that for all $\varepsilon > 0$, $\xi \in H^{\frac{1}{2}-\alpha-\varepsilon}(\ell_N^{-1}\mathbb{T})$ and note that $\frac{1}{2} - \alpha - \varepsilon = \frac{1-s}{2} + \frac{s-2\alpha}{2} - \varepsilon$. Therefore, by the estimate (5.2) in Lemma 5.1, there exists a constant $C > 0$ depending on s, β, ξ and the kernel K such that

$$|\sigma_{\ell_N}^2(\xi) - \sigma_{\ell_N}^2(\xi_{\text{reg}})| \leq C((N\ell_N)^{-1} + (N\ell_N)^{-(s-2\alpha-2\varepsilon)}). \quad (5.48)$$

Therefore, combining (5.46) and (5.48), we get that for all $\varepsilon > 0$,

$$\begin{aligned} & -\frac{1}{(N\ell_N)^{2(1-s)}} \mathbb{E}_{\mathbb{P}_{N,\beta}} [\beta \Psi \cdot \nabla^2 \mathcal{H}_N \Psi + \|D\Psi\|_F^2] + \frac{2}{(N\ell_N)^{1-s}} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi'_{\text{reg}}(\ell_N^{-1}x_i) \psi(\ell_N^{-1}x_i) \right] \\ & = (N\ell_N)^s \sigma_{\ell_N}^2(\xi) + O((N\ell_N)^{\max(2s-1, 2\alpha)+\varepsilon}), \end{aligned}$$

with $O(\cdot)$ depending on $s, \beta, \xi, \varepsilon$ and the kernel K .

Using this and (5.42), (5.43) and (5.44), we conclude that there exists $C > 0$ depending on $s, \beta, \xi, \varepsilon$ and the kernel K such that

$$\begin{aligned} & \left| \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Fluct}_N[\xi(\ell_N^{-1}\cdot)]] - (N\ell_N)^s \sigma_{\ell_N}^2(\xi) \right| \\ & \leq C \left((N\ell_N)^{2(s-1)} \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N}[\psi]] + (N\ell_N)^{\max(2s-1, 2\alpha)+\varepsilon} \right), \end{aligned}$$

which proves (5.41). $\square$

5.6. Control on the gradient of the loop equation term. In view of Proposition 5.4, expanding the variance of a linear statistic reduces to controlling the loop equation term (5.29). We first explain why a strategy based solely on local laws fails. Recall that for all $\psi : \ell_N^{-1}\mathbb{T} \rightarrow \mathbb{R}$ smooth enough,

$$\text{Loop}_{\ell_N}[\psi] = \iint_{\text{Diag}_{\mathbb{T}^2}^c} N\ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))N^{-(1+s)}g'(x-y)d\text{fluct}_N(x)d\text{fluct}_N(y).$$

By using local laws on gaps, one may control the above integral away from the diagonal. Nevertheless $\text{Loop}_{\ell_N}[\psi]$ contains local terms such as

$$\sum_{i=1}^N N\ell_N(\psi(\ell_N^{-1}x_{i+1}) - \psi(\ell_N^{-1}x_i))N^{-(1+s)}g'(x_{i+1} - x_i),$$

which is $O(N\ell_N\|\psi'\|_{L^\infty(\mathbb{T})})$ with overwhelming probability. Therefore, applying a local law estimate will give, in the best case, the bound

$$\text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N}[\psi]] = O\left((N\ell_N)^2\|\psi'\|_{L^\infty(\ell_N^{-1}\mathbb{T})}^2\right).$$

Inserting this into Proposition 5.4 gives an error term of order $O((N\ell_N)^{2s})$, which is larger than the expected variance.

Instead, we exploit the convexity of the Hamiltonian and bound the fluctuations of $\text{Loop}_{\ell_N}[\psi]$ by means of various concentration inequalities. As emphasized in Section 3, the variance of a smooth function under a log-concave probability measure is related to the norm of its gradient. The first task is therefore to control the gradient of the loop equation term (5.29).

We first define a localized version of $\text{Loop}_{\ell_N}[\psi]$. Fix $\gamma > 1$ and let

$$t_\gamma := \min\left(\frac{(N\ell_N)^\gamma}{N}, \frac{1}{2}\right). \quad (5.49)$$

Then define

$$I_\gamma := \{i \in \{1, \dots, N\} : d(i, 1) \leq Nt_\gamma\}. \quad (5.50)$$

For $\psi : \ell_N^{-1}\mathbb{T} \rightarrow \mathbb{R}$ smooth enough, we define a localized version of $\text{Loop}_{\ell_N}[\psi]$ by letting

$$\begin{aligned} \text{Loop}_{\ell_N, \gamma}[\psi] &:= \sum_{i \neq j \in I_\gamma} N\ell_N(\psi(\ell_N^{-1}x_i) - \psi(\ell_N^{-1}x_j))N^{-(1+s)}g'(x_i - x_j) \\ &\quad - 2N \sum_{i \in I_\gamma} \int_{|y| \leq t_\gamma} N\ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))N^{-(1+s)}g'(x_i - y)dy. \end{aligned} \quad (5.51)$$

Lemma 5.6 (Control on the gradient of the localized loop equation term). *Let $\psi \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$. Let $\gamma > 1$ and let t_γ and I_γ be as in (5.49) and (5.50). Let $\varepsilon > 0$. Define the good event*

$$\mathcal{A}_{\varepsilon,\gamma} := \bigcap_{\substack{i \in I_\gamma, k \in \{-\lfloor \frac{N}{2} \rfloor, \dots, \lfloor \frac{N}{2} \rfloor\}: \\ i+k \in I_\gamma}} \left\{ X_N \in D_N : N|x_{i+k} - x_i - \frac{k}{N}| \leq (N\ell_N)^\varepsilon k^{\frac{s}{2}} \right\} \\ \cap \bigcap_{i \in I_\gamma} \left\{ X_N \in D_N : (N\ell_N)^{-\varepsilon} \leq N|x_{i+1} - x_i| \leq (N\ell_N)^\varepsilon \right\}. \quad (5.52)$$

Let $\alpha \in [s-1, \frac{s}{2})$ and let

$$w_N^{(1)} : x \in \ell_N^{-1}\mathbb{T} \mapsto \frac{1}{\max(|x|, \frac{1}{N\ell_N})^{1-s+\alpha}} \mathbb{1}_{|x|<1} + \frac{1}{|x|^{2-s}} \mathbb{1}_{|x|\geq 1}, \quad (5.53)$$

$$w_N^{(2)} : x \in \ell_N^{-1}\mathbb{T} \mapsto \frac{1}{\max(|x|, \frac{1}{N\ell_N})^{2-s+\alpha}} \mathbb{1}_{|x|<1} + \frac{1}{|x|^{3-s}} \mathbb{1}_{|x|\geq 1}. \quad (5.54)$$

Suppose that there exists a constant $C > 0$ such that

$$|\psi'| \leq Cw_N^{(1)} \quad \text{and} \quad |\psi''| \leq Cw_N^{(2)}.$$

Let $\text{Loop}_{\ell_N,\gamma}[\psi]$ be as in (5.51).

Then, one can break $\nabla \text{Loop}_{\ell_N,\gamma}[\psi]$ into

$$\nabla \text{Loop}_{\ell_N,\gamma}[\psi] = V + W,$$

with $V, W \in L^2(\{1, \dots, N\}, \mathcal{C}^\infty(D_N))$ such that the following holds:

- (1) For every $i \notin I_\gamma$, $V_i = W_i = 0$.
- (2) There exist $C > 0$ and $\kappa > 0$ depending on s such that for every $i \in I_\gamma$ and $X_N \in \mathcal{A}_{\varepsilon,\gamma}$,

$$|V_i| \leq (N\ell_N)^{\kappa\varepsilon} \left(\ell_N^{-1}(w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}(x_i - t_\gamma)) + w_N^{(2)}(\ell_N^{-1}(x_i + t_\gamma)) \right. \\ \left. + (N\ell_N)^{1-s+\alpha} N^{-\frac{s}{2}} \frac{1}{\max(|x_i|, \frac{1}{N})^{1+\frac{s}{2}}} \mathbb{1}_{|x_i| \leq 2\ell_N} \right). \quad (5.55)$$

- (3) There exists $W^{\text{gap}} \in L^2(\{1, \dots, N\}, \mathcal{C}^\infty(D_N))$ such that for all $U_N \in \mathbb{R}^N$,

$$W \cdot U_N = - \sum_{i=1}^N W_i^{\text{gap}} N(u_{i+1} - u_i).$$

Moreover, there exist $C > 0$ and $\kappa > 0$ depending on s such that

$$\sup_{\mathcal{A}_{\varepsilon,\gamma}} |W^{\text{gap}}|^2 \leq C(N\ell_N)^{\kappa\varepsilon} \left(N\ell_N + (N\ell_N)^{2(\alpha+1-s)} \right). \quad (5.56)$$

The proof of Lemma 5.6 is deferred to Appendix B.

We now record some various auxiliary estimates that will be used in the main proof.

Lemma 5.7 (Auxiliary estimates). *Suppose that the assumption of Lemma 5.6 holds. Then, for γ large enough, there exists a constant $C > 0$ depending on β, s, γ , and ψ such that*

$$\text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N}[\psi] - \text{Loop}_{\ell_N,\gamma}[\psi]] \leq CN\ell_N. \quad (5.57)$$

Moreover, there exists a constant $C > 0$ depending on β, s, γ , and ξ such that

$$\left| \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N,\gamma}[\psi]] - \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N,\gamma}[\psi](0, x_2 - x_1, \dots, x_N - x_1)] \right| \leq CN\ell_N. \quad (5.58)$$

5.7. Control on the variance of the loop equation term. We now control the variance of the localized loop equation term (5.51). Let V and W be as in Lemma 5.6. By sub-additivity, it suffices to bound the A_1^{-1} -energy of V and W separately. For W , we use that, after the appropriate convexification, the measure becomes uniformly convex in gap coordinates. For V , we will use the comparison principle of Lemma 3.6, the a priori estimate of Lemma 5.2 and the rigidity estimate of Theorem 1.

Proposition 5.8. *Let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$ and (ℓ_N) satisfy Assumption 1.1 and $\alpha \in [s-1, \frac{s}{2})$ be as in (1.8). Let $h_N := (N\ell_N)^{-1}$ and $\xi_{\text{reg}} := \xi * K_{h_N}$ with K_{h_N} as in (5.1). Let ψ be as in (5.40).*

For all $\varepsilon > 0$, there exists $C > 0$ depending on β, ε, s and ξ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[\text{Loop}_{\ell_N}[\psi]] \leq C(N\ell_N)^{\varepsilon + \max(1, 2(\alpha+1-s))}. \quad (5.59)$$

Proof.

Step 1: reduction to the localized loop equation term. Recall the anchoring map

$$\text{Anch}_N : X_N \in D_N \mapsto (x_2 - x_1, x_3 - x_1, \dots, x_N - x_1) \in \mathbb{R}^{N-1}.$$

By combining the estimates (5.57) and (5.58) of Lemma 5.7, we get that there exists a constant $C > 0$ depending on β, s, γ and ψ such that

$$\left| \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Loop}_{\ell_N}[\psi]] - \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Loop}_{\ell_N,\gamma}[\psi] \circ (0, \text{Anch}_N)] \right| \leq CN\ell_N, \quad (5.60)$$

Denote for shorthand

$$\text{Loop} := \text{Loop}_{\ell_N,\gamma}[\psi] \quad \text{and} \quad \text{Loop}^*(y_1, \dots, y_{N-1}) := \text{Loop}_{\ell_N,\gamma}[\psi](0, y_1, \dots, y_{N-1}).$$

Step 2: multiplying by a cutoff function. Let $\varepsilon > 0$. Consider the good event (5.52) with 2γ in place of γ :

$$\begin{aligned} \mathcal{A}_{\varepsilon,2\gamma} := & \bigcap_{\substack{i \in I_\gamma, k \in \{-\lfloor \frac{N}{2} \rfloor, \dots, \lfloor \frac{N}{2} \rfloor\} \\ i+k \in I_{2\gamma}}} \left\{ X_N \in D_N : N|x_{i+k} - x_i - \frac{k}{N}| \leq (N\ell_N)^\varepsilon k^{\frac{s}{2}} \right\} \\ & \cap \bigcap_{i \in I_{2\gamma}} \left\{ X_N \in D_N : (N\ell_N)^{-\varepsilon} \leq N|x_{i+1} - x_i| \leq (N\ell_N)^\varepsilon \right\}. \end{aligned}$$

Let $\eta_{\varepsilon,2\gamma}$ be the cutoff function defined in (5.7). Note that $\eta_{\varepsilon,2\gamma}$ is supported on $\mathcal{A}_{\varepsilon,2\gamma}$. One can write

$$\begin{aligned} \text{Var}_{\mathbb{P}_{N,\beta}}[\text{Loop} \circ (0, \text{Anch}_N)] \\ \leq 2 \left(\text{Var}_{\mathbb{P}_{N,\beta}}[\eta_{\varepsilon,2\gamma} \text{Loop} \circ (0, \text{Anch}_N)] + \text{Var}_{\mathbb{P}_{N,\beta}}[(1 - \eta_{\varepsilon,2\gamma}) \text{Loop} \circ (0, \text{Anch}_N)] \right). \end{aligned}$$

Using the rigidity estimate of Theorem 1 and Lemma 4.5, we get that there exists $C > 0$ depending on $\beta, s, \varepsilon, \xi$ and γ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}}[(1 - \eta_{\varepsilon,2\gamma}) \text{Loop} \circ (0, \text{Anch}_N)] \leq \frac{C}{(N\ell_N)^{10}}. \quad (5.61)$$

Step 3: convexification. Let $\mathbb{Q}_{N,\beta}$ be as in (5.9). By (5.11), there exists $\delta > 0$ depending on β, s, ε and γ such that

$$\text{TV}(\mathbb{P}_{N,\beta}, \mathbb{Q}_{N,\beta}) \leq e^{-(N\ell_N)^\delta}. \quad (5.62)$$

Observe that on the support of $\eta_{\varepsilon,2\gamma}$, there exists a constant $C > 0$ depending on ξ and γ such that

$$|\text{Loop} \circ (0, \text{Anch}_N)| \leq C(N\ell_N)^2.$$

Therefore, by (5.62), there exists $\delta > 0$ depending on β, s, γ and ξ such that

$$\left| \text{Var}_{\mathbb{P}_{N,\beta}}[\eta_{\varepsilon,2\gamma} \text{Loop} \circ (0, \text{Anch}_N)] - \text{Var}_{\mathbb{Q}_{N,\beta}}[\eta_{\varepsilon,2\gamma} \text{Loop} \circ (0, \text{Anch}_N)] \right| \leq e^{-(N\ell_N)^\delta}. \quad (5.63)$$

We let $\mathbb{P}_{N,\beta}^* := \text{Anch}_N \# \mathbb{P}_{N,\beta}$ and $\mathbb{Q}_{N,\beta}^* := \text{Anch}_N \# \mathbb{Q}_{N,\beta}$. Let $\eta_{\varepsilon,2\gamma}^*$ be such that $\eta_{\varepsilon,2\gamma} = \eta_{\varepsilon,2\gamma}^* \circ \text{Anch}_N$. Notice that

$$\text{Var}_{\mathbb{Q}_{N,\beta}}[\eta_{\varepsilon,2\gamma} \text{Loop} \circ (0, \text{Anch}_N)] = \text{Var}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,2\gamma}^* \text{Loop}^*].$$

We now estimate the variance of $\eta_{\varepsilon,2\gamma}^* \text{Loop}^*$ under $\mathbb{Q}_{N,\beta}^*$.

Denote for shorthand $A_1 := A_1^{\mathbb{Q}_{N,\beta}^*}$. Since A_1 positive on $L^2(\{1, \dots, N\}, H^1(\mathbb{Q}_{N,\beta}^*))$,

$$\begin{aligned} \text{Var}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,2\gamma}^* \text{Loop}^*] &= \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[\nabla(\eta_{\varepsilon,2\gamma}^* \text{Loop}^*) \cdot A_1^{-1} \nabla(\eta_{\varepsilon,2\gamma}^* \text{Loop}^*)] \\ &\leq 2\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\eta_{\varepsilon,2\gamma}^* \nabla \text{Loop}^*) A_1^{-1} (\eta_{\varepsilon,2\gamma}^* \nabla \text{Loop}^*)] + 2\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\nabla \eta_{\varepsilon,2\gamma}^* \text{Loop}^*) A_1^{-1} (\nabla \eta_{\varepsilon,2\gamma}^* \text{Loop}^*)]. \end{aligned} \quad (5.64)$$

Let $V, W \in L^2(\{1, \dots, N\}, \mathcal{C}^\infty(D_N))$ be as in Lemma 5.6. Let $V^*, W^* \in L^2(\{1, \dots, N-1\}, \mathcal{C}^\infty(\text{Anch}_N(D_N)))$ be such that

$$(V_i)_{2 \leq i \leq N} \circ \text{Anch}_N = V^* \quad \text{and} \quad (W_i)_{2 \leq i \leq N} \circ \text{Anch}_N = W^*.$$

Notice that

$$\nabla \text{Loop}^* = V^* + W^*.$$

By sub-additivity again, we have

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\eta_{\varepsilon,2\gamma}^* \nabla \text{Loop}^*) A_1^{-1} (\eta_{\varepsilon,2\gamma}^* \nabla \text{Loop}^*)] &\leq 2\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\eta_{\varepsilon,2\gamma}^* V^*) \cdot A_1^{-1} (\eta_{\varepsilon,2\gamma}^* V^*)] \\ &\quad + 2\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\eta_{\varepsilon,2\gamma}^* W^*) \cdot A_1^{-1} (\eta_{\varepsilon,2\gamma}^* W^*)]. \end{aligned} \quad (5.65)$$

Step 4: Brascamp-Lieb-type estimate for $\eta_{\varepsilon,2\gamma}^* W^*$ and $\text{Loop}^* \nabla \eta_{\varepsilon,2\gamma}^*$. By Lemma 5.6, for all $U_{N-1} = (u_2, \dots, u_N) \in \mathbb{R}^{N-1}$, setting $u_1 = 0$,

$$(W^* \circ \text{Anch}_N) \cdot U_{N-1} = - \sum_{i=1}^N W_i^{\text{gap}} N(u_{i+1} - u_i)$$

with W^{gap} satisfying (5.56) on the event $\mathcal{A}_{\varepsilon,2\gamma}$ and $W_i^{\text{gap}} = 0$ for every $i \notin I_\gamma$.

Therefore, proceeding as in (5.13), we deduce that there exists a constant $\kappa > 0$ such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,2\gamma}^* W^* \cdot A_1^{-1} (\eta_{\varepsilon,2\gamma}^* W^*)] \leq (N\ell_N)^{\kappa\varepsilon} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\eta_{\varepsilon,2\gamma}^*)^2 |W^{\text{gap}}|^2].$$

Since $\eta_{\varepsilon,2\gamma}$ is supported on $\mathcal{A}_{\varepsilon,2\gamma}$, the estimate (5.56) gives

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,2\gamma}^* W^* \cdot A_1^{-1} (\eta_{\varepsilon,2\gamma}^* W^*)] \leq (N\ell_N)^{\kappa\varepsilon + \max(1, 2(\alpha+1-s))}. \quad (5.66)$$

Let Gap_N^* be such that $\text{Gap}_N = \text{Gap}_N^* \circ \text{Anch}_N$. Note that $\eta_{\varepsilon,\gamma}^* = \tilde{\eta}_\varepsilon \circ \text{Gap}_N^*$ for some smooth function $\tilde{\eta}_\varepsilon : \mathbb{R}^N \rightarrow \mathbb{R}$. Reasoning as in (5.14), we get

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\nabla \eta_{\varepsilon,2\gamma}^* \text{Loop}^*) A_1^{-1} (\nabla \eta_{\varepsilon,2\gamma}^* \text{Loop}^*)] \leq (N\ell_N)^{\kappa\varepsilon} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\text{Loop}^*)^2 |(\nabla \tilde{\eta}_{\varepsilon,2\gamma}) \circ \text{Gap}_N|^2].$$

Note that $(\nabla \tilde{\eta}_{\varepsilon,2\gamma}) \circ \text{Gap}_N = 0$ on $\mathcal{A}_{\varepsilon,2\gamma}^c$ and that Loop^* is uniformly bounded on $\mathcal{A}_{\varepsilon,2\gamma}$ by $(N\ell_N)^\kappa$ for some $\kappa > 0$. Thus, using Theorem 1 and Lemma 4.5, we deduce from the above display that there exists $C > 0$ depending on β, s and ξ such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\nabla \eta_{\varepsilon,2\gamma}^* \text{Loop}^*) A_1^{-1} (\nabla \eta_{\varepsilon,2\gamma}^* \text{Loop}^*)] \leq \frac{C}{(N\ell_N)^{10}}. \quad (5.67)$$

Step 5: using the comparison principle for $\eta_{\varepsilon,2\gamma}^* V$. Let $\chi_1 \in \mathcal{C}^0(\mathbb{T})$ be given by $\int \chi_1 = 0$ and for all $x \in \mathbb{T}$,

$$\chi_1'(x) = \ell_N^{-1} w_N^{(2)}(\ell_N^{-1} x).$$

Let $\chi_2 \in \mathcal{C}^0(\mathbb{T})$ be given by $\int \chi_2 = 0$ and for all $x \in \mathbb{T}$,

$$\chi_2'(x) = (N\ell_N)^{1-s+\alpha} (N\ell_N)^{-\frac{s}{2}} \ell_N^{-1} \frac{1}{\max(|\ell_N^{-1} x|, \frac{1}{N})^{1+\frac{s}{2}}} \mathbf{1}_{|x| \leq 2\ell_N}.$$

Let us define

$$G_N^* := \sum_{i \in I_\gamma} (\chi_1(x_i) + \chi_1(x_i - t_\gamma) + \chi_1(x_i + t_\gamma) + \chi_2(x_i)).$$

By Lemma 5.6, there exist constants $C > 0$ and $\kappa > 0$ such that

$$\eta_{\varepsilon,2\gamma}^* |V^*| \leq C(N\ell_N)^{\kappa\varepsilon} \eta_{\varepsilon,2\gamma}^* \nabla G_N^*.$$

Thus, by Lemma 3.6 and sub-additivity,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\eta_{\varepsilon,2\gamma}^* V^*) \cdot A_1^{-1} (\eta_{\varepsilon,2\gamma}^* V^*)] &\leq \mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[(\eta_{\varepsilon,2\gamma}^* \nabla G_N^*) \cdot A_1^{-1} (\eta_{\varepsilon,2\gamma}^* \nabla G_N^*)] \\ &\leq 2\text{Var}_{\mathbb{Q}_{N,\beta}^*}[\eta_{\varepsilon,2\gamma}^* G_N^*] + 2\mathbb{E}_{\mathbb{Q}_{N,\beta}^*}[G_N^* \nabla \eta_{\varepsilon,2\gamma}^* \cdot A_1^{-1} (G_N^* \nabla \eta_{\varepsilon,2\gamma}^*)]. \end{aligned}$$

Arguing as in (5.14), there exists $C > 0$ depending on $\beta, s, \varepsilon, \gamma$ and ξ such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [G_N^* \nabla \eta_{\varepsilon,2\gamma}^* \cdot A_1^{-1} (G_N^* \nabla \eta_{\varepsilon,2\gamma}^*)] \leq \frac{C}{(N\ell_N)^{10}}.$$

Moreover, applying the estimate (5.21) in the proof of Lemma 5.2, we get that there exist $C > 0$ and $\kappa > 0$ depending on $\beta, s, \gamma, \varepsilon$ and ξ such that

$$\begin{aligned} \text{Var}_{\mathbb{Q}_{N,\beta}} \left[\eta_{\varepsilon,2\gamma} \sum_{i \in I_\gamma} (\chi_1(x_i - x_1) + \chi_1(x_i - x_1 - t_\gamma) + \chi_1(x_i - x_1 + t_\gamma)) \right] \\ \leq C(N\ell_N)^{\kappa\varepsilon + \max(s, 2(1-s+\alpha))} \leq C(N\ell_N)^{\kappa\varepsilon + \max(1, 2(1-s+\alpha))} \end{aligned}$$

and

$$\text{Var}_{\mathbb{Q}_{N,\beta}} \left[\eta_{\varepsilon,2\gamma} \sum_{i \in I_\gamma} \chi_2(x_i - x_1) \right] \leq C(N\ell_N)^{\kappa\varepsilon}.$$

Combining the four above displays, we deduce that there exist $C > 0$ and $\kappa > 0$ depending on β, s, γ and ξ such that

$$\mathbb{E}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,2\gamma}^* V^* \cdot A_1^{-1} (\eta_{\varepsilon,2\gamma}^* V^*)] \leq C(N\ell_N)^{\kappa\varepsilon + \max(1, 2(1-s+\alpha))}. \quad (5.68)$$

Step 6: conclusion. Combining (5.64), (5.65), (5.66), (5.67) and (5.68), we deduce the existence of $\kappa > 0$ and $C > 0$ depending on $\beta, \varepsilon, s, \gamma$ and ξ such that

$$\text{Var}_{\mathbb{Q}_{N,\beta}^*} [\eta_{\varepsilon,2\gamma}^* \text{Loop}^*] \leq C(N\ell_N)^{\kappa\varepsilon} (N\ell_N)^{\max(1, 2(\alpha+1-s))}.$$

Together with (5.61), (5.63) and (5.7), we get that for all $\varepsilon > 0$, there exists $C > 0$ depending on $\beta, s, \gamma, \varepsilon$ and ξ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N, \gamma}[\psi]] \leq C(N\ell_N)^\varepsilon (N\ell_N)^{\max(1, 2(\alpha+1-s))}.$$

Combining this with (5.60), this concludes the proof of the proposition. $\square$

5.8. Proof of Theorem 2.

Proof. Piecing together Proposition 5.5 and Proposition 5.8 concludes the proof. $\square$

6. CENTRAL LIMIT THEOREM

6.1. Proof of the CLT. We prove the CLT stated in Theorem 3 using Stein's method, leveraging the variance estimates of Section 5.

Proof of Theorem 3. Let $\xi \in L^2(\ell_N^{-1}\mathbb{T})$ and (ℓ_N) satisfy Assumption 1.1. Let

$$F_N := (N\ell_N)^{-\frac{s}{2}} \text{Fluct}_N[\xi(\ell_N^{-1}\cdot)].$$

Let $h \in \mathcal{C}^1(\mathbb{R})$ such that $\|h\|_{L^\infty(\mathbb{R})} \leq 1$ and $\|h'\|_{L^\infty(\mathbb{R})} \leq 1$.

Step 1: regularization. Let $h_N := (N\ell_N)^{-1}$ and $\xi_{\text{reg}} = \xi * K_{h_N}$ with K_{h_N} as in (5.1). Let

$$G_N := (N\ell_N)^{-\frac{s}{2}} \text{Fluct}_N[\xi_{\text{reg}}(\ell_N^{-1}\cdot)].$$

We can write

$$\left| \mathbb{E}_{\mathbb{P}_{N,\beta}} [h(F_N)] - \mathbb{E}_{\mathbb{P}_{N,\beta}} [h(G_N)] \right| \leq \|h'\|_{L^\infty(\mathbb{R})} \mathbb{E}_{\mathbb{P}_{N,\beta}} [|F_N - G_N|] \leq \mathbb{E}_{\mathbb{P}_{N,\beta}} [|F_N - G_N|].$$

Moreover, by the Cauchy-Schwarz inequality,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} [|F_N - G_N|] \leq \mathbb{E}_{\mathbb{P}_{N,\beta}} [|F_N - G_N|^2]^{\frac{1}{2}} = \text{Var}_{\mathbb{P}_{N,\beta}} [F_N - G_N]^{\frac{1}{2}},$$

since both F_N and G_N are centered random variables under $\mathbb{P}_{N,\beta}$. Combining the two above displays, we deduce that

$$\begin{aligned} \left| \mathbb{E}_{\mathbb{P}_{N,\beta}} [h(F_N)] - \mathbb{E}_{\mathbb{P}_{N,\beta}} [h(G_N)] \right| &\leq \text{Var}_{\mathbb{P}_{N,\beta}} [F_N - G_N]^{\frac{1}{2}} \\ &= (N\ell_N)^{-\frac{s}{2}} \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Fluct}_N[(\xi - \xi_{\text{reg}})(\ell_N^{-1}\cdot)]]^{\frac{1}{2}}. \end{aligned}$$

By Lemma 5.3, there exists $C > 0$ depending on β, s, ε and ξ such that

$$\left| \mathbb{E}_{\mathbb{P}_{N,\beta}} [h(F_N)] - \mathbb{E}_{\mathbb{P}_{N,\beta}} [h(G_N)] \right| \leq C((N\ell_N)^{-\frac{s}{2}} + (N\ell_N)^{\alpha - \frac{s}{2} + \varepsilon}). \quad (6.1)$$

Step 2: Stein's method. Recall σ_{ℓ_N} from (1.10). Set $\sigma := \sigma_{\ell_N}(\xi_{\text{reg}})$ and let $Z \sim \mathcal{N}(0, \sigma^2)$. Our aim is to compare the law of G_N under $\mathbb{P}_{N,\beta}$ to the law of Z .

Consider the Stein ODE

$$\sigma^2 \eta'(x) - \eta(x)x = h(x) - \mathbb{E}[h(Z)], \quad (6.2)$$

The only bounded solution of the above equation is given by

$$\eta : x \in \mathbb{R} \mapsto \frac{1}{\sigma^2} e^{\frac{x^2}{2\sigma^2}} \int_{-\infty}^x e^{-\frac{y^2}{2\sigma^2}} (h(y) - \mathbb{E}[h(Z)]) dy.$$

By Lemma 2.4 of [CGS10], the solution η_1 of the unit-variance Stein equation, i.e. with $\sigma = 1$, belongs to $\mathcal{C}^1(\mathbb{R})$ and satisfies

$$\|\eta_1\|_{L^\infty(\mathbb{R})} \leq 2, \quad \|\eta_1'\|_{L^\infty(\mathbb{R})} \leq \sqrt{\frac{2}{\pi}}.$$

Using the exact scaling identity

$$\eta(x) = \frac{1}{\sigma} \eta_1\left(\frac{x}{\sigma}\right), \quad \eta'(x) = \frac{1}{\sigma^2} \eta_1'\left(\frac{x}{\sigma}\right),$$

we get

$$\|\eta\|_{L^\infty(\mathbb{R})} \leq \frac{2}{\sigma} \quad \text{and} \quad \|\eta'\|_{L^\infty(\mathbb{R})} \leq \sqrt{\frac{2}{\pi\sigma^2}}.$$

By (6.2),

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[h(G_N)] - \mathbb{E}_{\mathbb{P}_{N,\beta}}[h(Z)] = \mathbb{E}_{\mathbb{P}_{N,\beta}}[\sigma^2 \eta'(G_N)] - \mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)G_N]. \quad (6.3)$$

Our aim is now to control the right-hand side of the above display. Let $\psi \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$ satisfy

$$\psi' = -\frac{1}{2\beta c_s} \ell_N^{1-s} (-\Delta)^{\frac{1-s}{2}} (\xi_{\text{reg}}(\ell_N^{-1}\cdot))(\ell_N\cdot) \quad \text{and} \quad \int \psi = 0.$$

Define

$$\Psi : X_N \in D_N \mapsto \frac{1}{(N\ell_N)^{1-\frac{s}{2}}} \ell_N(\psi(\ell_N^{-1}x_1), \dots, \psi(\ell_N^{-1}x_N)),$$

which can be written as $\Psi = \nabla \Phi$ for some function $\Phi \in \mathcal{C}^\infty(D_N)$. Let $\mathcal{L} := \mathcal{L}^{\mathbb{P}_{N,\beta}}$. Recall that by (5.36),

$$\mathcal{L}\Phi - G_N = \frac{1}{(N\ell_N)^{1-\frac{s}{2}}} \left(\beta \text{Loop}_{\ell_N}[\psi] - \text{Fluct}_N[\psi'(\ell_N^{-1}\cdot)] \right).$$

Therefore,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)G_N] = \mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)\mathcal{L}\Phi] - \frac{1}{(N\ell_N)^{1-\frac{s}{2}}} \text{Cov}_{\mathbb{P}_{N,\beta}} \left[\eta(G_N), \beta \text{Loop}_{\ell_N}[\psi] - \text{Fluct}_N[\psi'(\ell_N^{-1}\cdot)] \right]. \quad (6.4)$$

By integration by parts,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)\mathcal{L}\Phi] = \mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta'(G_N)\nabla G_N \cdot \nabla \Phi].$$

Combining the two last displays, we get

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta(G_N)G_N] = \sigma_{\ell_N}^2(\xi_{\text{reg}}) \mathbb{E}_{\mathbb{P}_{N,\beta}}[\eta'(G_N)] + \text{Error}_N^1 + \text{Error}_N^2,$$

where

$$\begin{aligned} \text{Error}_N^1 &:= \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\eta'(G_N) \left(\sum_{i=1}^N \frac{1}{N\ell_N} \psi(\ell_N^{-1}x_i) \xi'_{\text{reg}}(\ell_N^{-1}x_i) - \sigma_{\ell_N}^2(\xi_{\text{reg}}) \right) \right], \\ \text{Error}_N^2 &:= -\frac{1}{(N\ell_N)^{1-\frac{s}{2}}} \text{Cov}_{\mathbb{P}_{N,\beta}} \left[\eta(G_N), \beta \text{Loop}_{\ell_N}[\psi] - \text{Fluct}_N[\psi'(\ell_N^{-1}\cdot)] \right]. \end{aligned}$$

Step 3: the error term Error_N^1 . One can bound Error_N^1 by

$$|\text{Error}_N^1| \leq \|\eta'\|_{L^\infty(\mathbb{R})} \frac{1}{N\ell_N} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left| \sum_{i=1}^N \psi(\ell_N^{-1}x_i) \xi'_{\text{reg}}(\ell_N^{-1}x_i) - N\ell_N \sigma_{\ell_N}^2(\xi_{\text{reg}}) \right| \right].$$

Since

$$\frac{1}{\ell_N^{1-s}} \psi'(\ell_N^{-1}\cdot) = -\frac{1}{2\beta c_s} (-\Delta)^{\frac{1-s}{2}} (\xi_{\text{reg}}(\ell_N^{-1}\cdot)),$$

we get using (2.14) that

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi'_{\text{reg}}(\ell_N^{-1}x_i) \psi(\ell_N^{-1}x_i) \right] &= \frac{(2\pi)^{1-s}}{2\beta c_s} N\ell_N^{1-s} \|\xi_{\text{reg}}(\ell_N^{-1}\cdot)\|_{H^{\frac{1-s}{2}}(\ell_N^{-1}\mathbb{T})}^2 \\ &= N\ell_N \sigma_{\ell_N}^2(\xi_{\text{reg}}). \end{aligned}$$

We therefore obtain

$$|\text{Error}_N^1| \leq C \|\eta'\|_{L^\infty(\mathbb{R})} \frac{1}{N\ell_N} \text{Var}_{\mathbb{P}_{N,\beta}} \left[\sum_{i=1}^N \xi'_{\text{reg}}(\ell_N^{-1}x_i) \psi(\ell_N^{-1}x_i) \right]^{\frac{1}{2}}.$$

By Lemma 4.2, there exists a constant $C > 0$ depending on β , s and ξ such that

$$|\text{Error}_N^1| \leq C(N\ell_N)^{-\frac{1}{2}}. \quad (6.5)$$

Step 4: the error term Error_N^2 . On the other hand, we have

$$\begin{aligned} |\text{Error}_N^2| &\leq \frac{2\beta}{(N\ell_N)^{1-\frac{s}{2}}} \|\eta\|_{L^\infty(\mathbb{R})} \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N}[\psi]]^{\frac{1}{2}} \\ &\quad + \frac{2}{(N\ell_N)^{1-\frac{s}{2}}} \|\eta\|_{L^\infty(\mathbb{R})} \text{Var}_{\mathbb{P}_{N,\beta}} [\text{Fluct}_N[\psi'(\ell_N^{-1}\cdot)]]^{\frac{1}{2}}. \end{aligned} \quad (6.6)$$

By (5.59), for all $\varepsilon > 0$, there exists a constant $C > 0$ depending on β , s , ε and ξ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}} [\text{Loop}_{\ell_N}[\psi]] \leq C(N\ell_N)^{\varepsilon + \max(1, 2(\alpha+1-s))}.$$

By (5.4) and Lemma 5.2, there exists a constant $C > 0$ depending on β , s and ξ such that

$$\text{Var}_{\mathbb{P}_{N,\beta}} [\text{Fluct}_N[\psi'(\ell_N^{-1}\cdot)]] \leq C(N\ell_N)^{\max(1, 2(\alpha+1-s))}.$$

It follows that for all $\varepsilon > 0$, there exists a constant $C > 0$ depending on β , s , ε and ξ such that

$$|\text{Error}_N^2| \leq C \|\eta\|_{L^\infty(\mathbb{R})} (N\ell_N)^{\varepsilon + \max(\frac{s-1}{2}, \alpha - \frac{s}{2})}. \quad (6.7)$$

Step 5: conclusion. Inserting (6.5), (6.7) into (6.4), we get that for all $\varepsilon > 0$, there exists $C > 0$ depending on β , s , ε and ξ such that

$$|\mathbb{E}_{\mathbb{P}_{N,\beta}}[h(G_N)] - \mathbb{E}[h(Z)]| \leq C(N\ell_N)^{\varepsilon + \max(\frac{s-1}{2}, \alpha - \frac{s}{2})}. \quad (6.8)$$

Let $Z' \sim \mathcal{N}(0, \sigma_{\ell_N}^2(\xi))$. We can write

$$\begin{aligned} &|\mathbb{E}_{\mathbb{P}_{N,\beta}}[h(F_N)] - \mathbb{E}[h(Z')]| \\ &\leq |\mathbb{E}_{\mathbb{P}_{N,\beta}}[h(G_N)] - \mathbb{E}[h(Z)]| + |\mathbb{E}_{\mathbb{P}_{N,\beta}}[h(F_N)] - \mathbb{E}[h(G_N)]| + |\mathbb{E}[h(Z)] - \mathbb{E}[h(Z')]|. \end{aligned} \quad (6.9)$$

Clearly,

$$|\mathbb{E}[h(Z)] - \mathbb{E}[h(Z')]| \leq \|\eta'\|_{L^\infty(\mathbb{R})} |\sigma_{\ell_N}(\xi) - \sigma_{\ell_N}(\xi_{\text{reg}})| \leq |\sigma_{\ell_N}(\xi) - \sigma_{\ell_N}(\xi_{\text{reg}})| \leq |\sigma_{\ell_N}^2(\xi) - \sigma_{\ell_N}^2(\xi_{\text{reg}})|^{\frac{1}{2}}.$$

Recall that $\xi \in H^{\frac{1}{2}-\alpha-\varepsilon}(\ell_N^{-1}\mathbb{T})$ and $\frac{1}{2} - \alpha = \frac{1-s}{2} + \frac{s-2\alpha}{2} - \varepsilon$. Thus, by Lemma 5.1, there exists a constant $C > 0$ depending on β , s , ξ and ε such that

$$|\sigma_{\ell_N}^2(\xi) - \sigma_{\ell_N}^2(\xi_{\text{reg}})| \leq C((N\ell_N)^{-1} + (N\ell_N)^{-(s-2\alpha-2\varepsilon)}),$$

which gives

$$|\mathbb{E}[h(Z)] - \mathbb{E}[h(Z')]| \leq C((N\ell_N)^{-\frac{1}{2}} + (N\ell_N)^{\alpha-\frac{s}{2}}). \quad (6.10)$$

Combining (6.9), (6.8), (6.1) and (6.10), we get that for all $\varepsilon > 0$, there exists $C > 0$ depending on β , s , ε and ξ such that

$$|\mathbb{E}_{\mathbb{P}_{N,\beta}}[h(F_N)] - \mathbb{E}[h(Z')]| \leq C(N\ell_N)^{\varepsilon + \max(-\frac{s}{2}, \frac{s-1}{2}, \alpha - \frac{s}{2} + \varepsilon)}.$$

This concludes the proof of Theorem 3. $\square$

6.2. Proof of Corollary 1.

Proof of Corollary 1. By Lemma 2.3, the function $\xi := \mathbf{1}_{(-a,a)}$ satisfies Assumption 1.1. Therefore, by Theorem 3,

$$(N\ell_N)^{-\frac{s}{2}} \sigma_{\ell_N}(\xi)^{-1} \text{Fluct}_N[\xi(\ell_N^{-1} \cdot)] \xrightarrow{\text{Law}} \mathcal{N}(0, 1).$$

Let $\ell = \lim_{N \rightarrow \infty} \ell_N \in \{0, 1\}$. We have

$$\lim_{N \rightarrow \infty} \sigma_{\ell_N}(\xi) = \sigma_\ell(\xi).$$

Moreover, by Lemma 2.3,

$$\sigma_1^2(\xi) = \frac{2\cotan(\frac{\pi}{2}s)}{\beta 4^s \pi s} \zeta(-s, 2a)$$

and

$$\sigma_0^2(\xi) = \frac{2\cotan(\frac{\pi}{2}s)}{\beta 4^s \pi s} (2a)^s.$$

These two cases combine into

$$\frac{1}{\sqrt{N^s \zeta(-s, 2a\ell_N)}} \text{Fluct}_N[\mathbf{1}_{(-a,a)}(\ell_N^{-1} \cdot)] \xrightarrow{\text{Law}} \mathcal{N}(0, \sigma^2), \quad (6.11)$$

where

$$\sigma^2 := \frac{2\cotan(\frac{\pi}{2}s)}{\beta 4^s \pi s}, \quad (6.12)$$

as claimed.

We now prove the second assertion. Let (k_N) be a sequence in $\{1, \dots, \lfloor \frac{N}{2} \rfloor\}$ such that $k_N \rightarrow \infty$. Let $i_0 \in \{1, \dots, N\}$. For $\varepsilon > 0$, define the event

$$\mathcal{A} := \bigcap_{i,j: d(i,i_0) \leq 2k_N, d(j,i_0) \leq 2k_N} \left\{ |N(x_j - x_i) - N \frac{j-i}{N}| \leq k_N^\varepsilon d(j,i)^{\frac{s}{2}} \right\}.$$

Set $\ell_N := \frac{k_N}{N}$. Let us define the error counting

$$\Delta_N := \left| \{i \in \{1, \dots, N\} : x_i \in [x_{i_0}, x_{i_0} + \ell_N]\} \right| - k_N.$$

We claim that there exists a constant $C > 0$ such that on the event $\mathcal{A}$,

$$\left| N(x_{k_N+i_0} - x_{i_0}) - k_N + \Delta_N \right| \leq C k_N^{2\varepsilon + \frac{s^2}{4}}. \quad (6.13)$$

Let us prove (6.13). Since $N(x_{i+1} - x_i) \leq (N\ell_N)^\varepsilon$ for every $i \in \{1, \dots, N\}$ such that $d(i, i_0) \leq 2N\ell_N$, we have

$$|N(x_{i_0+k_N+\Delta_N} - x_{i_0}) - k_N| \leq (N\ell_N)^\varepsilon.$$

Moreover, by the definition of $\mathcal{A}$,

$$|N(x_{i_0+k_N+\Delta_N} - x_{i_0+k_N}) - \Delta_N| \leq k_N^\varepsilon \Delta_N^{\frac{s}{2}}.$$

Combining the two above displays, we get

$$\begin{aligned} N(x_{i_0+k_N} - x_{i_0}) &= N(x_{i_0+k_N} - x_{i_0+k_N+\Delta_N}) + N(x_{i_0+k_N+\Delta_N} - x_{i_0}) \\ &= -\Delta_N + O\left(k_N^\varepsilon \Delta_N^{\frac{s}{2}}\right) + k_N + O(k_N^\varepsilon). \end{aligned}$$

Thus,

$$N(x_{i_0+k_N} - x_{i_0}) - k_N = -\Delta_N + O\left(k_N^\varepsilon (\Delta_N^{\frac{s}{2}} + 1)\right).$$

Since $\Delta_N = O(k_N^{\frac{s}{2}+\varepsilon})$, this proves the claim.

We now let i_0 be the index i (defined almost surely) of the point x_i closest to 0. We have

$$\Delta_N = \sum_{i=1}^N \mathbf{1}_{[0, \ell_N]}(x_i - x_{i_0}) - N\ell_N.$$

There exists a constant $\kappa > 0$ such that on the event $\mathcal{A} \cap \{|x_{i_0}| \leq (N\ell_N)^\varepsilon\}$,

$$\Delta_N = \sum_{i=1}^N \mathbb{1}_{[0, \ell_N]}(x_i) - N\ell_N + O((N\ell_N)^{\kappa\varepsilon}). \quad (6.14)$$

By Theorem 1, there exists $\delta > 0$ depending on β, s and ε such that

$$\mathbb{P}_{N, \beta} \left((\mathcal{A} \cap \{|x_{i_0}| \leq (N\ell_N)^\varepsilon\})^c \right) \leq e^{-k_N^\delta}.$$

Therefore, combining this with (6.14) and (6.11), we get

$$\frac{1}{\sqrt{N^s \zeta(-s, \ell_N)}} \Delta_N \xrightarrow[\text{Law}]{} \mathcal{N}(0, \sigma^2), \quad (6.15)$$

where σ^2 is as in (6.12). Therefore, by (6.13),

$$\frac{1}{\sqrt{N^s \zeta(-s, \ell_N)}} (N(x_{i_0+k_N} - x_{i_0}) - k_N) \xrightarrow[\text{Law}]{} \mathcal{N}(0, \sigma^2). \quad (6.16)$$

Since $N(x_{i_0+k_N} - x_{i_0})$ is independent from i_0 and since the variables $N(x_{i+k_N} - x_i)$, $i \in \{1, \dots, N\}$ are identically distributed, we get

$$\text{Law}_{\mathbb{P}_{N, \beta}}(N(x_{i_0+k_N} - x_{i_0})) = \text{Law}_{\mathbb{P}_{N, \beta}}(N(x_{1+k_N} - x_1)).$$

Thus, we conclude from (6.16) that

$$\frac{1}{\sqrt{N^s \zeta(-s, \ell_N)}} (N(x_{1+k_N} - x_1) - k_N) \xrightarrow[\text{Law}]{} \mathcal{N}(0, \sigma^2),$$

which concludes the proof. $\square$

APPENDIX A. WELL-POSEDNESS OF THE H.-S. EQUATION

A.1. Poincaré inequality. Recall the map

$$\text{Gap}_N : X_N \in D_N \mapsto N(x_2 - x_1, x_3 - x_2, \dots, x_N - x_{N-1}, x_1 - x_N) \in \mathbb{R}^N.$$

Lemma A.1 (Poincaré inequality). *Let μ satisfy Assumption 3.1 and denote $\mu' := \text{Gap}_N \# \mu$. Then, there exists a constant $C > 0$ such that for $\psi \in H^1(\mu')$,*

$$\text{Var}_\mu[(\psi \circ \text{Gap}_N)] \leq C \mathbb{E}_\mu[|\nabla(\psi \circ \text{Gap}_N)|^2]. \quad (\text{A.1})$$

Proof. Let $\phi = \psi \circ \text{Gap}_N$ with $\psi \in H^1(\mu')$. Observe that the measure μ' is uniformly log-concave:

$$d\mu'(x) = e^{-\tilde{H}(x)} dX_N,$$

where dX_N is the surface measure on $\text{Gap}_N(D_N)$ and $\tilde{H}$ is a function satisfying $\nabla^2 \tilde{H} \geq c \text{Id}$ for some constant $c > 0$. Therefore, by the Brascamp-Lieb inequality stated in Lemma 3.7, we have, defining $\partial_{N+1}\phi := \partial_1\phi$,

$$\text{Var}_\mu[\phi] = \text{Var}_{\mu'}[\psi] \leq c^{-1} \mathbb{E}_{\mu'}[|\nabla\psi|^2] = c^{-1} \mathbb{E}_\mu \left[\sum_{i=1}^N (N(\partial_{i+1}\phi - \partial_i\phi))^2 \right] \leq 4c^{-1} N^2 \mathbb{E}_\mu[|\nabla\phi|^2].$$

This proves (A.1). $\square$

A.2. Well-posedness for gradient vector fields. The formal adjoint with respect to μ of the derivative ∂_i , $i \in \{1, \dots, N\}$ is given by

$$\partial_i^* w = -\partial_i w + (\partial_i H)w,$$

meaning that for all $v, w \in \mathcal{C}^\infty(D_N)$ such that $\nabla v \cdot \vec{n} = 0$ on ∂D_N ,

$$\mathbb{E}_\mu[(\partial_i v)w] = \mathbb{E}_\mu[v \partial_i^* w]. \quad (\text{A.2})$$

The above formula can be shown by integration by parts under the Lebesgue measure on D_N .

Lemma A.2. *Let μ satisfy Assumption 3.1, let $\mu' := \text{Gap}_N \# \mu$, and let $F \in H^{-1}(\mu)$. Assume that F is of the form $F = G \circ \text{Gap}_N$ with $G \in H^{-1}(\mu')$. Then, there exists a unique, modulo an additive constant, $\phi \in H^1(\mu)$ such that*

$$\mathcal{L}^\mu \phi = F - \mathbb{E}_\mu[F] \quad \text{on } D_N. \quad (\text{A.3})$$

Moreover, the solution ϕ of (A.3) is the unique, modulo additive constant, minimizer of

$$\phi \mapsto \mathbb{E}_\mu[|\nabla \phi|^2 - 2\phi(F - \mathbb{E}_\mu[F])],$$

over functions $\phi \in H^1(\mu)$.

Lemma A.2 is a variation of the Lax–Milgram lemma.

Proof of Lemma A.2. We begin by considering the case where F is a function of the gaps, i.e. $F = G \circ \text{Gap}_N$ with $G \in H^{-1}(\mu')$. Let

$$E := \{\phi \in H^1(\mu) : \phi = \psi \circ \text{Gap}_N \text{ for some } \psi \in H^1(\mu'), \mathbb{E}_\mu[\phi] = 0\}$$

and

$$J : \phi \in E \mapsto \mathbb{E}_\mu[|\nabla \phi|^2] - 2\mathbb{E}_\mu[F\phi]. \quad (\text{A.4})$$

Let us prove that J admits a unique minimizer. First, for all $\phi \in E$,

$$|\mathbb{E}_\mu[F\phi]| \leq \|F\|_{H^{-1}(\mu)} \|\phi\|_{H^1(\mu)}.$$

By Lemma A.1, there exists a constant $C > 0$ such that for all $\phi \in E$,

$$\mathbb{E}_\mu[\phi^2] \leq C\mathbb{E}_\mu[|\nabla \phi|^2],$$

since ϕ is a function of the gaps and since $\mathbb{E}_\mu[\phi] = 0$. Therefore, there exists $C > 0$ such that for all $\phi \in E$,

$$\mathbb{E}_\mu[F\phi] \leq C\|F\|_{H^{-1}(\mu)} \mathbb{E}_\mu[|\nabla \phi|^2]^{\frac{1}{2}}. \quad (\text{A.5})$$

It follows that J is coercive with respect to the $H^1(\mu)$ norm and that J is bounded from below. Let (ϕ_k) be a sequence of elements of E such that $(J(\phi_k))$ converges to $\inf J$. Since (ϕ_k) is bounded in $H^1(\mu)$, there exists a sub-sequence converging weakly to a certain $\phi \in E$. It follows from (A.5) that J is l.s.c on $H^1(\mu)$. Since J is convex, J is l.s.c for the weak topology on $H^1(\mu)$. Therefore ϕ is a minimizer of J on E . The first-order minimality condition for ϕ reads

$$\mathbb{E}_\mu[\nabla \phi \cdot \nabla h] = \mathbb{E}_\mu[Fh],$$

for all $h \in E$. By integration by parts, for every $h \in E$, we have

$$\mathbb{E}_\mu[(\mathcal{L}^\mu \phi - F)h] + \int_{\partial D_N} (\nabla \phi \cdot \vec{n}) h e^{-H} = 0. \quad (\text{A.6})$$

Since $H(x) \rightarrow \infty$ when $d(x, \partial D_N) \rightarrow 0$, we get

$$\mathbb{E}_\mu[(\mathcal{L}^\mu \phi - F)h] = 0.$$

Moreover, since

$$\mathbb{E}_\mu[\mathcal{L}^\mu \phi - F + \mathbb{E}_\mu[F]] = 0,$$

we deduce, using the definition of E , that for all $h = \psi \circ \text{Gap}_N$ with $\psi \in H^1(\mu')$,

$$\mathbb{E}_\mu[(\mathcal{L}^\mu \phi - F + \mathbb{E}_\mu[F])h] = 0. \quad (\text{A.7})$$

Since $\mathcal{L}^\mu \phi - F + \mathbb{E}_\mu[F]$ is a function of the gaps (because ϕ and H are), we deduce that for all $h \in H^1(\mu)$,

$$\begin{aligned} \mathbb{E}_\mu[(\mathcal{L}^\mu \phi - F + \mathbb{E}_\mu[F])h] &= \mathbb{E}_\mu[\mathbb{E}_\mu[(\mathcal{L}^\mu \phi - F + \mathbb{E}_\mu[F])h \mid \text{Gap}_N(X_N)]] \\ &= \mathbb{E}_\mu[(\mathcal{L}^\mu \phi - F + \mathbb{E}_\mu[F])\mathbb{E}_\mu[h \mid \text{Gap}_N(X_N)]]. \end{aligned} \quad (\text{A.8})$$

Since $\mathbb{E}_\mu[h \mid \text{Gap}_N(X_N)]$ is a function of the gaps, we get from (A.7) that

$$\mathbb{E}_\mu[(\mathcal{L}^\mu \phi - F + \mathbb{E}_\mu[F])h] = 0.$$

We conclude that

$$\mathcal{L}^\mu \phi = F - \mathbb{E}_\mu[F],$$

as elements of $H^{-1}(\mu)$.

Let us finally prove uniqueness. Let $\phi \in H^1(\mu)$ solve

$$\mathcal{L}^\mu \phi = F - \mathbb{E}_\mu[F] \quad \text{on } D_N.$$

Since $\phi \in H^1(\mu)$ and since the equality holds in the $H^{-1}(\mu)$ sense, multiplying by ϕ and integrating by parts, we get

$$\mathbb{E}_\mu[|\nabla\phi|^2] = 0.$$

This proves that ϕ is constant a.e., which ensures the uniqueness of the solution of (A.3), modulo an additive constant. $\square$

We now complete the proof of Proposition 3.1.

Proof of Proposition 3.1. Note that if $F \in H^1(\mu)$, then $\nabla F \in L^2(\{1, \dots, N\}, H^{-1}(\mu))$. Indeed

$$\sum_{i=1}^N \|\partial_i F\|_{H^{-1}(\mu)}^2 \leq \sum_{i=1}^N \|\partial_i F\|_{L^2}^2 \leq \|F\|_{H^1(\mu)}^2.$$

By Lemma A.2, there exists $\phi \in H^1(\mu)$ such that $\mathcal{L}^\mu \phi = F - \mathbb{E}_\mu[F]$ as elements of $H^{-1}(\mu)$ and such that $\nabla\phi \cdot \vec{n} = 0$ a.e. on ∂D_N . Let $w \in \mathcal{C}^\infty(D_N)$ such that $\nabla w \cdot \vec{n} = 0$ a.e. on ∂D_N . For every $i \in \{1, \dots, N\}$, we have

$$\begin{aligned} \mathbb{E}_\mu[w \partial_i F] &= \mathbb{E}_\mu[\partial_i^* w (F - \mathbb{E}_\mu[F])] = \mathbb{E}_\mu[\partial_i^* w \mathcal{L}^\mu \phi] = \mathbb{E}_\mu[\nabla(\partial_i^* w) \cdot \nabla \phi] \\ &= \sum_{j=1}^N \mathbb{E}_\mu[(\partial_i^* \partial_j w) \partial_j \phi] + \sum_{j=1}^N \mathbb{E}_\mu[(\partial_j, \partial_i^*) w \partial_j \phi]. \end{aligned}$$

For the first term in the sum above, we have

$$\sum_{j=1}^N \mathbb{E}_\mu[(\partial_i^* \partial_j w) \partial_j \phi] = \sum_{j=1}^N \mathbb{E}_\mu[(\partial_j w) \partial_i \partial_j \phi] = \mathbb{E}_\mu[\nabla w \cdot \nabla(\partial_i \phi)] = \mathbb{E}_\mu[w \mathcal{L}^\mu(\partial_i \phi)].$$

For the second term, using the identity

$$[\partial_j, \partial_i^*] = \partial_j(-\partial_i + (\partial_i H)) - (-\partial_i + (\partial_i H))\partial_j = (\nabla^2 H)_{ij},$$

we get

$$\sum_{j=1}^N \mathbb{E}_\mu[(\partial_j, \partial_i^*) w \partial_j \phi] = \mathbb{E}_\mu[w e_i \cdot \nabla^2 H \nabla \phi].$$

We conclude by density that, for every $i \in \{1, \dots, N\}$,

$$\mathcal{L}^\mu(\partial_i \phi) + (\nabla^2 H \nabla \phi)_i = \partial_i F,$$

holds in the sense of $H^{-1}(\mu)$. This concludes the proof of existence of a solution to (3.6).

Let us prove uniqueness. Suppose that $\nabla\phi \in L^2(\{1, \dots, N\}, H^{-1}(\mu))$ solves (3.6). Then, ϕ solves (A.3) for $F = 0$, which has a unique solution such that $\mathbb{E}_\mu[\phi] = 0$, which is $\phi = 0$. Hence, uniqueness holds.

The formula (3.8) then easily follows from an integration by parts: letting $\nabla\phi$ be the solution of (3.6), we can write

$$\text{Var}_\mu[F] = \mathbb{E}_\mu[(F - \mathbb{E}_\mu[F]) \mathcal{L}^\mu \phi] = \mathbb{E}_\mu[\nabla F \cdot \nabla \phi].$$

The variational representation (3.7) is straightforward using that $\nabla\phi$ is the unique minimizer of the function J defined in (A.4). $\square$

A.3. Well-posedness for non gradient vector fields. We turn to the proof of Proposition 3.2.

Proof of Proposition 3.2. Let $v \in L^2(\{1, \dots, N\}, H^{-1}(\mu))$. Let

$$\begin{aligned} E := \{ \psi \in L^2(\{1, \dots, N\}, H^1(\mu)) : \psi = w \circ \text{Gap}_N \text{ for some } w \in L^2(\{1, \dots, N\}, H^1(\mu')), \\ \psi \cdot (e_1 + \dots + e_N) = 0 \} \end{aligned}$$

and

$$J : \psi \in E \mapsto \mathbb{E}_\mu[\|D\psi\|_F^2 + \psi \cdot \nabla^2 H \psi - 2v \cdot \psi].$$

Let us show that J admits a unique minimizer. First

$$|\mathbb{E}_\mu[v \cdot \psi]| \leq \sum_{i=1}^N \mathbb{E}_\mu[v_i^2]^{\frac{1}{2}} \mathbb{E}_\mu[\psi_i^2]^{\frac{1}{2}}.$$

Denoting $\psi_{N+1} := \psi_1$, since $\psi \cdot (e_1 + \dots + e_N) = 0$, there exists $c > 0$ such that

$$\psi \cdot \nabla^2 H \psi \geq c \sum_{i=1}^N (\psi_{i+1} - \psi_i)^2 \geq \frac{c}{N^2} \sum_{i=1}^N \psi_i^2. \quad (\text{A.9})$$

It follows that J is coercive with respect to the norm $\|\cdot\|_{L^2(\{1, \dots, N\}, H^1(\mu))}$. Arguing as in the proof of Lemma A.2, one can show that J admits a unique minimizer $\psi \in E$. Moreover, the minimizer ψ satisfies, for every $h \in E$,

$$\mathbb{E}_\mu \left[\sum_{i=1}^N \nabla \psi_i \cdot \nabla h_i + \psi \cdot \nabla^2 H h \right] = \mathbb{E}_\mu [v \cdot h].$$

By integration by parts, for every $h \in E$ and every $i = 1, \dots, N$, we have

$$\mathbb{E}_\mu [\nabla \psi_i \cdot \nabla h_i] = \mathbb{E}_\mu [\mathcal{L}^\mu \psi_i h_i] + \int_{\partial D_N} (\nabla \psi_i \cdot \vec{n}) h_i e^{-H} = \mathbb{E}_\mu [\mathcal{L}^\mu \psi_i h_i],$$

since for every i, j such that $i \neq j$, $\lim_{x \rightarrow 0} \chi_{ij}(x) = +\infty$. Therefore, for every $h \in E$,

$$\mathbb{E}_\mu [(A_1^\mu \psi - v)h] = 0. \quad (\text{A.10})$$

Since $(A_1^\mu \psi - v) \cdot (e_1 + \dots + e_N) = 0$, (A.10) holds for every $h \in L^2(\{1, \dots, N\}, H^1(\mu))$ such that for every i , h_i is a function of the gaps. Finally, since $A_1^\mu \psi - v$ is a function of the gaps, we get by proceeding as in (A.8) that (A.10) holds for any $h \in L^2(\{1, \dots, N\}, H^1(\mu))$. We conclude that ψ satisfies the Euler-Lagrange equation

$$\begin{cases} A_1^\mu \psi = v & \text{on } D_N \\ \psi \cdot (e_1 + \dots + e_N) = 0 & \text{on } D_N, \end{cases} \quad (\text{A.11})$$

where the first equality is an equality between elements of $L^2(\{1, \dots, N\}, H^{-1}(\mu))$.

We now turn to the proof of uniqueness. Let $\psi \in L^2(\{1, \dots, N\}, H^1(\mu))$ satisfy

$$\begin{cases} A_1^\mu \psi = 0 & \text{on } D_N \\ \psi \cdot (e_1 + \dots + e_N) = 0 & \text{on } D_N. \end{cases} \quad (\text{A.12})$$

Taking the scalar product of the equation $A_1^\mu \psi = 0$ with ψ and integrating by parts gives

$$\mathbb{E}_\mu \left[\sum_{i=1}^N |\nabla \psi_i|^2 \right] + \mathbb{E}_\mu [\psi \cdot \nabla^2 H \psi] = 0.$$

In particular,

$$\mathbb{E}_\mu [\psi \cdot \nabla^2 H \psi] = 0.$$

The above display and (A.9) imply that $\psi = 0$, which proves uniqueness. $\square$

APPENDIX B. AUXILIARY ESTIMATES

B.1. Proof of Lemma 5.6.

Lemma B.1. *Let*

$$\tilde{g} : x \in \mathbb{T} \setminus \{0\} \mapsto g'(x)x$$

and let $\psi \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$. Set

$$\zeta_0 : (x, y) \in \ell_N^{-1}(\mathbb{T} \setminus \text{diag}_{\mathbb{T}^2})^2 \mapsto \frac{\ell_N(\psi(\ell_N^{-1}y) - \psi(\ell_N^{-1}x))}{y - x}.$$

Recall t_γ and I_γ from (5.49) and (5.50). Recall $\text{Loop}_{\ell_N, \gamma}[\psi]$ from (5.51). Let V and $W \in L^2(\{1, \dots, N\}, \mathcal{C}^\infty(D_N))$ be given by $V_i = W_i = 0$ for every $i \notin I_\gamma$ and for every $i \in I_\gamma$,

$$\begin{aligned} V_i := & 2 \sum_{k: i+k \in I_\gamma} \partial_1 \zeta_0(x_i, x_{i+k}) N^{-s} \tilde{g}(x_{i+k} - x_i) - 2N \int_{|y| \leq t_\gamma} \partial_1 \zeta_0(x_i, y) N^{-s} \tilde{g}(y - x_i) dy \\ & + 2 \sum_{k: i+k \in I_\gamma} \zeta_0(x_i, x_{i+k}) N^{-s} \tilde{g}'(\frac{k}{N}) - 2N \int_{|y| \leq t_\gamma} \zeta_0(x_i, y) N^{-s} \tilde{g}'(y - x_i) dy, \end{aligned} \quad (\text{B.1})$$

$$W_i := 2 \sum_{k: i+k \in I_\gamma} \zeta_0(x_i, x_{i+k}) N^{-s} (\tilde{g}'(x_{i+k} - x_i) - \tilde{g}'(\frac{k}{N})). \quad (\text{B.2})$$

Then, we have

$$\nabla \text{Loop}_{\ell_N, \gamma}[\psi] = V + W.$$

Proof. For all $x, y \in \mathbb{T}$, $x \neq y$, we have

$$N \ell_N(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y)) N^{-(s+1)} g'(x - y) = \zeta_0(x, y) N^{-s} \tilde{g}(x - y).$$

Therefore,

$$\text{Loop}_{\ell_N, \gamma}[\psi] = \sum_{i \neq j \in I_\gamma} \zeta_0(x_i, x_j) N^{-s} \tilde{g}(x_i - x_j) - 2N \sum_{i \in I_\gamma} \int_{|y| \leq t_\gamma} \zeta_0(x_i, y) N^{-s} \tilde{g}(x_i - y) dy.$$

Thus, we can decompose the gradient of $\text{Loop}_{\ell_N, \gamma}[\psi]$ into $\nabla \text{Loop}_{\ell_N, \gamma}[\psi] = V + W$ with V and W as in (B.1) and (B.2). $\square$

Lemma B.2. Let $\psi \in C^\infty(\ell_N^{-1}\mathbb{T})$. Let $\alpha \in [s - 1, \frac{s}{2})$. Let

$$\zeta_0 : (x, y) \in \ell_N^{-1}(\mathbb{T} \setminus \text{diag}_{\mathbb{T}^2})^2 \mapsto \frac{\ell_N(\psi(\ell_N^{-1}y) - \psi(\ell_N^{-1}x))}{y - x}.$$

Let $w_N^{(1)}$ and $w_N^{(2)}$ be as in (5.53) and (5.54). Suppose that

$$|\psi'| \leq C w_N^{(1)} \quad \text{and} \quad |\psi''| \leq C w_N^{(2)}.$$

Then, for all $x, y \in \mathbb{T}$, we have

$$|\zeta_0(x, y)| \leq C(w_N^{(1)}(\ell_N^{-1}x) + w_N^{(1)}(\ell_N^{-1}y)), \quad (\text{B.3})$$

$$|\partial_x \zeta_0(x, y)| \leq C \ell_N^{-1}(w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}y)), \quad (\text{B.4})$$

$$|\partial_y \zeta_0(x, y)| \leq C \ell_N^{-1}(w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}y)) \quad (\text{B.5})$$

$$|\partial_{xy} \zeta_0(x, y)| \leq \frac{C \ell_N^{-1}}{|y - x|} (w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}y)). \quad (\text{B.6})$$

For all $x, y, a \in \mathbb{T}$, we have

$$\begin{aligned} & |\partial_x \zeta_0(x, y + a) - \partial_x \zeta_0(x, y)| \\ & \leq \frac{C \ell_N^{-1} |a|}{\min(|x - y|, |x - y - a|)} (w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}y) + w_N^{(2)}(\ell_N^{-1}(y + a))). \end{aligned} \quad (\text{B.7})$$

Proof. One can notice that

$$\zeta_0(x, y) = \int_0^1 \psi'(\ell_N^{-1}((1 - \theta)x + \theta y)) d\theta. \quad (\text{B.8})$$

Therefore, since $1 - s + \alpha < 1$, we get (B.3).

Let us now prove (B.4). We have

$$\partial_x \zeta_0(x, y) = \frac{-(y - x)\psi'(\ell_N^{-1}x) + \ell_N(\psi(\ell_N^{-1}y) - \psi(\ell_N^{-1}x))}{(y - x)^2}.$$

Suppose that $x \in (-\frac{1}{4}, 0)$ and $y \in (0, \frac{1}{4})$. We use

$$|\partial_x \zeta_0(x, y)| \leq \frac{|\psi'(\ell_N^{-1}x)|}{|y - x|} + \frac{\ell_N |\psi(\ell_N^{-1}y) - \psi(\ell_N^{-1}x)|}{(y - x)^2}.$$

Since $|y - x| \geq |x|$, we get

$$\frac{|\psi'(\ell_N^{-1}x)|}{|y - x|} \leq \frac{|\psi'(\ell_N^{-1}x)|}{|x|} \leq C \ell_N^{-1} w_N^{(2)}(\ell_N^{-1}x).$$

Moreover, using $|y - x| \geq \max(|x|, |y|)$ and (B.3), we get

$$\frac{\ell_N |\psi(\ell_N^{-1}y) - \psi(\ell_N^{-1}x)|}{(y - x)^2} \leq C \left(\frac{w_N^{(1)}(\ell_N^{-1}x)}{|x|} + \frac{w_N^{(1)}(\ell_N^{-1}y)}{|y|} \right) \leq C \ell_N^{-1} (w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}y)).$$

Combining the two above displays shows (B.4) in the case $x < 0$ and $y > 0$.

Now assume that $x \in (0, \frac{1}{4})$ and $y \in (0, \frac{1}{4})$. Then by (B.8),

$$\partial_x \zeta_0(x, y) = -\ell_N^{-1} \int_0^1 \psi''(\ell_N^{-1}((1-\theta)x + \theta y)) \theta d\theta.$$

Hence,

$$|\partial_x \zeta_0(x, y)| \leq \ell_N^{-1} \int_0^1 |\psi''(\ell_N^{-1}((1-\theta)x + \theta y))| \theta d\theta.$$

Since $0 \notin [x, y]$, we get (B.4). The proof of (B.5) is similar.

We finally prove (B.7). We compute

$$\partial_{xy} \zeta_0(x, y) = -\frac{\psi'(\ell_N^{-1}y) - \psi'(\ell_N^{-1}x)}{(y-x)^2} - 2\frac{-(y-x)\psi'(\ell_N^{-1}x) + \ell_N(\psi(\ell_N^{-1}y) - \psi(\ell_N^{-1}x))}{(y-x)^3}.$$

Proceeding as above, we get

$$|\partial_{xy} \zeta_0(x, y)| \leq \frac{C\ell_N^{-1}}{|y-x|} (w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}y)),$$

which gives (B.7). $\square$

Lemma B.3. *Let $\psi \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$ satisfy the assumption of Lemma B.2. Let $V \in L^2(\{1, \dots, N\}, \mathcal{C}^\infty(D_N))$ be as in (B.1). Let $\mathcal{A}_{\varepsilon, \gamma}$ be the event defined in (5.52). There exists a constant $\kappa > 0$ depending on s such that for every $i \in I_\gamma$, we have*

$$|V_i| \leq (N\ell_N)^{\kappa\varepsilon} \left(\ell_N^{-1}(w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}(x_i - t_\gamma)) + w_N^{(2)}(\ell_N^{-1}(x_i + t_\gamma)) \right. \\ \left. + (N\ell_N)^{1-s+\alpha} N^{-\frac{s}{2}} \frac{1}{\max(|x_i|, \frac{1}{N})^{1+\frac{s}{2}}} \mathbf{1}_{|x_i| \leq 2\ell_N} \right).$$

Proof. We assume that $\ell_N \rightarrow 0$. The case $\ell_N \equiv 1$ is similar and easier. Recall the event $\mathcal{A}_{\varepsilon, \gamma}$ defined in (5.52). Throughout this proof, we work on the event $\mathcal{A}_{\varepsilon, \gamma}$, and all bounds below are understood to hold on this event.

Step 1: splitting of V . We can split V into $V = V^{(1)} + V^{(2)}$, where for every $i \in I_\gamma^c$, $V_i^{(1)} = V_i^{(2)} = 0$, and for every $i \in I_\gamma$,

$$V_i^{(1)} := 2 \sum_{k: i+k \in I_\gamma} \partial_x \zeta_0(x_i, x_{i+k}) N^{-s} \tilde{g}(x_{i+k} - x_i) - 2 \sum_{k: i+k \in I_\gamma} \partial_x \zeta_0(x_i, x_i + \frac{k}{N}) N^{-s} \tilde{g}(\frac{k}{N}) \\ + 2 \sum_{k: i+k \in I_\gamma} (\zeta_0(x_i, x_{i+k}) - \zeta_0(x_i, x_i + \frac{k}{N})) N^{-s} \tilde{g}'(\frac{k}{N}), \\ V_i^{(2)} := 2 \sum_{k: i+k \in I_\gamma} (\partial_x \zeta_0(x_i, \frac{k}{N}) \tilde{g}(\frac{k}{N}) + \zeta_0(x_i, \frac{k}{N}) \tilde{g}'(\frac{k}{N})) \\ - 2N \int_{|y| \leq \frac{(N\ell_N)^\gamma}{N}} (\partial_x \zeta_0(x_i, y) \tilde{g}(y) + \zeta_0(x_i, y) \tilde{g}'(y)) dy. \quad (\text{B.9})$$

Step 2: control on $V^{(1)}$. For every $i \in I_\gamma$, one may write

$$V_i^{(1)} = 2 \sum_{k: i+k \in I_\gamma} \partial_x \zeta_0(x_i, x_{i+k}) N^{-s} (\tilde{g}(x_{i+k} - x_i) - \tilde{g}(\frac{k}{N})) \\ + 2 \sum_{k: i+k \in I_\gamma} (\partial_x \zeta_0(x_i, x_{i+k}) - \partial_x \zeta_0(x_i, x_i + \frac{k}{N})) N^{-s} \tilde{g}(\frac{k}{N}) \\ + 2 \sum_{k: i+k \in I_\gamma} (\zeta_0(x_i, x_{i+k}) - \zeta_0(x_i, x_i + \frac{k}{N})) N^{-s} \tilde{g}'(\frac{k}{N}). \quad (\text{B.10})$$

By Lemma B.2,

$$|\partial_x \zeta_0(x_i, x_{i+k})| \leq C\ell_N^{-1} (w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}(x_{i+k}))),$$

$$|\partial_x \zeta_0(x_i, x_{i+k}) - \partial_x \zeta_0(x_i, x_i + \frac{k}{N})| \leq \frac{C\ell_N^{-1}}{\min(\frac{k}{N}, x_{i+k} - x_i)} (w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}(x_{i+k}))).$$

$$|\partial_x \zeta_0(x_i, x_{i+k}) - \partial_x \zeta_0(x_i, x_i + \frac{k}{N})| \leq |x_{i+k} - x_i - \frac{k}{N}| C\ell_N^{-1} (w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}(x_{i+k}))).$$

Therefore, we conclude that there exists a constant $\kappa > 0$ such that for every $i \in I_\gamma$,

$$|V_i^{(1)}| \leq \ell_N^{-1} (N\ell_N)^{\kappa\varepsilon} \sum_{j \in I_\gamma: j \neq i} (w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}x_j) + w_N^{(2)}(\ell_N^{-1}(x_i + \frac{j-i}{N}))) \frac{1}{d(i, j)^{1+\frac{s}{2}}}.$$

Notice that for all $\gamma \in \mathbb{R}$,

$$\frac{\ell_N^{-1}}{\max(\ell_N^{-1}x, \frac{1}{N\ell_N})^\gamma} = \frac{\ell_N^{-1}(N\ell_N)^\gamma}{\max(Nx, 1)^\gamma}.$$

Therefore,

$$|V_i^{(1)}| \leq (N\ell_N)^{\kappa\varepsilon} (A_i + B_i) \quad (\text{B.11})$$

with

$$A_i := \ell_N^{-1} (N\ell_N)^{2-s+\alpha} \sum_{j \in I_\gamma: j \neq i} \frac{1}{|j|^{2-s+\alpha}} \frac{1}{|i-j|^{1+\frac{s}{2}}} \mathbf{1}_{|j| \leq N\ell_N},$$

$$B_i := \ell_N^{-1} (N\ell_N)^{3-s} \sum_{j \in I_\gamma: j \neq i} \frac{1}{|j|^{3-s}} \frac{1}{|i-j|^{1+\frac{s}{2}}} \mathbf{1}_{|j| > N\ell_N}.$$

Step 3: control on A_i and B_i . Suppose that $|i| \leq 2N\ell_N$. Since $2-s+\alpha > 1$, we have

$$|A_i| \leq \ell_N^{-1} (N\ell_N)^{2-s+\alpha} \sum_{j \in I_\gamma: j \neq i} \frac{1}{|j|^{2-s+\alpha}} \frac{1}{|j-i|^{1+\frac{s}{2}}} \leq C\ell_N^{-1} (N\ell_N)^{2-s+\alpha} \left(\frac{1}{|i|^{2-s+\alpha}} + \frac{1}{|i|^{1+\frac{s}{2}}} \right). \quad (\text{B.12})$$

Moreover, we have

$$|B_i| \leq C\ell_N^{-1} (N\ell_N)^{3-s} \sum_j \frac{1}{(N\ell_N)^{3-s}} \frac{1}{|i-j|^{1+\frac{s}{2}}} \leq C\ell_N^{-1}. \quad (\text{B.13})$$

Therefore, if $|i| \leq 2N\ell_N$, then by combining (B.11), (B.12) and (B.13), we get

$$|V_i^{(1)}| \leq C\ell_N^{-1} (N\ell_N)^{2-s+\alpha+\kappa\varepsilon} \left(\frac{1}{|i|^{2-s+\alpha}} + \frac{1}{|i|^{1+\frac{s}{2}}} \right).$$

Suppose that $|i| > 2N\ell_N$. Then, we have

$$|A_i| \leq C\ell_N^{-1} (N\ell_N)^{2-s+\alpha} \frac{1}{|i|^{1+\frac{s}{2}}}. \quad (\text{B.14})$$

Moreover, we have

$$|B_i| \leq C\ell_N^{-1} (N\ell_N)^{3-s} \sum_{j: |j| \leq \frac{i}{2}} \frac{1}{|j|^{3-s}} \frac{1}{|i-j|^{1+\frac{s}{2}}} \mathbf{1}_{|j| \geq N\ell_N} + C\ell_N^{-1} (N\ell_N)^{3-s} \sum_{j: |j| > \frac{i}{2}} \frac{1}{|j|^{3-s}} \frac{1}{|i-j|^{1+\frac{s}{2}}} \mathbf{1}_{|j| \geq N\ell_N}.$$

The first term is bounded by

$$\ell_N^{-1} (N\ell_N)^{3-s} \sum_{j: |j| \leq \frac{i}{2}} \frac{1}{|j|^{3-s}} \frac{1}{|i-j|^{1+\frac{s}{2}}} \mathbf{1}_{|j| \geq N\ell_N} \leq C\ell_N^{-1} (N\ell_N)^{3-s} \frac{1}{|i|^{1+\frac{s}{2}}} \frac{1}{(N\ell_N)^{2-s}}.$$

For the second term, we have

$$C\ell_N^{-1} (N\ell_N)^{3-s} \sum_{j: |j| > \frac{i}{2}} \frac{1}{|j|^{3-s}} \frac{1}{|i-j|^{1+\frac{s}{2}}} \mathbf{1}_{|j| \geq N\ell_N} \leq C\ell_N^{-1} (N\ell_N)^{3-s} \frac{1}{|i|^{3-s}}.$$

Combining the above displays gives

$$|B_i| \leq C\ell_N^{-1} \left(\frac{N\ell_N}{|i|^{1+\frac{s}{2}}} + \frac{(N\ell_N)^{3-s}}{|i|^{3-s}} \right). \quad (\text{B.15})$$

Hence, when $|i| > 2N\ell_N$, then by combining (B.11), (B.14) and (B.15) and using that $2 - s + \alpha \geq 1$, we get

$$|V_i^{(1)}| \leq C\ell_N^{-1} (N\ell_N)^{\kappa\varepsilon} \left(\frac{(N\ell_N)^{2-s+\alpha}}{|i|^{2-s+\alpha}} + \frac{(N\ell_N)^{3-s}}{|i|^{3-s}} \right).$$

We conclude that

$$|V_i^{(1)}| \leq C(N\ell_N)^{\kappa\varepsilon} \left(\ell_N^{-1} w_N^{(2)}(\ell_N^{-1} x_i) + (N\ell_N)^{1-s+\alpha} N^{-\frac{s}{2}} \frac{1}{\max(|x_i|, \frac{1}{N})^{1+\frac{s}{2}}} \mathbf{1}_{|x_i| \leq 2\ell_N} \right). \quad (\text{B.16})$$

Step 4: control on $V^{(2)}$. Fix $x \in \mathbb{T}$ and define

$$\phi_x : y \in N\mathbb{T} \mapsto \zeta_0(x, x + \frac{y}{N}) N^{-s} \tilde{g}'(\frac{y}{N}) + \partial_1 \zeta_0(x, x + \frac{y}{N}) N^{-s} \tilde{g}(\frac{y}{N}).$$

Recalling the definition (B.9) of $V^{(2)}$, we have

$$V_i^{(2)} = \sum_{\substack{k \neq 0: \\ -Nt_\gamma \leq i+k \leq Nt_\gamma}} \phi_x(k).$$

By the Euler-Maclaurin formula,

$$\begin{aligned} \sum_{\substack{k \neq 0: \\ -Nt_\gamma \leq i+k \leq Nt_\gamma}} \phi_x(k) &= \int_{[-t_\gamma-i, t_\gamma-i] \setminus [-1, 1]} \phi_x(y) dy \\ &+ \frac{1}{2} (\phi_x(Nt_\gamma - i) + \phi_x(-Nt_\gamma - i) - \phi_x(1) - \phi_x(-1)) + O \left(\int_{-\frac{N}{2}}^{-1} |\phi'_x(y)| dy + \int_1^{\frac{N}{2}} |\phi'_x(y)| dy \right). \end{aligned}$$

We compute

$$\begin{aligned} \phi_x(1) + \phi_x(-1) &= \left(\zeta_0(x, x + \frac{1}{N}) - \zeta_0(x, x - \frac{1}{N}) \right) N^{-s} \tilde{g}'(\frac{1}{N}) \\ &+ \left(\partial_1 \zeta_0(x, x + \frac{1}{N}) + \partial_1 \zeta_0(x, x - \frac{1}{N}) \right) N^{-s} \tilde{g}(\frac{1}{N}). \end{aligned}$$

By Lemma B.2,

$$|\phi_x(1) + \phi_x(-1)| \leq C\ell_N^{-1} w_N^{(2)}(\ell_N^{-1} x). \quad (\text{B.17})$$

We compute

$$\phi'_x(y) = N^{-(s+1)} \left(\zeta_0(x, x + \frac{y}{N}) \tilde{g}''(\frac{y}{N}) + (\partial_x + \partial_y) \zeta_0(x, x + \frac{y}{N}) \tilde{g}'(\frac{y}{N}) + \partial_{xy} \zeta_0(x, x + \frac{y}{N}) \tilde{g}(\frac{y}{N}) \right).$$

By using Lemma B.2, one can check that

$$\int_1^{\frac{N}{2}} |\phi'_x(y)| dy \leq C\ell_N^{-1} \int_1^{\frac{N}{2}} \left(w_N^{(2)}(\ell_N^{-1} x) + w_N^{(2)} \left(\ell_N^{-1} \left(x + \frac{y}{N} \right) \right) \right) \frac{1}{y^{1+s}} dy.$$

After some computations, we get

$$\int_1^{\frac{N}{2}} |\phi'_x(y)| dy \leq C \left(\ell_N^{-1} w_N^{(2)}(\ell_N^{-1} x_i) + (N\ell_N)^{1-s+\alpha} N^{-\frac{s}{2}} \frac{1}{\max(|x_i|, \frac{1}{N})^{1+\frac{s}{2}}} \mathbf{1}_{|x_i| \leq 2\ell_N} \right).$$

It follows that

$$\begin{aligned} |V_i^{(2)}| &\leq C \left(\ell_N^{-1} w_N^{(2)}(\ell_N^{-1} x_i) + (N\ell_N)^{1-s+\alpha} N^{-\frac{s}{2}} \frac{1}{\max(|x_i|, \frac{1}{N})^{1+\frac{s}{2}}} \mathbf{1}_{|x_i| \leq 2\ell_N} \right) \\ &+ \frac{1}{2} (|\phi_x(Nt_\gamma - i)| + |\phi_x(-Nt_\gamma - i)|). \quad (\text{B.18}) \end{aligned}$$

It remains to bound the second term in the above display. We have

$$\phi_x(Nt_\gamma - i) = \zeta_0(x, x + t_\gamma - \frac{i}{N}) N^{-s} \tilde{g}'(t_\gamma - \frac{i}{N}) + \partial_1 \zeta_0(x, x + t_\gamma - \frac{i}{N}) N^{-s} \tilde{g}(t_\gamma - \frac{i}{N}). \quad (\text{B.19})$$

By Lemma B.2,

$$|\zeta_0(x, x + t_\gamma - \frac{i}{N})| \leq C \left(w_N^{(1)}(\ell_N^{-1}x) + w_N^{(1)}(\ell_N^{-1}(x + t_\gamma - \frac{i}{N})) \right).$$

It follows that

$$|\zeta_0(x, x + t_\gamma - \frac{i}{N}) N^{-s} \tilde{g}'(t_\gamma - \frac{i}{N})| \leq \frac{C \ell_N^{-1}}{|N t_\gamma - i|^{1+s}} \left(w_N^{(1)}(\ell_N^{-1}x) + w_N^{(1)}(\ell_N^{-1}(x + t_\gamma - \frac{i}{N})) \right).$$

Therefore, by the definition of $\mathcal{A}_{\varepsilon, \gamma}$, there exists $\kappa > 0$ such that

$$\begin{aligned} |\zeta_0(x_i, x_i + t_\gamma - \frac{i}{N}) N^{-s} \tilde{g}'(t_\gamma - \frac{i}{N})| &\leq \frac{C \ell_N^{-1} (N \ell_N)^{\kappa \varepsilon}}{|N t_\gamma - i|^{1+s}} \left(w_N^{(1)}(\ell_N^{-1}x_i) + w_N^{(1)}(\ell_N^{-1}t_\gamma) \right) \\ &\leq 2 \frac{C \ell_N^{-1} (N \ell_N)^{\kappa \varepsilon}}{|N t_\gamma - i|^{1+s}} w_N^{(1)}(\ell_N^{-1}x_i) \\ &\leq 2 \frac{C \ell_N^{-1} (N \ell_N)^{\kappa' \varepsilon}}{|N(x_i - t_\gamma)|^{1+s}} w_N^{(1)}(\ell_N^{-1}x_i). \end{aligned}$$

One can notice that if $|N(x_i - t_\gamma)| \geq N \ell_N$, then

$$\frac{1}{|N(x_i - t_\gamma)|} w_N^{(1)}(\ell_N^{-1}x_i) \leq (N \ell_N)^{\kappa \varepsilon} w_N^{(2)}(\ell_N^{-1}x_i).$$

Now, if $|N(x_i + t_\gamma)| \leq N \ell_N$, then

$$\ell_N^{-1} \frac{1}{|N(x_i + t_\gamma)|} w_N^{(1)}(\ell_N^{-1}x_i) \leq (N \ell_N)^{\kappa \varepsilon} \ell_N^{-1} (N \ell_N)^{(2-s)(1-\gamma)}.$$

Therefore, taking γ large enough,

$$\ell_N^{-1} \frac{1}{|N(x_i - t_\gamma)|} w_N^{(1)}(\ell_N^{-1}x_i) \leq (N \ell_N)^{\kappa \varepsilon} \ell_N^{-1} w_N^{(2)}(\ell_N^{-1}(x_i + t_\gamma)).$$

Combining the two above displays shows that

$$|\zeta_0(x_i, x_i + t_\gamma - \frac{i}{N}) N^{-s} \tilde{g}'(t_\gamma - \frac{i}{N})| \leq (N \ell_N)^{\kappa \varepsilon} \ell_N^{-1} \left(w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}(x_i + t_\gamma)) \right). \quad (\text{B.20})$$

For the second term in (B.19),

$$|\partial_x \zeta_0(x, x + t_\gamma - \frac{i}{N})| \leq C \ell_N^{-1} \left(w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}(x + t_\gamma - \frac{i}{N})) \right).$$

Therefore,

$$|\partial_x \zeta_0(x, x + t_\gamma - \frac{i}{N}) N^{-s} \tilde{g}(t_\gamma - \frac{i}{N})| \leq \frac{C \ell_N^{-1}}{|N t_\gamma - i|^s} \left(w_N^{(2)}(\ell_N^{-1}x) + w_N^{(2)}(\ell_N^{-1}(x + t_\gamma - \frac{i}{N})) \right).$$

Hence, by the definition of $\mathcal{A}_{\varepsilon, \gamma}$, there exists $\kappa > 0$ such that

$$\begin{aligned} |\partial_x \zeta_0(x_i, x_i + t_\gamma - \frac{i}{N}) N^{-s} \tilde{g}(t_\gamma - \frac{i}{N})| &\leq \frac{(N \ell_N)^{\kappa \varepsilon} \ell_N^{-1}}{|N t_\gamma - i|^s} \left(w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}t_\gamma) \right) \\ &\leq (N \ell_N)^{\kappa \varepsilon} \ell_N^{-1} \left(w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}t_\gamma) \right) \\ &\leq C (N \ell_N)^{\kappa \varepsilon} \ell_N^{-1} w_N^{(2)}(\ell_N^{-1}x_i). \end{aligned} \quad (\text{B.21})$$

We conclude by combining (B.20), (B.21), (B.19) and (B.18) that

$$\begin{aligned} |V_i^{(2)}| &\leq (N \ell_N)^{\kappa \varepsilon} \left(\ell_N^{-1} \left(w_N^{(2)}(\ell_N^{-1}x_i) + w_N^{(2)}(\ell_N^{-1}(x_i + t_\gamma) + w_N^{(2)}(\ell_N^{-1}(x_i - t_\gamma)) \right) \right. \\ &\quad \left. + (N \ell_N)^{1-s+\alpha} N^{-\frac{s}{2}} \frac{1}{\max(|x_i|, \frac{1}{N})^{1+\frac{s}{2}}} \mathbb{1}_{|x_i| \leq 2\ell_N} \right). \end{aligned} \quad (\text{B.22})$$

Step 5: conclusion. Combining (B.16) and (B.22) concludes the proof. $\square$

Lemma B.4. *Let $\psi \in \mathcal{C}^\infty(\ell_N^{-1}\mathbb{T})$ satisfy the assumption of Lemma B.2. Let $W \in L^2(\{1, \dots, N\}, \mathcal{C}^\infty(D_N))$ be as in (B.2). Let $\mathcal{A}_{\varepsilon, \gamma}$ be the event defined in (5.52). For all $U_N \in \mathbb{R}^N$, we have*

$$W \cdot U_N = - \sum_{i=1}^N W_i^{\text{gap}} N(u_{i+1} - u_i),$$

where W^{gap} is such that there exists $\kappa > 0$ depending on s such that

$$\sup_{\mathcal{A}_{\varepsilon, \gamma}} |W^{\text{gap}}|^2 \leq (N\ell_N)^{\kappa\varepsilon} \left(N\ell_N + (N\ell_N)^{2(\alpha+1-s)} \right). \quad (\text{B.23})$$

Proof. Let $W^{\text{gap}} \in L^2(\{1, \dots, N\}, \mathcal{C}^\infty(D_N))$ be the vector field given for every $i \in I_\gamma$ by

$$W_i^{\text{gap}} := N^{-(1+s)} \sum_{k=1}^{N-1} \left(\delta_{k \neq N/2} + \frac{1}{2} \delta_{k=N/2} \right) \sum_{\substack{l \in I_\gamma \\ l+k \in I_\gamma \\ i-k < l \leq i}} \zeta_0(x_{l+k}, x_l) \left(\tilde{g}'(x_{l+k} - x_l) - \tilde{g}'\left(\frac{k}{N}\right) \right)$$

and $W_i^{\text{gap}} = 0$ for every $i \notin I_\gamma$. For all $U_N \in \mathbb{R}^N$, we have

$$W \cdot U_N = - \sum_{i=1}^N W_i^{\text{gap}} N(u_{i+1} - u_i).$$

Let us now bound $|W^{\text{gap}}|$ on the good event $\mathcal{A}_{\varepsilon, \gamma}$. Throughout this proof, we work on the event $\mathcal{A}_{\varepsilon, \gamma}$, and all bounds below are understood to hold on this event.

By the definition of $\mathcal{A}_{\varepsilon, \gamma}$, there exists a constant $\kappa > 0$ depending on s such that for every $l, l+k \in I_\gamma$,

$$N^{-(1+s)} |\tilde{g}'(x_{l+k} - x_k) - \tilde{g}'\left(\frac{k}{N}\right)| \leq \frac{C(N\ell_N)^{\kappa\varepsilon}}{|k|^{2+\frac{s}{2}}}. \quad (\text{B.24})$$

By Lemma B.2, we have

$$|\zeta_0(x_{l+k}, x_l)| \leq C(w_N^{(1)}(\ell_N^{-1}x_{l+k}) + w_N^{(1)}(\ell_N^{-1}x_l)).$$

Therefore,

$$|W_i^{\text{gap}}| \leq (N\ell_N)^{\kappa\varepsilon} \sum_{k=1}^{N-1} \frac{1}{k^{2+\frac{s}{2}}} \sum_{l=i-k+1}^i \left(|w_N^{(1)}(\ell_N^{-1}x_l)| + |w_N^{(1)}(\ell_N^{-1}x_{l+k})| \right).$$

Reindexing the sum gives

$$|W_i^{\text{gap}}| \leq (N\ell_N)^{\kappa\varepsilon} \sum_{j=1}^N |w_N^{(1)}(\ell_N^{-1}x_j)| \frac{1}{d(j, i)^{1+\frac{s}{2}}}. \quad (\text{B.25})$$

On the event $\mathcal{A}_{\varepsilon, \gamma}$, we have

$$|W_i^{\text{gap}}| \leq (N\ell_N)^{\kappa\varepsilon} (A_i + B_i) \quad (\text{B.26})$$

where

$$A_i := (N\ell_N)^{1-s+\alpha} \sum_j \frac{1}{|j|^{1-s+\alpha}} \mathbb{1}_{|j| \leq N\ell_N} \frac{1}{|j-i|^{1+\frac{s}{2}}},$$

$$B_i := (N\ell_N)^{2-s} \sum_j \frac{1}{|j|^{2-s}} \mathbb{1}_{|j| \geq N\ell_N} \frac{1}{|j-i|^{1+\frac{s}{2}}}.$$

Suppose that $|i| \leq 2N\ell_N$. Since $1-s+\alpha < 1$, we have

$$|A_i| \leq C(N\ell_N)^{1-s+\alpha} \left(\frac{1}{|i|^{1-s+\alpha}} + \frac{1}{|i|^{1+\frac{s}{2}-s+\alpha}} \right) \leq 2C(N\ell_N)^{1-s+\alpha} \frac{1}{|i|^{1-s+\alpha}}.$$

Moreover, we have $|B_i| \leq C$.

Suppose that $|i| > 2N\ell_N$. Since $1-s+\alpha < 1$, we have

$$|A_i| \leq C \frac{N\ell_N}{|i|^{1+\frac{s}{2}}} \quad \text{and} \quad |B_i| \leq C \frac{N\ell_N}{|i|^{1+\frac{s}{2}}}.$$

Therefore,

$$\sum_{i:|i|\leq N\ell_N} |A_i|^2 \leq C \left(N\ell_N \mathbb{1}_{1-s+\alpha<\frac{1}{2}} + N\ell_N \log(N\ell_N) \mathbb{1}_{1-s+\alpha=\frac{1}{2}} + (N\ell_N)^{2(1-s+\alpha)} \mathbb{1}_{1-s+\alpha>\frac{1}{2}} \right)$$

and

$$\sum_{i:|i|\geq N\ell_N} |B_i|^2 \leq CN\ell_N.$$

Inserting the two above displays into (B.26) concludes the proof. $\square$

Combining Lemma B.1, Lemma B.3 and Lemma B.4 concludes the proof of Lemma 5.6.

B.2. Additional useful estimates.

Lemma B.5. *Let $\psi \in C^\infty(\ell_N^{-1}\mathbb{T})$ be as in Lemma B.3. Let $B_{\ell_N}[\psi]$ be as in (5.45). For all $\varepsilon > 0$, there exists a constant $C > 0$ depending on ε , s and β such that*

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[B_{\ell_N}[\psi]] \leq C(N\ell_N)^\varepsilon (N\ell_N + (N\ell_N)^{2(1-s+\alpha)}). \quad (\text{B.27})$$

Proof. Let $w_N^{(1)}$ be as in (5.53). Let us recall that

$$B_{\ell_N}[\psi] = \iint_{\text{Diag}_{\mathbb{T}^2}^c} N^{-(s+2)} g''(x-y) (N\ell_N)^2 (\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))^2 d\text{fluct}_N(x) d\text{fluct}_N(y).$$

Denote

$$\Phi : (x, y) \in (\ell_N^{-1}\mathbb{T})^2 \setminus \text{Diag}_{\mathbb{T}^2} \mapsto N^{-(s+2)} g''(x-y) (N\ell_N)^2 (\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y))^2,$$

so that

$$B_{\ell_N}[\psi] = \iint_{\text{Diag}_{\mathbb{T}^2}^c} \Phi(x, y) d\text{fluct}_N(x) d\text{fluct}_N(y).$$

Notice that

$$B_{\ell_N}[\psi] = - \iint \Phi(x, y) (N dx) d\text{fluct}_N(y) + \iint_{\text{Diag}_{\mathbb{T}^2}^c} \Phi(x, y) d(N d\mu_N)(x) d\text{fluct}_N(y).$$

Since the first marginal x_1 of $\mathbb{P}_{N,\beta}$ is uniformly distributed on $\mathbb{T}$, we have

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{N,\beta}}[B_{\ell_N}[\psi]] &= \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\iint_{\text{Diag}_{\mathbb{T}^2}^c} \Phi(x, y) d(N d\mu_N)(x) d\text{fluct}_N(y) \right] \\ &= \sum_{i=1}^N \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\sum_{j:j \neq i} \Phi(x_i, x_j) - N \int \Phi(x_i, y) dy \right]. \end{aligned}$$

Let us write

$$\sum_{j:j \neq i} \Phi(x_i, x_j) - N \int \Phi(x_i, y) dy = E_i^{(1)} + E_i^{(2)}, \quad (\text{B.28})$$

where

$$\begin{aligned} E_i^{(1)} &:= \sum_{j:j \neq i} (\Phi(x_i, x_j) - \Phi(x_i, x_i + \frac{j-i}{N})) \\ E_i^{(2)} &:= \sum_{j:j \neq i} \Phi(x_i, x_i + \frac{j-i}{N}) - N \int \Phi(x_i, x_i + y) dy. \end{aligned}$$

Let $x, y, a \in \mathbb{T}$. We have

$$\begin{aligned} &\left(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y) \right)^2 - \left(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}(y+a)) \right)^2 \\ &= \left(\psi(\ell_N^{-1}(y+a)) - \psi(\ell_N^{-1}y) \right) \left(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y) + \psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}(y+a)) \right). \end{aligned}$$

Therefore, by Lemma B.2, there exists a constant $C > 0$ such that

$$\begin{aligned} \left| \left(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}y) \right)^2 - \left(\psi(\ell_N^{-1}x) - \psi(\ell_N^{-1}(y+a)) \right)^2 \right| &\leq C|a| \left(w_N^{(1)}(\ell_N^{-1}(y+a)) + w_N^{(1)}(\ell_N^{-1}y) \right) \\ &\quad \times \left((w_N^{(1)}(\ell_N^{-1}x) + w_N^{(1)}(\ell_N^{-1}y))|x-y| + (w_N^{(1)}(\ell_N^{-1}(y+a)) + w_N^{(1)}(\ell_N^{-1}y))|x-y-a| \right). \end{aligned}$$

Moreover, by Lemma B.2, there exists a constant $C > 0$ such that

$$|\Phi(x, y)| \leq C|x-y|^2(w_N^{(1)}(\ell_N^{-1}x) + w_N^{(1)}(\ell_N^{-1}y)).$$

Thus, combining the two last displays, we get that there exists a constant $C > 0$ such that

$$\begin{aligned} |\Phi(x_i, x_j) - \Phi(x_i, x_i + \frac{j-i}{N})| &\leq C \left(w_N^{(1)}(\ell_N^{-1}x_i) + w_N^{(1)}(\ell_N^{-1}x_j) + w_N^{(1)}(\ell_N^{-1}(x_i + \frac{j-i}{N})) \right)^2 \\ &\quad \times \frac{|N(x_j - x_i) - N\frac{j-i}{N}|}{|N(x_j - x_i)|^{s+1}}. \end{aligned}$$

Using Theorem 1 and Lemma 4.5, we get that for all $\varepsilon > 0$, there exists a constant $C > 0$ depending on β , s and ε such that

$$\begin{aligned} \sum_{i=1}^N \mathbb{E}_{\mathbb{P}_{N,\beta}}[|E_i^{(1)}|] &\leq C(N\ell_N)^\varepsilon \sum_{i=1}^N \sum_{j \neq i: |j| \leq 2N\ell_N} (N\ell_N)^{2(1-s+\alpha)} \frac{1}{|j|^{2(1-s+\alpha)}} \frac{1}{|j-i|^{1+\frac{s}{2}}} \\ &\quad + C(N\ell_N)^\varepsilon \sum_{j: j \neq i} (N\ell_N)^{2(2-s)} \frac{1}{(j+N\ell_N)^{2(2-s)}} \frac{1}{|j-i|^{1+\frac{s}{2}}}, \end{aligned}$$

After some computations similar to the proof of Lemma 5.6, we find that for all $\varepsilon > 0$, there exists a constant $C > 0$ depending on β , s and ε such that

$$\sum_{i=1}^N \mathbb{E}_{\mathbb{P}_{N,\beta}}[|E_i^{(1)}|] \leq C(N\ell_N)^{\varepsilon + \max(2(1-s+\alpha), 1)}. \quad (\text{B.29})$$

For the discretization error, proceeding as in the proof of Lemma 5.6, one can write

$$\begin{aligned} |E_i^{(2)}| &\leq \frac{1}{N} \int_1^{\frac{N}{2}} |\partial_y \Phi(x_i, x_i + \frac{y}{N})| dy \\ &\leq C \int_1^{\frac{N}{2}} \frac{1}{y^{1+s}} \left(w_N^{(1)}(\ell_N^{-1}x_i)^2 + w_N^{(1)}(\ell_N^{-1}(x_i + \frac{y}{N}))^2 \right) dy \\ &\leq C w_N^{(1)}(\ell_N^{-1}x_i)^2. \end{aligned}$$

Summing the above estimate gives that for all $\varepsilon > 0$, there exists a constant $C > 0$ depending on β , s and ε such that

$$\sum_{i=1}^N \mathbb{E}_{\mathbb{P}_{N,\beta}}[|E_i^{(2)}|] \leq C(N\ell_N)^{\varepsilon + \max(2(1-s+\alpha), 1)}. \quad (\text{B.30})$$

In combination with (B.28) and (B.29), this gives (B.27). $\square$

We now turn to the proof of Lemma 5.7.

proof of Lemma 5.7. Let us denote

$$\text{Loop}_{\ell_N}^{\text{err}}[\psi] = \text{Loop}_{\ell_N}[\psi] - \text{Loop}_{\ell_N, \gamma}[\psi].$$

Let $w_N^{(1)}$ be as in (5.53). By Lemma B.2, there exists a constant $C > 0$ such that

$$|\text{Loop}_{\ell_N}^{\text{err}}[\psi]| \leq C \sum_{i=1}^N \sum_{j \in I_\gamma^c} (w_N^{(1)}(\ell_N^{-1}x_i) + w_N^{(1)}(\ell_N^{-1}x_j)) \frac{1}{|N(x_j - x_i)|^{1+\frac{s}{2}-\varepsilon}}. \quad (\text{B.31})$$

Using Lemma 4.5, we get that for all $\varepsilon > 0$, there exists a constant $C > 0$ depending on ε such that

$$\begin{aligned} \mathbb{E}_{\mathbb{P}_{N,\beta}}[\text{Loop}_{\ell_N}^{\text{err}}[\psi]^2] \\ \leq C(N\ell_N)^\varepsilon \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i=1}^N \sum_{j \in I_\gamma^c} (w_N^{(1)}(\ell_N^{-1}x_i) + w_N^{(1)}(\ell_N^{-1}x_j)) \frac{1}{|N(j-i)|^{1+\frac{s}{2}-\varepsilon}} \right)^2 \right]. \end{aligned}$$

Next, we use that for ε small enough, there exists a constant $C > 0$ such that

$$\begin{aligned} \sum_{i=1}^N \sum_{j \in I_\gamma^c} (w_N^{(1)}(\ell_N^{-1}x_i) + w_N^{(1)}(\ell_N^{-1}x_j)) \frac{1}{|N(j-i)|^{1+\frac{s}{2}-\varepsilon}} \\ \leq C \sum_{i=1}^N w_N^{(1)}(\ell_N^{-1}x_i) \frac{1}{1+d(i, I_\gamma^c)^{\frac{s}{2}-\varepsilon}} + C \sum_{j \in I_\gamma^c} w_N^{(1)}(\ell_N^{-1}x_j). \end{aligned}$$

Since the singularity of $w_N^{(1)}$ is in L^1 (since $1-s+\alpha < 1$), we get that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{i \in I_\gamma} w_N^{(1)}(\ell_N^{-1}x_i) \right)^2 \frac{1}{1+d(i, I_\gamma^c)^{\frac{s}{2}-\varepsilon}} \right]^{\frac{1}{2}} \leq C(N\ell_N)^{\kappa\varepsilon+1-\gamma(\frac{s}{2}-\varepsilon)}.$$

Moreover, since $|w_N^{(1)}(x)| \leq C|x|^{-(2-s)}$ for $|x| > 1$, we get that for all $\varepsilon > 0$, there exists $C > 0$ depending on β , s and ε such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\left(\sum_{j \in I_\gamma^c} w_N^{(1)}(\ell_N^{-1}x_j) \right)^2 \right]^{\frac{1}{2}} \leq C \frac{(N\ell_N)^{2-s+\varepsilon}}{(N\ell_N)^{\gamma(1-s)}}.$$

Combining the two above displays, we get that for all $\varepsilon > 0$, there exists $C > 0$ depending on β , s and ε such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\text{Loop}_{\ell_N}^{\text{err}}[\psi]^2]^{\frac{1}{2}} \leq C(N\ell_N)^\varepsilon \left((N\ell_N)^{2-s-\gamma(1-s-\varepsilon)} + (N\ell_N)^{1-\gamma(\frac{s}{2}-\varepsilon)} \right).$$

Therefore, for γ large enough, we get the existence of a constant $C > 0$ depending on β , s and γ such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}}[\text{Loop}_{\ell_N}^{\text{err}}[\psi]^2]^{\frac{1}{2}} \leq C(N\ell_N)^{\frac{1}{2}},$$

which concludes the proof of (5.57).

The proof of (5.58) is similar to the proof of (5.22), since that $1-s+\alpha < 1$. $\square$

B.3. Energy estimate.

Lemma B.6. *For every $q > 1$, there exists a constant $C > 0$ depending on β , s and q such that*

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[\max \left\{ \left(\sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} N^{-s} (g(x_{i+k} - x_i) - g(\frac{k}{N})) \right), 0 \right\}^q \right]^{\frac{1}{q}} \leq CN.$$

Proof. Let

$$F_N : X_N \in D_N \mapsto N^{-s} \iint_{\text{Diag}_{\mathbb{T}^2}^c} g(x-y) d\text{fluct}_N(x) d\text{fluct}_N(y).$$

Observe that

$$\mathcal{H}_N = N^{2-s} \iint g(x-y) dx dy + F_N.$$

Define, on the set of non-ordered particles, the partition function

$$K_{N,\beta} := \int_{\mathbb{T}^N} e^{-\beta F_N(X_N)} dX_N.$$

We now prove, following [Ser24, Ch. 5], that $\log K_{N,\beta} = O(N)$.

First, by [Ser24, Cor. 4.2.4], we have

$$F_N \geq -CN,$$

which gives

$$\log K_{N,\beta} \leq CN.$$

On the other hand, by Jensen's inequality,

$$\begin{aligned} \log K_{N,\beta} &\geq -\beta \int F_N(X_N) dx_1 \dots dx_N \\ &= -\beta N^{-s} \int \left(\sum_{i \neq j} g(x_i - x_j) - 2N \sum_{i=1}^N \int g(x_i - y) dy + N^2 \iint g(x - y) dx dy \right) dx_1 \dots dx_N \\ &= -\beta N^{-s} (N(N-1) - 2N^2 + N^2) \iint g(x - y) dx dy = \beta N^{1-s} \iint g(x - y) dx dy \geq 0. \end{aligned}$$

We deduce that there exists a constant $C > 0$ depending on β and s such that

$$|\log K_{N,\beta}| \leq CN. \quad (\text{B.32})$$

It follows that there exists a constant $C > 0$ depending on β and s such that

$$\log \mathbb{E}_{\mathbb{P}_{N,\beta}} \left[e^{\frac{\beta}{2} F_N} \right] = \log K_{N,\frac{\beta}{2}} - \log K_{N,\beta} \leq CN. \quad (\text{B.33})$$

Let

$$X := \sum_{i=1}^N \sum_{k=1}^{\lfloor \frac{N}{2} \rfloor} N^{-s} \left(g(x_{i+k} - x_i) - g\left(\frac{k}{N}\right) \right).$$

Notice that there exists $C > 0$ depending on s such that

$$F_N \geq -CN + X. \quad (\text{B.34})$$

We deduce from (B.33) and (B.34) that there exists $C > 0$ depending on β and s such that

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[e^{\frac{\beta}{2} X} \right] \leq e^{CN}.$$

Therefore, for every $u > 0$, we have

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} \left[e^{\frac{\beta}{2} X} \mathbb{1}_{X > u} \right] \leq e^{CN}. \quad (\text{B.35})$$

Let $q > 1$ and $u_q := (q-1)^q$. Let us define $\phi : u \in [u_q, \infty] \mapsto e^{u^{\frac{1}{q}}}$. One can notice that ϕ is convex and increasing. Therefore, by Jensen's inequality,

$$\mathbb{E}_{\mathbb{P}_{N,\beta}} [\exp(\max(X, u_q))] = \mathbb{E}_{\mathbb{P}_{N,\beta}} [\phi(\max(X, u_q)^q)] \geq \phi(\mathbb{E}_{\mathbb{P}_{N,\beta}} [\max(X, u_q)^q]).$$

Hence,

$$\log \mathbb{E}_{\mathbb{P}_{N,\beta}} [\exp(\max(X, u_q))] \geq \mathbb{E}_{\mathbb{P}_{N,\beta}} [\max(X, u_q)^q]^{\frac{1}{q}}.$$

Together with (B.35), this concludes the proof. $\square$

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